

IDENTIFICATION

PRODUCT CODE: MAINDEC-08-DHKMA-D-D
PRODUCT NAME: PDP-8E EXTENDED MEMORY DATA & CHECKERBOARD TEST
RELEASE DATE MAY 1978
MAINTAINER: DIAGNOSTIC ENGINEERING
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1. **ABSTRACT**

MODIFIED TO RUN ON APT SYSTEMS, APRIL 1975.
SEE NOTES AT END OF DOCUMENT.

MODIFIED TO RUN ON CLASSIC 8 SYSTEMS (CONSOLE PACKAGE).
SEE SECTION 10.

MODIFIED TO RUN ON SYSTEMS WITH NO CONSOLE TERMINAL.
REFER TO SECTIONS STARTING AT SECTION 11 FOR PROGRAM INITIALIZATION,
OPERATING PROCEDURES, SWITCH REGISTER SETTINGS AND ERROR REPORTING.

THE DIAGNOSTIC HAS BEEN MODIFIED TO ACCOMMODATE THE KT8-A
MEMORY MANAGEMENT OPTION WITH MEMORY ADDRESSING OF UP TO 128K
WORD OF READ WRITE MEMORY.

THE PDP-8E EXTENDED MEMORY DATA & CHECKERBOARD TEST IS
DESIGNED TO DETECT MEMORY FAILURE DUE TO SENSE-LINE
NOISE UNDER WORST CASE CONDITIONS. THE FOUR WORST CASE
PATTERNS PROVIDED WILL GENERATE WORST CASE
NOISE CONDITIONS IN ALL STANDARD AND SPECIALLY PURCHASED
PDP-8E CORE STACKS, AND WILL TEST SYSTEMS EQUIPPED WITH
FROM 8K TO 128K WORDS OF CORE MEMORY. THE ALL 0'S AND ALL
1'S PATTERNS ARE PROVIDED TO IDENTIFY BASIC MEMORY FAILURES.
AUTOMATIC PROGRAM RELOCATION IS PROVIDED IN ORDER TO TEST
ALL MEMORY FIELDS FROM EACH MEMORY FIELD. TELETYPE PRINTOUTS
ARE PROVIDED FOR ERROR IDENTIFICATION, AND THE OPERATOR
IS GIVEN A DEGREE OF CONTROL OVER THE PROGRAM BY VARIOUS
SWITCH REGISTER SETTINGS.

2. **REQUIREMENTS**

2.1 **EQUIPMENT**

A PDP-8E COMPUTER EQUIPPED WITH AT LEAST 8K OF CORE MEMORY.

2.2 **STORAGE**

THE PROGRAM OCCUPIES CORE LOCATIONS 0000 TO 7577 IN THE PRESENT FIELD.

2.3 **PRELIMINARY PROGRAMS**

THE BINARY LOADER MUST BE IN MEMORY. ALSO, ALL DIAGNOSTICS
FOR A BASIC 4K PDP-8E MUST HAVE BEEN PREVIOUSLY RUN
SUCCESSFULLY. IF A KT8-A IS AVAILABLE, THE KT8-A DIAGNOSTIC
MUST BE RUN SUCCESSFULLY.

3. **LOADING PROCEDURE**

LOAD THE PROGRAM WITH THE BINARY LOADER (BIN). THE PROGRAM
MAY BE LOADED INTO ANY FIELD.

4. **OPERATING PROCEDURE**

SR00-02 SR02, IF SET, WILL RING THE TTY BEL' ONCE FOR EACH ERROR. SR00 AND SR01 HAVE NO EFFECT WITH SR02 SET.

SR03 SR03 MAY BE SET OR RESET AT ANY TIME AND THE PROGRAM
WILL ACT ACCORDINGLY.
SR04 SR04 ALLOWS THE OPERATOR TO CHANGE THE FIELD LIMITS
AS DEFINED BY SR06-11.
SR05 SR05 IS NORMAL HALT FOR PROGRAM.
SR06-08 THESE SWITCHES DEFINE THE STARTING FIELD LIMIT

4.2 EXAMPLE OF SELECTING FIELDS FOR TEST

EXAMPLE 1: SR = 0037, 28K SYSTEM
FIELDS SELECTED FOR TESTING ARE 0:6543210

EXAMPLE 2: SR = 0004, 28K SYSTEM
FIELDS SELECTED FOR TESTING ARE 0:43210

EXAMPLE 3: SR = 0202, 28K SYSTEM
FIELDS SELECTED FOR TESTING ARE 0:2 (NO RELOCATION
WILL OCCUR).

EXAMPLE 4: SR = 0401, 28K SYSTEM
FIELDS SELECTED FOR TESTING ARE 0:64310

NOTE 1: FIELDS NOT IN THE SYSTEM ARE AUTOMATICALLY
DESELECTED AS IN EXAMPLE 1. FIELD 7 IS NOT
PRESENT, THEREFORE, NOT SELECTED.

NOTE 2: DO NOT SELECT A FIELD THAT CONTAINS A ROM.

NOTE 3: A SINGLE FIELD CAN BE SELECTED FOR TESTING
PROVIDING THE PROGRAM IS NOT IN THAT FIELD
AS IN EXAMPLE 3.

NOTE 4: ANY FIELD OR GROUP OF FIELDS CAN BE BY-PASSED
AS IN EXAMPLE 4. FIELDS 2 AND 3 ARE NOT SELECTED,
FIELD 7 IS NOT PRESENT.

NOTE 5: THE ABOVE INFORMATION ALSO APPLIES TO SYSTEMS WITH UPTO
128K WORDS OF READ WRITE MEMORY. FOR EACH ADDITIONAL
BANK OF MEMORY THE PROGRAM WILL PRINT THE FOLLOWING DEPENDING
ON THE AMOUNT OF MEMORY AVAILABLE.

"FIELDS SELECTED FOR TESTING 3:76543210 2:76543210 :76543210 0:76543210"

5. ERRORS

A TEST ERROR WILL OCCUR ANYTIME THE DATA WRITTEN DOES NOT
MATCH THE DATA READ. A RELOCATION ERROR WILL OCCUR IF THE
RELOCATION COMPARISON CHECK FAILS.

5.1 TEST ERROR TYPEOUTS

FOR THE FIRST ERROR ENCOUNTERED A HEADER WILL BE TYPED OUT
FOLLOWED BY THE PERTINENT DATA. FOR ALL SUBSEQUENT ERRORS,
ONLY THE PERTINENT DATA WILL BE TYPED. THE FORMAT IS AS
FOLLOWS:

PR.LOC.. FAIL. ADR..GOOD..BAD..PATTERN

PR LOC = THE PROGRAM ADDRESS WHERE THE ERROR JMS OCCURRED.
(INCLUDES FIELD).

FAIL ADR = THE ADDRESS OF THE LOCATION IN ERROR.
(INCLUDES FIELD).

GOOD = THE DATA THAT WAS WRITTEN.

BAD = THE DATA THAT WAS READ.

PATTERN= THE PRESENT TEST PATTERN AND THE NUMBER
OF TIMES IT WAS COMPLEMENTED.
NC (NOT COMPLEMENTED).
1C (ONE COMPLEMENT).
2C (TWO COMPLEMENTS).

5.2 RELOCATION ERROR TYPEOUTS

ALL RELOCATION ERRORS ARE IN THE FOLLOWING FORMAT:

XXXXXX RELOCATION ERROR AT LOCATION YYYYYY

XXXXXX = THE PROGRAM ADDRESS WHERE THE ERROR JMS OCCURRED.
(INCLUDES FIELD).

YYYYYY = THE ADDRESS OF THE LOCATION IN ERROR.
(INCLUDES FIELD).

NOTE: AFTER EACH ERROR PRINT-OUT THE PROGRAM CONTINUES
ON WITH THE NEXT SEQUENTIAL MEMORY LOCATION.

5.3 PARITY ERROR TYPEOUTS

IF THE BE SYSTEM CONTAINS A PARITY OPTION THE INTERRUPT WILL
BE TURNED ON TO ALLOW PARITY ERRORS WHEN THE PROGRAM IS
EXECUTING FROM FIELD 0. THE FOLLOWING 3 TYPEOUTS CAN OCCUR
WITH A PARITY OPTION:

A. PARITY ERROR, LOC 0=XXXX TSTAD=XXXXXX (PRESENT PATTERN)

B. INTERRUPT FROM KEYBOARD

C. UNWANTED INTERRUPT OCCURRED

6. RESTRICTIONS

6.1 STARTING RESTRICTIONS

THE PROGRAM MAY BE RESTARTED AT ANY TIME FROM LOCATION 0200
OR 0202 OF THE FIELD THE PROGRAM IS PRESENTLY IN.

6.2 OPERATING RESTRICTIONS

THE PARITY ERROR TYPEOUT CAN NOT BE INHIBITED.

7. EXECUTION TIME

THE TIME TO WRITE AND READ ALL SIX PATTERNS IN ONE FIELD IS APPROXIMATELY 6 SECONDS.

DURING PROGRAM EXECUTION A 15 WILL BE TYPED ON THE TTY APPROXIMATELY EVERY 15 MINUTES OF PROGRAM RUN TIME. THIS ALLOWS THE OPERATOR TO DETERMINE APPROXIMATE RUN TIME BEFORE A FAILURE OCCURRED.

NOTE: IT SHOULD BE NOTED THAT IF THE PROGRAM IS RELOCATING THE DIAGNOSTIC WILL PRINT AN END OF PASS MESSAGE AT THE COMPLETION OF ALL FIELDS SELECTED FOR RELOCATION. THE MESSAGE WILL BE 'S FOLLOWS:

"END OF PASS XXXX"

WHERE XXXX= THE OCTAL VALUE OF THE NUMBER OF TIMES THE PROGRAM HAS RELOCATED THROUGH ALL SELECTED FIELDS.

8. SCOPE LOOPS

8.1 SCOPE LOOP 1

THIS SCOPE LOOP DOES A READ, COMPLEMENT, WRITE ON THE ADDRESS SPECIFIED BY THE SR. THE ADDRESS BEING LOOPED ON CAN BE CHANGED SIMPLY BY CHANGING THE SWITCH SETTING. THE PREVIOUS ADDRESS WILL BE LEFT WITH ITS ORIGINAL CONTENT.

- A. SET THE SR TO THE INSTRUCTION FIELD THAT THE PROGRAM IS IN AND THE DATA FIELD WANTED TO TEST.
- B. PRESS KEY EXTD ADDR LOAD.
- C. SET THE SR EQUAL TO 5200.
- D. PRESS KEY ADDR LOAD.
- E. SET THE SR EQUAL TO THE ADDRESS TO TEST.
- F. PRESS KEYS CLEAR, AND CONT.

8.2 SCOPE LOOP 2

THIS SCOPE LOOP DOES A READ, COMPLEMENT, WRITE ON THE TWO ADDRESSES INPUT VIA THE SR. TO CHANGE THE ADDRESSES, THE LOOP MUST BE RESTARTED.

- A. SET THE SR TO THE INSTRUCTION FIELD THAT THE PROGRAM IS IN THE DATA FIELD WANTED TO TEST.
- B. PRESS KEY EXTD ADDR LOAD.
- C. SET THE SR EQUAL TO 5400.
- D. PRESS KEYS ADDR LOAD, CLEAR, AND CONT.

E. FOLLOW DIRECTIONS THAT ARE TYPED OUT.

8.3 SCOPE LOOP 3

THIS SCOPE LOOP DOES A READ, COMPLEMENT, WRITE ON THE GROUP OF ADDRESSES INPUT VIA THE SR. THE STARTING ADDRESS SPECIFIED MUST BE LESS THAN THE ENDING ADDRESS SPECIFIED.

- A. SET THE SR TO THE INSTRUCTION FIELD THAT THE PROGRAM IS IN AND THE DATA FIELD WANTED TO TEST.
- B. PRESS KEY EXT D ADL. LOAD.
- C. SET THE SR EQUAL TO 5600.
- D. PRESS KEYS ADDR LOAD, CLEAR, AND CONT.
- E. FOLLOW DIRECTIONS THAT ARE TYPED OUT.

8.4 SCOPE LOOP 4

THIS SCOPE LOOP DOES A READ, COMPLEMENT, WRITE ON THE ADDRESS INPUT VIA THE SR USING THE DATA SPECIFIED BY THE SR. THE DATA CAN BE CHANGED SIMPLY BY CHANGING THE SWITCH SETTING.

- A. SET THE SR TO THE INSTRUCTION FIELD THAT THE PROGRAM IS IN AND THE DATA FIELD WANTED TO TEST.
- B. PRESS KEY EXT D ADDR LOAD.
- C. SET THE SR EQUAL TO 6000.
- D. PRESS KEYS ADDR LOAD, CLEAR, AND CONT.
- E. A MESSAGE WILL BE TYPED OUT TO SET THE SR TO THE SELECTED ADDRESS.
- F. SET SR TO THE SELECTED ADDRESS AND DEPRESS CONT.
- G. SET S: TO SELECTED DATA (SCOPE LOOP IS CYCLING).

8.5 SCOPE LOOP 5

THIS SCOPE LOOP DOES A READ, COMPLEMENT, WRITE ON THE GROUP OF ADDRESSES INPUT VIA THE SR USING THE DATA SPECIFIED BY THE SR. THE STARTING ADDRESS SPECIFIED MUST BE LESS THAN THE ENDING ADDRESS SPECIFIED.

- A. SET THE SR TO THE INSTRUCTION FIELD THAT THE PROGRAM IS IN AND THE DATA FIELD WANTED TO TEST.
- B. PRESS KEY EXT D ADDR LOAD.
- C. SET THE SR EQUAL TO 5700.
- D. PRESS KEYS ADDR LOAD, CLEAR, AND CONT.

E. FOLLOW THE TYPED OUT MESSAGE THAT INPUTS THE ADDRESS
SELECTION.

F. SET SR TO SELECTED DATA (SCOPE LOOP IS CYCLING).

NOTE 1: THE ADDRESS(S) SPECIFIED WILL BE LOOPED UNTIL STOPPED
BY THE OPERATOR WITH KEY HALT. NO ERROR CHECKING IS DONE.
TO RESUME NORMAL OPERATION, RESTART PROGRAM AT
ADDRESS 0200 OR 0202 OF THE CURRENT INSTRUCTION FIELD.

9. PROGRAM DESCRIPTION

9.1 TEST PATTERNS

THE FOLLOWING TEST PATTERNS ARE EMPLOYED BY THE PROGRAM:

- A. BASIC ALL 0'S PATTERN.
- B. BASIC ALL 1'S PATTERN.
- C. 0000-7777 WORST CASE CHECKERBOARD PATTERN.
- D. 7777-0000 WORST CASE CHECKERBOARD PATTERN.
- E. 2525-5252 WORST CASE CHECKERBOARD PATTERN.
- F. 5252-2525 WORST CASE CHECKERBOARD PATTERN.

9.2 PROGRAM RELOCATION

PROGRAM RELOCATION IS GOVERNED BY THE STATUS OF SR BIT 3 OR BY THE FACT THAT ONLY ONE FIELD IS SELECTED FOR TESTING. WITH SR BIT 3 DOWN (0 POSITION) PROGRAM RELOCATION OCCURS EACH TIME THE TEST PATTERN AND ITS COMPLEMENT HAVE BEEN COMPLETELY TESTED IN EACH SELECTED FIELD. THE PROGRAM FIRST RELOCATES TO THE HIGHEST ORDER 4K FIELD UNDER TEST. THE PROGRAM KEEPS RELOCATING TO THE NEXT LOWER FIELD UNDER TEST UNTIL IT REACHES THE LOWEST ORDER FIELD UNDER TEST. THE TESTING AND RELOCATION CYCLE IS THEN REPEATED. THE CONTENTS OF THE ENTIRE FIELD ARE RELOCATED WHICH ENABLES ANY OTHER INFORMATION (RIM-BIN) TO BE CARRIED WITH THE PROGRAM.

THE PROGRAM PROVIDES A DEGREE OF PROTECTION FOR ITSELF BY REMEMBERING ALL FIELDS WHERE ERRORS OCCUR. WHEN A FAULTY FIELD IS NEXT IN SEQUENCE TO CONTAIN THE PROGRAM, THE PROGRAM WILL SKIP THE FAULTY FIELD AND RELOCATE TO THE FIRST LOWER ORDER FIELD WHICH IS ERROR FREE. IF ALL OTHER SELECTED FIELDS ARE FAULTY, PROGRAM RELOCATION WILL NOT TAKE PLACE.

DURING RELOCATION A COMPARISON CHECK IS MADE TO INSURE NO PROGRAM LOSS.

9.3 TEST PROCEDURE

- A. WRITE THE PATTERN IN ALL SELECTED FIELDS (EACH LOCATION IS THEN TREATED AS FOLLOWS):

- B. READ-WRITE THE LOCATION 11 TIMES.
- C. READ-WRITE-TEST THE LOCATION (NC).
- D. READ-WRITE THE LOCATION 11 TIMES.
- E. READ-COMPLEMENT-WRITE THE LOCATION.
- F. READ-WRITE THE LOCATION 11 TIMES.
- G. READ-WRITE-TEST THE LOCATION (1C).
- H. READ-WRITE THE LOCATION 11 TIMES.
- I. READ-COMPLEMENT-WRITE THE LOCATION.
- J. READ-WRITE THE LOCATION 11 TIMES.
- K. READ-WRITE-TEST THE LOCATION (2C).
- L. GO ON TO NEXT LOCATION REPEATING B-K.
- M. GO ON TO NEXT PATTERN REPEATING A-L WHEN ALL LOCATIONS OF ALL SELECTED FIELDS ARE COMPLETED.

FOR FURTHER UNDERSTANDING OF HOW THE TEST IS PERFORMED, REFER TO THE LISTING.

THE WORST CASE CHECKERBOARD PATTERN CONSISTS OF ALTERNATING 4 MEMORY CORES CONTAINING 0000 AND 4 MEMORY CORES CONTAINING 1111 ON A MEMORY PLANE. THIS PATTERN IS REVERSED EVERY 400 OCTAL LOCATIONS. (THIS TEST PATTERN IS GENERATED ACCORDING TO THE STRINGING OF THE STACK AND THE WIRING OF THE MEMORY SYSTEM. IT IS THE SAME PATTERN FOR ALL 8K STACKS).

Y LINES (MA6L THRU MA11L)

ADDRESS BIT 9 HIGH					ADDRESS BIT 9 LOW								
	00	01	02	03	04	05	06	07	10	11	→	76	77
ADDRESS	00	1	1	1	1	0	0	0	0	1	1	0	0
BIT 3 HIGH	01	1	1	1	1	0	0	0	0	1	1	0	0
	02	1	1	1	1	0	0	0	0	1	1	0	0
	03	1	1	1	1	0	0	0	0	1	1	0	0

ADDRESS	04	0	0	0	0	1	1	1	1				
BIT 3 LOW	05	0	0	0	0	1	1	1	1				
	06	0	0	0	0	1	1	1	1				
	07	0	0	0	0	1	1	1	1				

	10	1	1	1	1								
	11	1	1	1	1								

	76	0	0	0	0								
	77	0	0	0	0								

	176	0	0	0	0								
	177	0	0	0	0								

EMA2L USED IF AN 8K MEMORY

EMA2L USED IF AN 8K MEMORY

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THE ABOVE REPRESENTS ONE MEMORY PLANE.

8102400 1.01

1910200 1.02

1904 1.03

1904 1.04

1904 1.05

1904 1.06

1904 1.07

1904 1.08

1904 1.09

1904 1.10

1904 1.11

1904 1.12

1904 1.13

1904 1.14

1904 1.15

1010100 2.01

1004 1.16

1004 1.17

1004 1.18

1004 1.19

1004 1.20

1004 1.21

1004 1.22

1004 1.23

1004 1.24

1004 1.25

1004 1.26

10. CONSOLE PACKAGE ADDENDUM

10.1. DESCRIPTION

THE CONSOLE PACKAGE HAS BEEN ADDED TO THIS DIAGNOSTIC TO ALLOW THE PROGRAM TO RUN WITH NO HARDWARE SWITCH REGISTER AND TO HAVE COMMUNICATIONS WITH THE DIAGNOSTIC VIA A TERMINAL. THE DIAGNOSTIC CAN BE RUN IN TWO MODES WITH THE CONSOLE PACKAGE . 1) RUNNING WITH THE CONSOLE PACKAGE ACTIVE - THIS ALLOWS THE OPERATOR CONTROL OF THE DIAGNOSTIC THROUGH THE TERMINAL. THE DIAGNOSTIC WILL ASK FOR THE VALUE OF THE PSEUDO SWITCH REGISTER, BEFORE CONTINUING WITH EXECUTION OF THE DIAGNOSTIC. ALL ERRORS AND PASS COMPLETES WILL BE PRINTED AT THE TERMINAL. NO HALTS WILL BE EXECUTED. 2) CONSOLE PACKAGE NOT ACTIVE-THIS WILL RESULT IN THE NORMAL STANDALONE OPERATION OF THE PROGRAM AS DISCRIBED IN SECTIONS 1 THROUGH 9 OF THIS DOCUMENT.

10.2 RESTRICTIONS

- 1) WHEN RUNNING THE CONSOLE PACKAGE SOME SUBTESTS MAY NOT BE EXECUTED.
- 2) THE CONSOLE PACKAGE WILL USE EITHER THE HARDWARE OR PSEUDOSWITCH REGISTER DEPENDING UPON THE CONDITION OF BIT 0 OF ADDRESS 21(HCW1). (SEE SECTION 10.8 FOR EXPLANATION)
- 3) ONCE RUNNING THE CONSOLE PACKAGE NONACTIVE AND NOW DESIRE TO RUN IT ACTIVE. ONE MUST RELOAD THE DIAGNOSTIC AND INITILIZE FOR A ACTIVE CONSOLE PACKAGE.

10.3 INITIALIZATION

FOR A ACTIVE CONSOLE PACKAGE

- 1.) SET LOCATION 22 BIT3=1 TO INDICATE CONSOLE PACKAGE ACTIVE.
- 2.) SET LOCATION 21 BIT0=0 TO INDICATE USE PSEUDO SWITCH REGISTER.
- 3.) SET LOCATION 21 BIT0=1 TO INDICATE NOT TO USE PSEUDO SWITCH REGISTER, BUT TO USE HARDWARE SWITCHES.

10.4 CONTROL CHARACTERS

CONTROL CHARACTERS ARE USED TO GIVE THE OPERATOR THE ABILITY TO PERFORM THE FOLLOWING FUNCTIONS.

NOTE: THE PROGRAM WILL RESPOND TO THE CONTROL CHARACTER IN FIVE (5) SECONDS OR LESS.

CONTROL C

THIS RESETS THE LOADER (PGS 37 OF FLD 0 & 1) AND STARTS IT AT LOC 7600 OF FLD 0.

CONTROL S

THIS WILL STOP ANY OUTPUT TO A CONSOLE TERMINAL. THE ONLY WAYS TO CONTINUE IS TO TYPE CONTROL Q TO RESUME PRINTING OR CONTROL C TO ABORT THE PROGRAM COMPLETELY. THIS IS A NONPRINTING CHARACTER.

CONTROL Q

THIS IS TO CONTINUE A PROGRAM AFTER A CONTROL S IS TYPED. THIS IS A NONPRINTING CHARACTER.

CONTROL G

THE CONTROL G ALLOWS THE OPERATOR TO CHANGE THE VALUE OF THE PSEUDO SWITCH REGISTER. UPON TYPING "CTRL" AND A "G" SIMULTANEOUSLY THE KEYBOARD WILL RESPOND WITH "G" AND PRINT THE SWITCH REGISTER QUESTION. (SEE SECTION 10.5 FOR DETAILS) AT THIS POINT THE OPERATOR MAY CHANGE THE VALUE OF THE PSEUDO SWITCHES OR TYPE A TERMINATING CHARACTER. IN ANY EVENT ONLY THE PSEUDO SWITCH REGISTER IS CHANGED. IT HAS NO EFFECT UPON THE HARDWARE SWITCHES.

10.5 SWITCH REGISTER MESSAGE

THIS MESSAGE IS USED TO SETUP THE PSEUDO SWITCH REGISTER BEFORE PROGRAM EXECUTION TAKES PLACE OR TYPING A CONTROL G. THE PSEUDO SWITCH REGISTER IS SET UP UPON TYPING A TERMINATOR. THE TERMINATORS ARE AS FOLLOWS:

<CR> CARRIAGE RETURN: THIS CAUSES THE PROGRAM TO RESUME TESTING FROM WHERE IT LEFT OFF.

<LF> LINEFEED: THIS CAUSES THE PROGRAM TO RESTART THE TESTING FROM THE BEGINNING.

SR=0000 4000<CR> CARRIAGE RETURN
PROGRAM RESUMES TESTING FROM
POINT OF INTERRUPTION.

SR=0000 4000<LF> LINEFEED
CAUSES PROGRAM TO RESTART.

UNDER SCORING INDICATES OPERATOR RESPONSE

10.6 ERRORS

THE STANDARD ERROR REPORTS AS DESCRIBED IN SECTION 6 OF THIS DOCUMENT WILL BE USED.

10.7 SWITCH REGISTER SETTINGS

THE STANDARD SWITCH SETTINGS AS DESCRIBED IN SECTION 5 OF THIS DOCUMENT WILL BE USED.

10.8 PARAMETER CONTROL WORDS

THE CONSOLE PACKAGE USES THE LOCATIONS 20 21 22 FOR THE FOLLOWING PURPOSES.

LOCATION 20
PSEUDO SWITCH REGISTER

LOCATION 21
HARDWARE IDENTIFIER 1

LOCATION 22
HARDWARE IDENTIFIER 2

LOCATION 002

<u>BIT</u>	<u>OCTAL VALUE</u>	<u>FUNCTION WHEN 0</u>	<u>FUNCTION WHEN 1</u>
0	4000	USE PSEUDO SWITCHES	USE HARDWARE SWITCHES
1	2000	NO OPTION 1	HAS OPTION 1
2	1000	NO OPTION 2	HAS OPTION 2
3	400	NO 8A SIMULATOR	HAS 8A SIMULATOR
4	200	NO OPTION SIMULATOR	HAS OPTION SIMULATOR
5	100	NOT ON 8A XOR	ON 8A XOR
6	40	NOT PDP8-E TYPE CPU	PDP8-E TYPE CPU
7-11		8A MEMORY SIZE EX. 1K=00 2K=01 7K=06 32K=31	

LOCATION 0022

<u>BIT</u>	<u>OCTAL VALUE</u>	<u>FUNCTION WHEN 0</u>	<u>FUNCTION WHEN 1</u>
0	4000	NOT ON ACT8A LINE	ON ACT 8A LINE
1	2000	NOT ON ACT 8E LINE	ON ACT 8E LINE
2	1000	NOT YET DEFINED	
3	400	DEACTIVE CONSOLE PACKAGE	ACTIVE CONSOLE PACKAGE

11.0. NON CONSOLE TERMINAL SYSTEM ADDENDUM

11.1. DESCRIPTION

THE PROGRAM HAS BEEN MODIFIED TO RUN WITHOUT A CONSOLE TERMINAL BY MEANS OF A SPECIAL STARTING ADDRESS AND OPERATING PROCEDURES. THIS ALLOWS THE DIAGNOSTIC TO BE RUN ON THOSE SYSTEMS WITHOUT A CONSOLE TERMINAL. ALL ERRORS AND FIELD LIMIT CHANGES WILL RESULT IN A HALT OR HALTS INSTEAD OF TYPEOUTS ON THE CONSOLE TERMINAL.

11.2. RESTRICTIONS

1. IF THE CONSOLE PACKAGE WAS ENABLED, THE PROGRAM WILL DISABLE IT AT THE START OF THE PROGRAM.
2. FIELD LIMITS MUST BE SET AT PROGRAM START, OTHERWISE, THE PROGRAM WILL HALT TO ALLOW THE OPERATOR TO SET THE FIELD

LIMITS IN THE SWITCH REGISTER.

3. TO RUN THIS PROGRAM, A MINIMUM OF 8K OF MEMORY IS REQUIRED.
4. MEMORIES TO BE TESTED MUST BE IN SEQUENTIAL ORDER STARTING AT FIELD 0.

11.3 INITIALIZATION

THE PROGRAM WHEN LOADED IS INITIALIZED TO USE THE HARDWARE SWITCH REGISTER. IF NO HARDWARE SWITCH REGISTER IS AVAILABLE, DO THE FOLLOWING TO DISABLE THE SWITCH REGISTER SELECTION FROM HARDWARE TO A SOFTWARE PSEUDO SWITCH REGISTER (LOCATION 0020).

1. SET BIT 0 EQUAL TO A 0 IN LOCATION 21 TO INDICATE TO THE PROGRAM THAT LOCATION 20 WILL BE USED AS THE PSEUDO SWITCH REGISTER. THE PROGRAM WHEN STARTED WILL THEN SET THE PSEUDO SWITCH REGISTER TO FIELD LIMITS FOR A NORMAL SYSTEM STARTUP. PSEUDO SWITCH REGISTER WILL EQUAL XX00 WHERE XX EQUALS SWITCH REGISTER BITS, PREVIOUSLY SET, 0 EQUALS STARTING FIELD LIMIT AND 7 EQUALS ENDING FIELD LIMITS.

IF IT IS DESIRED TO INITIALIZE THE FIELD LIMITS TO OTHER THAN THE ABOVE DO THE NEXT STEP.

2. SET LOCATION 0021 TO 00XX WHERE XX IS THE MEMORY SIZE IN 4K INCREMENTS.

11.4 OPERATING PROCEDURES

TO START THE PROGRAM:

- A. SET THE IF AND DF TO THE FIELD THAT CONTAINS THE PROGRAM
- B. LOAD ADDRESS TO 0201
- C. IF THE HARDWARE SWITCH REGISTER IS USED, SET THE SWITCH REGISTER TO 0037.
- D. PRESS "INIT" AND THEN "RUN".
- E. THE PROGRAM WILL AGAIN HALT. AT THIS TIME SELECT THE DESIRED TEST PARAMETERS.
- F. PRESS "INIT" AND "RUN".
- G. THE PROGRAM WILL NOW RUN UNTILL AN ERROR IS ENCOUNTERED OR A SWITCH REGISTER OPTION IS SELECTED TO CAUSE THE PROGRAM TO HALT REFER TO LISTING FOR ALL HALTS.
- H. SETTING THE SWITCH REGISTER TO 0100, WILL CAUSE THE PROGRAM TO HALT AFTER THE CURRENT TEST-REFER TO LISTING FOR HALT.
- I. SETTING THE SWITCH REGISTER TO 0200 WILL CAUSE THE PROGRAM TO HALT FOR FIELD LIMIT CHANGES VIA THE SWITCH REGISTER. REFER TO LISTING FOR ADDRESS OF THE HALT.

11.5 SWITCH REGISTER SETTINGS

SR0=1	HALT AFTER ERROR
SR1=1	INHIBIT ERROR HALTS EXCEPT HALT AFTER ERROR SWITCH
SR2=1	INHIBIT OPERATION OF SR0 AND SR1
SR3=1	INHIBIT PROGRAM RELOCATION
SR4=1	HALT PROGRAM FOR FIELD LIMIT CHANGES
SR5=1	HALT AFTER CURRENT TEST

11.6 ERRORS

ALL ERRORS ENCOUNTERED WILL RESULT IN A ERROR HALT WITH ERROR INFORMATION IN THE AC. REFER TO THE LISTING FOR THE TYPE OF ERROR HALT AND GO TO THE APPROPRIATE PARAGRAPH BELOW.

A TEST ERROR WILL OCCUR ANYTIME THE DATA WRITTEN DOES NOT MATCH THE DATA READ. A RELOCATION ERROR WILL OCCUR IF THE RELOCATION COMPARISON CHECK FAILS.

11.7 TEST ERROR HALTS

FOR ERRORS ENCOUNTERED TESTING MEMORIES, THE PROGRAM WILL HALT WITH PERTINENT INFORMATION IN THE AC. REFER TO THE STEPS BELOW FOR THE TEST ERROR INFORMATION

- A. PRESS "CONT". THE PROGRAM WILL HALT AT 3115 WITH THE AC EQUAL TO THE PROGRAM ADDRESS OF THE ERROR JMS.
- B. PRESS "CONT". THE PROGRAM WILL HALT AT 3121 WITH THE CONTENTS OF AC BITS 7,8,9,10 AND 11 EQUAL TO THE FIELD BEING TESTED.
- C. PRESS "CONT". THE PROGRAM WILL HALT AT 3124 WITH THE AC EQUAL TO THE FAILING ADDRESS IN THE FIELD BEING TESTED.
- D. PRESS "CONT". THE PROGRAM WILL HALT AT 3127 WITH THE AC EQUAL TO THE EXPECTED DATA THAT WAS PUT INTO THE FAILING ADDRESS.
- E. PRESS "CONT". THE PROGRAM WILL HALT AT 3132 WITH THE AC EQUAL TO THE ACTUAL DATA THAT WAS READ FROM THE FAILING ADDRESS.
- F. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3136 WITH THE PATTERN NUMBER IN THE AC. THE PATTERN NUMBER CORRESPONDS TO THE PATTERNS BELOW.

AC=0	-	NO TEST PATTERN
AC=1	-	BASIC ALL 0'S PATTERN
AC=2	-	BASIC ALL 1'S PATTERN
AC=3	-	0000-7777 WCP PATTERN
AC=4	-	7777-0000 WCP PATTERN
AC=5	-	2525-5252 WCP PATTERN
AC=6	-	5252-2525 WCP PATTERN

- H. PRESS "CONT" TO CONTINUE THE PROGRAM ON TO THE NEXT SEQUENTIAL TEST MEMORY ADDRESS.
- I. ERROR HALTS MAY BE INHIBITED BY SETTING SR1 TO A 1

11.8 RELOCATION ERROR HALTS

ALL RELOCATION ERRORS WILL RESULT IN A HALT WITH PERTINENT INFORMATION IN THE AC. REFER TO THE STEPS BELOW FOR THE ERROR INFORMATION.

- A. THE PROGRAM WILL HALT AT ADDRESS 2735 WITH THE CONTENTS OF THE AC EQUAL TO THE PROGRAM LOCATION OF THE ERROR JMS.
- B. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 2741 WITH THE CONTENTS OF AC BITS 7,8,9,10, AND 11 EQUAL TO THE FIELD THAT PROGRAM TRIED TO PUT THE INSTRUCTION INTO.
- C. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 2744 WITH THE CONTENTS OF AC EQUAL TO THE LOCATION IN THE FAILING FIELD IN ERROR.
- D. PRESSING "CONTINUE" AGAIN WILL RESULT IN THE PROGRAM CONTINUING WITH THE NEXT SEQUENTIAL MEMORY LOCATION.

11.9 PARITY ERROR HALTS

IF THE SYSTEM CONTAINS A PARITY OPTION, THE INTERRUPT WILL BE TURNED ON TO ALLOW PARITY ERRORS WHEN THE PROGRAM IS EXECUTING FROM FIELD 0. THERE ARE 3 TYPES OF FAILURES UNDER THIS ERROR, REFER TO THE APPROPRIATE PARAGRAPH BELOW FOR THE FAILING ADDRESS.

11.9.1 PARITY ERROR

- A. THE PROGRAM WILL HALT AT ADDRESS 3355 WITH THE CONTENTS OF THE AC EQUAL TO THE INTERRUPTED PC.
- B. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3361 WITH THE CONTENTS OF THE AC EQUAL TO THE DATA FIELD AT THE TIME OF THE PARITY ERROR.
- C. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3364 WITH THE CONTENTS OF THE AC EQUAL TO THE ADDRESS IN THE TEST FIELD BEING TESTED.
- D. PRESS "CONT". THE PROGRAM WILL HALT AT ADDRESS 3136 WITH THE CONTENTS OF THE AC EQUAL TO THE PATTERN NUMBER. REFER TO STEP G FOR PATTERN BEING EXECUTED IN SECTION 11.7.
- E. PRESS "CONT". THE PROGRAM WILL CONTINUE UNTIL ANOTHER ERROR IS ENCOUNTERED OR THE PROGRAM IS STOPPED.

11.9.2 INTERRUPT FROM KEYBOARD

THE PROGRAM WILL HALT AT ADDRESS 3404. THIS SIGNIFIES THAT THE PROGRAM DETECTED A PARITY OPTION AND TURNED THE INTERRUPT ON. UPON TURNING THE INTERRUPT ON, A INTERRUPT WAS RECEIVED FROM THE CONSOLE TERMINAL. TO COVER FROM THIS ERROR PRESS CONTINUE.

11.9.3 UNWANTED INTERRUPT OCCURRED

THE PROGRAM WILL HALT AT ADDRESS 3425 FOR THIS ERROR. THIS
ERROR SIGNIFIES THAT AN INTERRUPT OCCURED FROM SOME
OTHER DEVICE THAN THE PARITY OPTION OR THE CONSOLE
KEYBOARD. PRESS "CONTINUE" TO RECOVER FROM THIS ERROR.

12.0 APT HOOKS

12.1 DESCRIPTION

THE APT INTERFACES PROVIDES A MEANS OF COMMUNICATING WITH THE APT MOTHER AND THE SYSTEM UNDER TEST. IT FURTHER PROVIDES A MEANS OF LOADING DIAGNOSTICS.

TWO INTERFACES ARE PROVIDED FOR THIS COMMUNICATION. THEY ARE:

1. TIMING
2. ERROR REPORTING

EACH WILL BE DESCRIBED AT A LATER TIME.

12.2 APT INITIALIZATION

SHOULD BIT ZERO OF HCW2(ADDRESS 22) BE SET TO A ONE(1), APT IS ASSUMED TO BE PRESENT. THE PROGRAM WILL SET THE PSEUDO-SWITCH REGISTER TO 0037 AND THE NOTTY INDICATOR IS SET SO AS TO DISABLE ALL TERMINAL COMMUNICATION. AT THIS POINT AN EXIT BACK TO MAIN LINE C: E TO DETERMINE IF A KT8-A IS PRESENT IN THE SYSTEM. AFTER DETERMINING THIS THE PROGRAM WILL AUTO-SIZE MEMORY AND SET UP THE APPROPRIATE FIELD STATUS BITS. AFTER AUTO-SIZING MEMORY THE CONTENTS OF FIELD 7, ADDRESSES 6000-7777 ARE MOVED TO FIELD ZERO.

SHOULD LESS THAN 32K OF MEMORY BE FOUND BANK ZERO FIELD 7 OF THE FIELD STATUS WORDS IS SET SO AS NOT TO ENABLE TESTING FIELD 7 OF BANK ZERO.

FROM THIS POINT ON ALL APT INTERFACING IS DONE FROM THE PROGRAM FIELD. THIS ALLOWS TEST OF A FULL 32 K SYSTEM. IT ALSO PROVIDES A MEANS OF TESTING UP TO 128K OF MEMORY.

12.3 APT INTERFACES

12.3.1 TIMING

THE TIMING INTERFACE PROVIDES THE NECESSARY INFORMATION TO THAT THE DIAGNOSTIC IS RUNNING ERROR FREE. THE TIME INTERVAL IS BETWEEN 2 AND 5 SECONDS.

12.3.2 ERRORS

THE ERROR INFORMATION THAT IS PASSED TO THE APT HOST CONSISTS OF THE FIELD THE PROGRAM IS CURRENTLY IN AND THE PC OF THE ERROR CALL.

NOTE:

IT SHOULD BE NOTED THAT THIS PROGRAM NO LONGER SUPPORTS THE APT PROM. THE NEW INTERFACE MUST BE USED. THE NEW INTERFACE CONSISTS OF A BOOT ROM TO LOAD IN THE ACTUAL ROM CODE FOR APT INTO MEMORY. IF FIELD 7 IS NOT PRESENT THEN THE MS8-A MUST BE USED IN PLACE OF MEMORY.

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48
49
50
51
52
53
54
55
```

/PDP-8E EXTENDED MEMORY DATA AND CHECKERBOARD TEST
/MAINDEC-88-DMA-C-L
/COPYRIGHT (C) 1972, 1975, 1976 DIGITAL EQUIPMENT CORPORATION, MAYNARD, MASS. 01754
/PROGRAMMER, VERNON FREY
/

/SR00=1 HALT AFTER ERROR
/SR01=1 INHIBIT ERROR TYPEOUT
/SR02=1 BELL ON ERROR (USEFUL FOR MAINTENANCE)
/SR03=1 INHIBIT PROGRAM RELOCATION
/SR04=1 CHANGE FIELD LIMITS
/SR05=1 HALT AFTER CURRENT TEST
/

/PROGRAM STARTING ADDRESS
/C200
/

/W-CRD
/

/DEFINE NPAGE
/K JMP I (..20087800)
/

/PDP-8E IOT COMMANDS & MICRO INSTRUCTIONS
/

/EXPANDED MODE COMMANDS USED IN THIS TEST
/

6204 CUF=6204
6274 SUF=6274
0001 K18A=1
8200 LAM= 8200 /LOAD EXPANDED MODE REGISTER.
8230 REG= 8230
8280 LLS1= 8280
8240 LRR= 8240
8250 RRR= 8250
8203 C01=8203 /CHANGE TO DF & IF 0
8107 SPO=8107 /SKIP ON PARITY OPTION
8101 SKP=8101 /SKIP IF NO PARITY ERROR
8104 CVP=8104 /CLEAR PARITY ERROR FLAG
8004 GTF=8004 /GET INTERRUPT FLAGS
8005 RTF=8005 /RESTORE INTERRUPT FLAGS
7701 ACL=7701 /LOAD M0 IN M0 AC
7002 BS4=7002 /SWAP BYTES IN AC
7421 MQL=7421 /LOAD M0 FROM AC THEN CLR AC
7521 S4P=7521 /SWAP AC AND M0
8000 SKO=8000 /SKIP IF INTERRUPT ON, & TURN OFF
8007 CAF=8007 /CLEAR ALL FLAGS
0000 *0

```
56 0000 0304 *D /INTERRUPT ADDRESS
57 0001 3574 DCA SAC /SAVE AC
58 0002 7701 ACL /SAVE M0
59 0003 3075 DCA SWD
60 0004 5777 JMP INTRCU
61
62
63 0005 5600 IAPTR, APTER /APT/
64 0006 5652 IAPTCK, APTOK /APT/
65
66
67 0020 *20
68
69
70 0020 0007 PSR, 7 /APT/
71 0021 4000 MCW1, 4000 /APT/
72 0022 0000 MCW2, 0 /APT/
73
74 0023 0000 INMODE, 0
75
76 /PAGE 0 CONSTANTS AND POINTERS
77 /
78 0024 0000 BANK, 0
79 0025 0000 ENVFLG, 0
80 0026 4000 SR00, 4000 /HALT AFTER ERROR
81 0027 2000 SR01, 2000 /INHIBIT ERROR TYPEOUT
82 0030 1000 SR02, 1000 /BELL ON ERROR
83 0031 4000 SR03, 4000 /INHIBIT PROGRAM RELOCATION
84 0032 2000 SR04, 2000 /CHANGE FIELD LIMITS
85 0033 1000 SR05, 1000 /HALT AFTER CURRENT TEST
86 0034 7070 SR06, 70 /STARTING FIELD (0-7)
87 0035 0037 SR011, 7 /ENDING FIELD (0-7)
88 0036 0000 CS, 0 /COMPLEMENT STATUS
89
90
91
92 0037 2000 TS, 0 /0000=MC (NO COMPLEMENT)
93 /BIT 1=1C (ONE COMPLEMENT)
94 /BIT 2=2C (TWO COMPLEMENTS)
95 /TEST STATUS
96 /0000=ND TEST
97 /BIT 0=ALL ZEROS TEST
98 /BIT 1=ALL ONES TEST
99 /BIT 2 = 0050-7777 MCP TEST
100 /BIT 3 = 7777-0100 MCP TEST
101 /BIT 4 = 5525-5252 MCP TEST
102 /BIT 5 = 5252-2525 MCP TEST
103 /BIT 6 = MARK-1 PATTERN
104 /FIELD STATUS BANK0
105 /FIELD STATUS BANK 1
106 /FIELD STATUS BANK 2
107 /FIELD STATUS BANK 3
108 /BITS 0-7 COINCIDE WITH FIELDS
109 /0-7 FOR EACH FIELD NOT IN
110 /THE SYSTEM THE EQUIVALENT BIT
/IS SET.
0044 0000 RS, 0 /RELOCATION STATUS BANK 0
0045 0000 RS1, 0 /RELOCATION STATUS BANK1
```

```

111 0046 0000 RS2, 0
112 0047 0000 RS3, 0
113
114
115
116
117
118 0050 0000 CRELO, 0
119 0051 0000 PROF, 0
120 0052 0000 TSTFLD, 0
121 0053 0000 COUNT, 0
122 0054 0000 MOVE, 0
123 0055 0000 HEAD1, 0
124 0056 0000 TEMP, 0
125 0057 0000 TSTAC, 0
126 0060 0000 FCNT, 0
127 0061 0000 RELCNT, 0
128 0062 0000 STARTF, 0
129 0063 0000 ENDF, 0
130 0064 0000 INSAME, 0
131 0065 0000 LEGALD, 0
132 0066 0000 A, 0
133 0067 0000 B, 0
134 0070 0000 P2, 0
135 0071 0000 W4, 0
136 0072 0000 CDATA, 0
137 0073 0000 BDATA, 0
138 0074 0000 SAC, 0
139 0075 0000 SMQ, 0
140 0076 0000 NOTTY, 0
141 0077 0000 NUMFLD, 0
142 0100 0000 FIVE, 0
143 0101 6570 MINS, 6570
144
145
146
147
148 4502 ENDRHT=JMS I
149 0102 0341 KENDHL
150 4503 PRINT=JMS I
151 0103 5400 KPRINT
152 4504 GETSR=JMS I
153 0104 4747 XGETSR
154 4505 CBLAL=JMS I
155 0105 4723 XCBLAL
156 4506 SETFS=JMS I
157 0106 0522 XSETFS
158 4507 SETRS=JMS I
159 0107 0532 XSETRS
160 4510 RACA=JMS I
161 0110 5117 XRACA
162 4511 RACB=JMS I
163 0111 5142 XRACB
164
165 0200 *200

```

```

/RELOCATION STATUS BANK 2
/RELOCATION STATUS BANK 3
/BITS 0-7 COINCIDE WITH FIELDS
/G-7. FS IS XFERRED INTO RS.
/EACH FIELD THAT FAILS SETS THE
/EQUIVALENT BIT SO THAT PROGRAM
/WILL NOT RELOCATE TO A FAILING FIELD.
/O = INHIBIT PROGRAM RELOCATION
/PROGRAM IN FIELD 00X0
/TESTING FIELD 00X0
/MOVE ERROR COUNTER
/MOVE ADDRESS COUNTER
/7777 MEANS TYPEOUT ERROR HEADING
/TEMP STORAGE LOCATION
/TEST ADDRESS COUNTER
/COUNT # OF FIELDS PRESENT
/END OF PASS COUNTER.
/STARTING FIELD 00X0
/ENDING FIELD 00X0
/PROGRAM IN SELECTED FIELD
/LEGAL FIELD SELECTION CONTROL
/A REG TO WRITE/READ
/B REG TO WRITE/READ
/CONTROLS 2 PAGES
/CONTROLS 4 WORDS
/GOOD DATA = DATA WRITTEN
/BAD DATA = DATA READ
/SAVE AC (INT)
/SAVE SO (INT)
/PROGRAM FLAG FOR NO TELETYPE TO AVOID PRINTOUTS
/NUMBER OF FIELDS BEING TESTED.
/THIS IS NOW 15 MINUTE COUNTER

```

/END OF TEST LAS

```

166
167
168 0200 5325 JMP START+1
169 0201 5324 JMP START
170
171 0202 4251 JMS DFEIF
172 0203 7410 SKP
173
174 0204 4771 JMS TITLE
175 0205 6002 PATA, IOF
176 0206 1376 TAD (7200)
177 0207 6200 LXM
178 0210 7200 CLA
179 0211 6230 REM
180 0212 7710 SPA CLA
181 0213 7340 CLL CLA CMA
182 0214 3625 DCA EMNFLG
183 0215 4775 JMS PNTOPT
184 0216 4774 JMS SETSW
185 0217 3636 DCA CS
186 0220 3637 DCA TS
187 0221 3040 DCA FS
188 0222 3044 DCA RS
189 0223 7240 STA
190 0224 3050 DCA CRELO
191 0225 3657 DCA TSTAD
192 0226 1101 TAD MINS
193 0227 3100 DCA FIVE
194 0230 7240 STA
195 0231 3655 DCA HEAD1
196 0232 4773 JMS FSSET
197 0233 4772 JMS APFL
198 0234 7240 STA
199 0235 1060 TAD FCNT
200 0236 3660 DCA FCNT
201 0237 4771 JMS LEGAL
202 0240 1077 TAD NUMFLD
203 0241 7041 CIA
204 0242 3061 DCA RELCNT
205 0243 1050 TAD CRELO
206 0244 7650 SNA CLA
207 0245 5261 JMP PATN
208 0246 4770 JMS CSRO3
209 0247 5301 JMP PATO
210 0250 5270 JMP PATN
211
212
213 /
214 /MAKE DF = IF
215 /
216 0251 0000 DFEIF, 0
217 0252 6002 IOF
218 0253 7300 CLA CLL
219 0254 6224 RIF
220 0255 1176 TAD (6201)

```

```

/APT/200 = START ADDRESS.
/STARTING ADDRESS IF NO TTY AVAILABLE
/202 = RESTART ADDRESS
/TYPOUT PROG TITLE
/LOAD EXPANDED MODE REGISTER
/MAKE SURE AC CLEAR
/SKIP IF KDBA NOT ENABLED
/SET KDBA ACTV
/SETUP SR
/CLEAR INH RELO
/CLEAR TEST ADDRESS COUNTER
/SET UP COUNTER
/RESET ERROR HEADING
/SET FIELD STATUS & TYPE SELECTION
/AC=-:
/-1 TO FIELDS IN SYSTEM
/CHECK FOR LEGAL FIELD SELECTION
/SET UP PASS COUNTER
/NEGATE IT
/SAVE IT.
/NO RELOCATE & TEST ONLY 1 FIELD
/RELOCATION PROGRAM
/INHIBIT PROGRAM RELOCATION

```

```
221 0256 3257 DCA .+1
222 0257 6301 CDF 0
223 0260 5651 JMP 1 DFEIF
224
225
226
227 /
228 /NO PROGRAM RELOCATION AND TEST ONLY 1 FIELD
229 0261 6224 PATM, RIF
230 0262 3051 DCA PRCFLO
231
232 0263 4767 JMS PAREL /TYPEOUT NO RELOCATION
233 0264 4766 PATMO, JMS TEST
234
235 0265 4502 ENDMLT /TEST FOR END OF PASS INFORMATION
236 0266 5205 JMP PATA /YES
237
238
239 0267 9264 JMP PATMO /NO
240
241
242 /
243 /NO PROGRAM RELOCATION BUT TEST ALL SELECTED FIELDS
244 /
245 0270 6224 PATN, RIF
246 0271 3051 DCA PRCFLO
247
248 0272 4767 JMS PAREL /TYPEOUT NO RELOCATION
249 0273 4766 PATNO, JMS TEST
250
251 0274 4502 ENDMLT /TEST FOR END OF PASS INFORMATION
252 0275 5205 JMP PATA /YES
253
254 0276 4770 JMS CSRO3 /NO
255 0277 9301 JMP PATO /RELOCATE PROGRAM
256 0300 5273 JMP PATNO /CONTINUE
257
258
259 /
260 /CHECK ALL SELECTED FIELDS FROM EACH SELECTED FIELD
261 /
262 0301 6224 PATO, RIF
263 0302 3051 DCA PRCFLO
264 0303 1040 TAD FS
265 0304 3044 DCA RS /SETUP RELO STATUS
266 0305 1041 TAD FS1
267 0306 3045 DCA RS1
268 0307 1042 TAD FS2
269 0310 1046 DCA RS2
270 0311 1043 TAD FS3
271 0312 3047 DCA RS3
272
273 0313 4785 JMS PREL /TYPEOUT RELOCATION
274 0314 4786 PATOO, JMS TEST
275
```

```
276 0315 4502 ENDMLT /TEST FOR END OF PASS INFORMATION
277 0316 5205 JMP PATA /YES
278
279 0317 4770 JMS CSRO3 /NO
280 0320 7410 SKP
281 0321 9270 JMP PATH /INHIBIT PROGRAM RELOCATION
282 0322 4764 JMS SETREL /GO TEST RELOCATION
283 0323 5314 JMP PATOO /CONTINUE
284
285
286 0324 5334 START, JMP .+13
287 0325 3076 DCA NOTTY /SAVE TTY FLAG STATUS
288 0326 4763 JMS APTIZ /GO SETUP MEMORY SIZE FROM PSR OR SR
289 0327 1076 TAD NOTTY /GET TTY FLAG
290 0330 7650 SNA CLA /TTY AVAILABLE
291 0331 5204 JMP PATA-1 /YES GO PRINT TITLE
292 0332 6002 IOF /NO, TURN THE INTERRUPT OFF
293 0333 5205 JMP PATA /GO START WITHOUT TITLE
294 0334 1022 TAD MCW2 /GET HARDWARE WORD 2
295 0335 0362 AND (7377 /MASK OUT CLB BIT
296 0336 3022 DCA MCW2
297 0337 7240 CLA CMA
298 0340 5325 JMP START+1 /GO SET FLAG FOR NO TELETYPE
299
300
301 /
302 /THIS ROUTINE DETERMINES IF HALT AFTER TEST SELECTED
303 /AND/OR CHANGE FIELD LIMITS SELECTED
304 /IF FIELD LIMITS ARE TO CHANGE RETURN CALL +1 IF THEY ARE NOT TO
305 /CHANGE RETURN CALL +2.
306 /
307 0341 0000 XENDML, 0
308 0342 4504 GETSR /GET CURRENT SWITCH VALUE
309 0343 0033 AND SRO5 /HALT AFTER TEST
310 0344 7650 SNA CLA /SKIP IF HALT AFTER TEST SELECTED
311 0345 5350 JMP .+3 /DON'T HALT
312 0346 4505 CBCL /TEST FOR ACTIVE CONSOLE
313 0347 7402 MLT /NON ACTIVE CONSOLE, JUST HALT
314 0350 4504 GETSR /GET SWITCHES AGAIN
315 0351 0032 AND SRO4 /ISOLATE CHANGE FIELD LIMITS SWITCH
316 0352 7330 SNA CLA /SKIP IF FIELD LIMITS ARE TO CHANGE
317 0353 2341 ISZ XENDML
318 0354 5741 JMP I XENDML
319
320 0362 7377 /
321 0363 5622
322 0364 0600
323 0365 4154
324 0366 0600
325 0367 4115
326 0370 1607
327 0371 1470
328 0372 5704
329 0373 1615
330 0374 4071
331 0375 1726
```

```
331 0376 7200
332 0377 4043
      0400
      PAGE
      /DETERMINE WHICH RELOCATION ROUTINE TO USE.
333
334
335 0400 0000
336 0401 1025
337 0402 7110
338 0403 1034
      SETREL, 0
      TAD EXWFLD
      SZA CLA
      JMP KTBREL
      /SKIP IF EXPANDED MODE NOT ENABLED
340
341
342
343
344
345
      /THIS ROUTINE IS USED ONLY IF KTB OR KTB8 FUNCTIONS ARE TO BE TESTED
346 0404 7200
347 0405 6224
348 0406 3151
349 0407 1051
350 0410 7112
351 0411 7010
352 0412 3777
353 0413 7240
354 0414 1777
355 0415 0035
356 0416 3777
357 0417 1777
358 0420 1376
      KTBREL, CLA
      RIF
      DCA PROFLD
      TAD PROFLD
      CLL RTR
      RAR
      DCA FLOCNT
      STA
      TAD FLOCNT
      AND SR911
      DCA FLOCNT
      TAD FLOCNT
      TAD (TRSTAB)
      /ESTABLISH POINTER OF STATUS
      /ROUTINE
      DCA TEMP
      TAD I TEMP
      DCA TRS
      JMS I TRS
      JMP KTBREL+7
      TAD FLOCNT
      AND SR911
      CLL RTR
      RAL
      JMP CSAME
      /SET UP FOR INSTRUCTION
      /GO TEST FOR SAME FIELD
360 0421 3056
361 0422 1456
362 0423 3272
363 0424 4672
364 0425 5213
365 0426 1777
366 0427 0035
367 0430 7106
368 0431 7004
369 0432 5263
      /
371 0433 5213
      JMP KTBREL+7
372
373
374
375
      /THIS ROUTINE WILL TEST THE KTB8 FUNCTIONS.
376 0434 7200
377 0435 6224
378 0436 3051
379 0437 1051
380 0440 4510
381 0441 3777
382 0442 7240
383 0443 1777
384 0444 0375
      KTBREL, CLA
      RIF
      DCA PROFLD
      TAD PROFLD
      RACA
      DCA FLOCNT
      STA
      TAD FLOCNT
      AND
      (37
      /MAKE SURE AC IS CLEAR
      /GET PROGRAM FIELD
      /AND SAVE IT
      /SAVE THE BINARY COUNT VALUE OF FIELD
```

```
385 0445 3777
386 0446 1777
387 0447 0035
388 0450 1376
389 0451 3056
390 0452 1456
391 0453 3272
392 0454 4672
393 0455 5242
394 0456 1777
395 0457 0375
396 0460 4511
397 0461 5263
398 0462 5242
      DCA FLOCNT
      TAD FLOCNT
      AND SR911
      TAD (TRSTAB)
      DCA TEMP
      TAD I TEMP
      DCA TRS
      JMS I TRS
      JMP KTBREL+6
      TAD FLOCNT
      AND
      (37
      RACA
      JMP CSAME
      KTBREL+6
      /DECREMENT FIELD VALUE
      /ISOLATE 5 BITS
      /REARRANGE INTO PROPER FORMAT
      /GO TEST FOR VALID FIELD SELECTION
400 0463 3052
401 0464 4774
402 0465 5800
403 0466 4773
404 0467 6224
405 0470 3051
406 0471 5600
      CSAME, DCA
      JMS SAME
      JMS I SETREL
      JMS RELO
      RIF
      DCA PROFLD
      JMP I SETREL
      /SAVE FIELD VALUE TO TEST
      /PCFLD+TSTFLD
      /YES
      /GO RELOCATE PROGRAM
      /GET PROGRAM LOCATION
      /EXIT RELOCATION ROUTINE
407
408 0472 0000
409 0473 3703
410 0474 3712
411 0475 3722
412 0476 3732
413 0477 3743
414 0500 3754
415 0501 3765
416 0502 4000
      TRS, 0
      TRSTAB, TRS0
      TRS1
      TRS2
      TRS3
      TRS4
      TRS5
      TRS6
      TRS7
      /STORAGE FOR TEST STATUS ROUTINE
417
418
419
420
421
422 0503 0000
423 0504 7200
424 0505 1030
425 0506 7450
426 0507 5703
427 0510 2303
428 0511 7106
429 0512 7430
430 0513 5703
431 0514 2303
432 0515 7710
433 0516 1113
434 0517 7000
      TCS, 0
      CLA
      TAD CS
      SNA
      JMP I TCS
      ISZ TCS
      CLL RTR
      SZL
      JMP I TCS
      ISZ TCS
      SPA CLA
      JMP I TCS
      APTEOB, NOP
      HLT
      JMP -1
      /TEST COMPLEMENT STATUS
      /RETURN IF NC, RETURN+1 IF IC, RETURN+2 IF ZC
      /NC
      /IC
      /ZC
      /APY/
      /ERRONEOUS STATUS BITS SET
      /SET UP PROPER FIELD STATUS BITS
```

```
440
441 0522 0000 /
442 0523 1777' XSETFS, 0 FLDONT /GET FIELD 10 CO
443 0524 7112 CLL RTR
444 0525 7010 RAR /MOVE BANK SELECT INTO POSITION
445 0526 0372 AND 13
446 0527 1371 TAD 1FS /ESTABLISH POINTER TO ADDRESS OF STATUS
447 0530 3056 DCA TEMP /AND SAVE THE POINTER
448 0531 5722 JMP I XSETFS /AND EXIT
449
450 /ESTABLISH RELOCATION STATUS POINTER
451
452 0532 0000 XSETFS, 0
453 0533 1777' TAD FLDONT /GET FIELD 11 CO
454 0534 7112 CLL RTR
455 0535 7010 RAR
456 0536 0372 AND 13
457 0537 1371 TAD 1FS /SAVE THE POINTER
458 0540 3056 DCA TEMP
459 0541 5732 JMP I XSETFS
460
461 /
462 0571 0040
463 0572 0003
464 0573 4200
465 0574 1600
466 0575 0037
467 0576 0073
468 0577 : -6
469
470 PAGE
471
472 /TEST PATTERN CONTROL
473
474 0600 0000 TEST, 0
475 0601 4777' JMS PAR
476 0602 7200 CLA
477 0603 3066 DCA A
478 0604 3147 DCA B
479 0605 4776' JMS STS0 /ALL ZEROS TEST
480 0606 4253 JMS TEST0
481 0607 7240 STA
482 0610 3066 DCA A
483 0611 7240 STA
484 0612 3067 DCA B
485 0613 4775' JMS STS1 /ALL ONES TEST
486 0614 4253 JMS TEST0
487 0615 7240 STA
488 0616 3067 DCA B
489 0617 3066 DCA A
490 0620 4774' JMS STS2 /0000-7777 MCP TEST
491 0621 4253 JMS TEST0
492 0622 7240 STA
493 0623 3066 DCA A
```

```
494 0624 3067 DCA B
495 0625 4773' JMS STS3 /7777-0000 MCP TEST
496 0626 4253 JMS TEST0
497 0627 7200 CLA
498 0630 1175 TAD [2525
499 0631 3066 DCA A
500 0632 1174 TAD [5252
501 0633 3067 DCA B
502 0634 1172' JMS STS4 /2525-5252 MCP TEST
503 0635 4253 JMS TEST0
504 0636 7200 CLA
505 0637 1174 TAD [5252
506 0640 3066 DCA A
507 0641 1175 TAD [2525
508 0642 3067 DCA B
509 0643 4771' JMS STS5 /5252-2525 MCP TEST
510 0644 4253 JMS TEST0
511 0645 7200 CLA
512 0646 3067 DCA TS
513 0647 6002 IOF
514 0650 4770' JMS M0STST /GO PERFORM MARCH PATTERN
515 0651 5600 JMP I TEST
516
517 /TEST ALL FIELDS SELECTED FOR TEST
518
519
520 0652 5653 KTEST, JMP I TEST0
521 0653 0000 TEST0, 0
522 0654 3767' DCA FLDONT /CLEAR FIELD INDICATOR
523 0655 3074' DCA BANK /CLEAR BANK VALUE
524 0656 4307 JMS N0FLD /GO TEST FOR VALID SELECTION
525 0657 5261 JMP T0LPO /NO A VALID FIELD, UPDATE AND TRY AGAIN
526 0660 4766' JMS N0FLD /GO WRITE SELECTED FIELD
527 0661 1767' TAD FLDONT
528 0662 7041 CIA
529 0663 1060 TAD FCNT /COMPLEMENT CURRENT VALUE
530 0664 7550 SNA CLA /SEE IF AT MAXIMUM FOUND
531 0665 5270 JMP TEST1 /SKIP IF NO AT MAX AVAILABLE
532 0666 2767' ISZ TEST1
533 0667 8256 JMP TEST0+3 /UPDATE FIELD TO DO
534
535 /
536
537
538 0670 3767' TEST1, DCA FLDONT /CLEAR FIELD INDICATOR
539 0671 3074' DCA BANK /CLEAR BANK INDICATOR
540 0672 4307 JMS N0FLD
541 0673 5500 JMP T1UPD /GO SET UP PROPER ERROR ROUTINE
542 0674 4765' JMS R0FLD /SET UP TO TEST FOR MAX FIELDS
543 0675 1053 TAD CCUNT /TEST FOR ANY READ ERRORS
544 0676 7640 SZA CLA
545 0677 4784' JMS SETERR
546 0700 1767' TAD FLDONT
547 0701 7041 CIA
548 0702 1060 TAD FCNT /GET NUMBER OF FIELDS FOUND
```

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549 0703 7650 SNA CLA /SKIP IF NOT AT MAX
550 0704 5763 JNO TEST8 /GO CMTD NEXT TEST
551 0705 2767 ISZ FLOCNT /GO BACK AND DO NEXT VALUE
552 0706 5272 JNP TEST1+2
553 /
554 /
555 /
556 /
557 /
558 /
559 /THIS ROUTINE DETERMINES IF A VALID FIELD IS SELECTED.
560 /A VALID SELECTION RETURNS+2
561 /NON-VALID SELECTION RETURNS +1
562 /
563 0707 0000 NUFLO, 0
564 0710 1767 TAD FLOCNT /GET FIELD TO TEST
565 0711 0035 AND SR911 /ISOLATE FIELD
566 0712 1262 TAD (TFSTAB /GET TO POINTER OF FIELD STATUS
567 0713 3337 DCA TFS /SAVE THE VALUE
568 0714 1737 TAD I TFS /GET ROUTINE TO EXECUTE
569 0715 3337 DCA TFS /AND SAVE IT
570 0716 4737 JMS I TFS /GO TEST FOR VALID FIELD SELECTION
571 0717 5707 JNP I NUFLO /NOT A VALID FIELD
572 0720 1025 TAD ENVFLG
573 0721 7650 SNA CLA /SKIP IF EXPANDED MODE ENABLED
574 0722 5232 JNP NCEMM
575 0723 1767 TAD FLOCNT
576 0724 4811 RACB /REARRANGE AC INTO PROPER FORMAT
577 0725 3052 SETFLD, DCA TSTFLD
578 0726 4761 JMS SAME /PROFLD=TSTFLD
579 0727 7410 SKP
580 0730 2307 ISZ NUFLO
581 0731 5707 JNP I NUFLO /GET A VALID FIELD
582 /
583 0732 1767 NCEMM, TAD FLOCNT
584 0733 0035 AND SR911 /ISOLATE FIELD TO DO
585 0734 7106 CLL RTL
586 0735 7604 RAL /MOVE INTO BITS 6-8
587 0736 5325 JNP SETFLD /GO TEST FOR FIELD SELECTION
588 /
589 0737 0000 TFS, 0
590 /
591 0740 3600 TFSTAB, TFS0
592 0741 3607 TFS1
593 0742 3617 TFS2
594 0743 3627 TFS3
595 0744 3640 TFS4
596 0745 3651 TFS5
597 0746 3662 TFS6
598 0747 3672 TFS7
599 /
600 /
601 0761 1600
602 0762 0743
603 0763 1000
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604 0764 4510
605 0765 1200
606 0766 1005
607 0767 2346
608 0770 5200
609 0771 3262
610 0772 2255
611 0773 3250
612 0774 3243
613 0775 3237
614 0776 3233
615 0777 4267
616 /
617 /
618 1000 4504 TEST8, GETSR
619 1001 0032 AND SR04 /CHANGE FIELD LIMITS?
620 1002 7640 SZA CLA
621 1003 5777 JNP PAT1 /YES
622 /
623 1004 5778 JNP KTEST
624 /
625 /
626 /WRITE A & B REG PATTERN INTO SELECTED FIELD
627 /
628 /
629 1005 0000 WRFLD, 0
630 1006 1773 TAD [-40
631 1007 1770 DCA P2 /WRITE 2 PAGES
632 1010 241 JMS WRA /WRITE 4 WORDS FROM A REG
633 1011 4263 JMS WRB /WRITE 4 WORDS FROM B REG
634 1012 2070 ISZ P2
635 1013 5210 JNP -3
636 1014 1773 TAD [-40
637 1015 3070 DCA P2
638 1016 4263 JMS WRB
639 1017 4241 JMS WRA
640 1020 2070 ISZ P2
641 1021 5216 JNP -3
642 1022 1757 TAD TSTAD
643 1023 7640 SZA CLA
644 1024 5206 JNP WRFLD+1
645 /
646 1025 2100 ISZ FIVE /SEE IF READY TO PRINT IS YET
647 1026 5605 JNP I WRFLD
648 1027 1076 TAD NOTTY /GET THE TELETYPE FLAG
649 1030 7710 SPA CLA /IS THERE ONE AVAILABLE
650 1031 5605 JNP I WRFLD /NO-ABORT THE FIVE MINUTE TYPEOUT
651 1032 1101 TAD MINS /RESET FIFTEEN MINUTE COUNTER
652 1033 3100 DCA FIVE
653 /
654 1034 4775 JMS MES
655 1035 4543
656 1036 6165 /TYPE A 15
657 1037 0000 0000
```

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658 1040 5605      JMP I  WRFLD      /END OF MEMORY REACHED
659 1041 0000      WRA, 0
660 1042 1172      TAD  [-4
661 1043 3071      DCA  W4      //WRITE 4 WORDS FROM A REG
662 1044 1052      TAD  TSTFLD
663 1045 1176      TAD  [6201
664 1046 3247      DCA  .-1
665 1047 8711      CDF  0      /TEST DF
666 1050 1138      WRA1, TAD  A
667 1051 3457      DCA I  TSTAD
668 1052 2057      ISZ  TSTAD
669 1053 7000      NOP
670 1054 2071      ISZ  W4
671 1055 5250      JMP  W4A1
672 1056 1051      TAD  PROFLO
673 1057 1176      TAD  [6201      /4 WORDS ARE WRITTEN
674 1060 3261      DCA  .-1
675 1061 6201      CDF  0      /PROGRAM DF
676 1062 5441      JMP I  WRA
677 1063 0000      WRB, 0
678 1064 1172      TAD  [-4
679 1065 3071      DCA  W4      /WRITE 4 WORDS FROM B REG
680 1066 1052      TAD  TSTFLD
681 1067 1176      TAD  [6201
682 1070 3271      DCA  .-1
683 1071 6201      CDF  0      /TEST DF
684 1072 1067      WRB1, TAD  B
685 1073 3457      DCA I  TSTAD
686 1074 2057      ISZ  TSTAD
687 1075 7000      NOP
688 1076 2071      ISZ  W4
689 1077 5272      JMP  WRB1
690 1100 1051      TAD  PROFLO      /4 WORDS ARE WRITTEN
691 1101 1176      TAD  [6201
692 1102 3303      DCA  .-1
693 1103 6201      CDF  0      /PROGRAM DF
694 1104 5683      JMP I  WRB
695
696
697 /TYPEOUT FIELDS SELECTED FOR TESTING
698
699 1105 0000      TOSEL, 0
700 1106 7346      CLL  CLA  CMA  RTL      /AC=-3
701 1107 3024      DCA  BANK      /SAVE BANK VALUE
702 1110 1024      TAD  BANK
703 1111 7041      CIA
704 1112 7106      CLL  RTL      /MAKE BANK VALUE POSITIVE
705 1113 7004      RAL
706 1114 3774      DCA  FLD CNT      /SHIFT OVER FOR OTHER ROUTINES
707 1115 1024      TAD  BANK
708 1116 7041      CIA
709 1117 1171      TAD  [PS
710 1120 3056      DCA  TEMP      /ESTABLISH STATUS CONTROL WORD
711 1121 1456      TAD I  TEMP      /GET ADDRESS OF CONTROL WORD
712 1122 1170      TAD  [-7760      /SEE IF THERE IS A FIELD TO DO

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713 1123 7650      SNA  CLA
714 1124 5336      JMP  TOSEL1      /IS THIS BANK TO BE TESTED
715 1125 1024      TAD  BANK      /UPDATE BANK SELECT VALUE
716 1126 7041      CIA
717 1127 1167      TAD  [280
718 1130 4773      JMS  TYPE      /MAKE ASCII VALUE FOR BANK NUMBER
719 1131 1372      TAD  [-1
720 1132 4773      JMS  TYPE      /TYPE IT
721 1133 4771      JMS  TYPE
722 1134 1370      TAD  [240
723 1135 4773      JMS  TYPE      /PRINT A**
724 1136 2024      TOSEL1, ISZ  BANK      /UPDATE BANK VALUE
725 1137 7000      NOP
726 1140 1024      TAD  BANK
727 1141 7740      SNA  SZA  CLA      /SKIP IF LAST BANK NOT DONE
728 1142 5705      JMP I  TOSEL      /EXIT ROUTINE
729 1143 5310      JMP  TOSEL+3      /GO BACK AND TRY AGAIN
730
731 1170 0240
732 1171 2102
733 1172 0272
734 1173 5035
735 1174 2346
736 1175 2240
737 1176 0652
738 1177 0205
739 1200
740
741 /
742 /READ & TEST A & B REG PATTERN FROM SELECTED FIELD
743 1200 0000      RDFLD, 0
744
745 1201 4406      JMS I  IAPTRK      /APT/
746
747 1202 7200      CLA
748 1203 1052      TAD  TSTFLD
749 1204 1176      TAD  [6201
750 1205 3210      DCA  RDA2
751 1206 1210      TAD  RDA2
752 1207 3263      DCA  RCB2
753 1210 6201      RDA2, CDF  0      /TEST DF
754 1211 1166      TAD  [-100
755 1212 3070      DCA  P2
756 1213 1172      RDFLDA, TAD  [-4
757 1214 3071      DCA  W4
758 1215 3036      RDAC,  OCA  CS      /READ & TEST 4 WORDS
759 1216 4777      JMS  READ      /NO COMPLEMENT
760 1217 1141      CIA
761 1220 1066      TAD  A
762 1221 7440      SZA
763
764 1222 4777      JMS  ERRA      /A REG ERROR - NC
765 1223 4327      JMS  READ
766 1224 7040      CMA

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767 1225 3457 DCA I TSTAD
768 1226 4778 JMS SC51 /1 COMPLEMENT
769 1227 4327 JMS READ
770 1230 7001 IAC
771 1231 1066 TAD A
772 1232 7440 SZA
773
774 1233 4775 JMS ERROR1
775 1234 4327 JMS READ
776 1235 7040 CMA
777 1236 3457 DCA I TSTAD
778 1237 4774 JMS SC52 /2 COMPLEMENTS
779 1240 4327 JMS READ
780 1241 7041 CIA
781 1242 1166 TAD A
782 1243 7440 SZA
783
784 1244 4777 JMS ERROR
785 1245 2057 ISZ TSTAD /A REG ERROR - 2C
786 1246 7000 NOP
787 1247 2071 ISZ W4
788 1250 5215 JMP RDAC /COMPLETE 4 WORDS
789 1251 2070 ISZ P2
790 1252 5266 JMP RDFLODB /COMPLETE CURRENT 2 PAGES
791 1253 1061 TAD PROFLO
792 1254 1176 TAD [6201
793 1255 3256 DCA -1
794 1256 6111 CDF 2 /PROGRAM OF
795 1257 107 TAD TSTAD
796 1260 1640 SZA CLA
797 1261 5210 JMP PCA2 /READ ANOTHER 2 PAGES
798 1262 5600 JMP I RDFLO /END OF MEMORY REACHED
799
800 1263 6201 RDB2, CCF 0 /TEST OF
801 1264 1166 TAD [-100
802 1265 3070 DCA P2 /READ & TEST 2 PAGES
803 1266 1172 RDFLODB, TAD [-4
804 1267 3071 DCA -2 /READ & TEST 4 WORDS
805 1270 3036 RDB2, DCA CS /NO COMPLEMENT
806 1271 4327 JMS READ
807 1272 7041 CIA B
808 1273 1067 TAD
809 1274 7440 SZA
810
811 1275 4773 JMS ERROR
812 1276 4327 JMS READ /B REG ERROR - NC
813 1277 7040 CMA
814 1300 3457 DCA I TSTAD
815 1301 4778 JMS SC51 /1 COMPLEMENT
816 1302 4327 JMS READ
817 1303 7001 IAC
818 1304 1067 TAD B
819 1305 7440 SZA
820
821 1306 4772 JMS ERROR1 /B REG ERROR - 1C
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822 1307 4327 JMS READ
823 1310 7040 CMA
824 1311 3457 DCA I TSTAD
825 1312 4774 JMS SC52 /2 COMPLEMENTS
826 1313 4327 JMS READ
827 1314 7041 CIA
828 1315 1067 TAD B
829 1316 7110 SZA
830
831 1317 4773 JMS ERROR
832 1320 2057 ISZ TSTAD /B REG ERROR - 2C
833 1321 7000 NOP
834 1322 2071 ISZ A1
835 1323 5270 JMP RDB2 /COMPLETE 4 WORDS
836 1324 2070 ISZ P2
837 1325 5213 JMP RFLDA /COMPLETE CURRENT 2 PAGES
838 1326 5263 JMP RDB2
839
840 /READ TEST ADDRESS SUBROUTINE
841 /
842 1327 0000 READ, 0
843 1330 1457 TAD I TSTAD
844 1331 1457 TAD I TSTAD
845 1332 1457 TAD I TSTAD
846 1333 1457 TAD I TSTAD
847 1334 1457 TAD I TSTAD
848 1335 1457 TAD I TSTAD
849 1336 1457 TAD I TSTAD
850 1337 1457 TAD I TSTAD
851 1340 1457 TAD I TSTAD
852 1341 1457 TAD I TSTAD
853 1342 1457 TAD I TSTAD
854 1343 7000 CLA
855 1344 1457 TAD I TSTAD
856 1345 5727 JMP I READ
857
858
859
860 /KEYBOARD INTERRUPT OCCURRED
861 /
862 1346 0000 KBINT, 0
863 1347 1070 TAD NOTV /GET THE TELETYPE FLAG
864 1350 7000 SZA CLA /IS THERE A TELETYPE AVAILABLE
865 1351 5134 JMP -3 /YES GO PREPARE THE ERROR
866 1352 7002 RLT /NO, -LT- INTERRUPTED FROM THE KEY BOARD
867 1353 5164 JMP KBINTC /GO CLEAR FLAG AND CONTINUE
868 1354 4771 JMS YES
869 1355 4643 JMS YES
870 1356 1116 TEXT "KBINT FROM KB"
871 1357 2440
872 1358 0000
873 1359 1115
874 1362 4013
875 1363 0000
876 1364 6232 KBINTC, ACC
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071 1365 7240 STA
072 1366 7255 DCA -EAD1
073 1367 5745 JWP I KEINT
074
075 1371 2243
076 1372 1437
077 1373 1425
078 1374 3272
079 1375 1411
080 1376 3266
081 1377 1410
082
083 1401 3002 ERRA, 0
084 1401 7041 CIA
085 1402 1065 TAD A
086 1403 3073 DCA BDATA /DATA READ
087 1404 1066 TAD A
088 1405 4254 JWS GERRC /GO TO ERRC SETUP ROUTINE
089 1406 1066 TAD I
090 1407 3457 DCA I TSTAD /RE-WRITE B/D LOCATION
091 1408 5623 JWP I ERRA
092 1411 3000 ERRA1, 0
093 1412 3056 DCA TEMP
094 1413 1066 TAD A
095 1414 7040 CMA
096 1415 1056 TAD TEMP
097 1416 3073 DCA BDATA /DATA READ
098 1417 1066 TAD A
099 1420 7040 CMA
100 1421 4254 JWS GERRC /GO TO ERRC SETUP ROUTINE
101 1422 1066 TAD A
102 1423 7040 CMA
103 1424 3457 DCA I TSTAD /RE-WRITE B/D LOCATION
104 1425 5611 JWP I ERRA1
105 1426 3000 ERRA, 0
106 1427 7041 CIA
107 1430 1067 TAD B
108 1431 3073 DCA BDATA /DATA READ
109 1432 1067 TAD B
110 1433 4254 JWS GERRC /GO TO ERRC SETUP ROUTINE
111 1434 1067 TAD B
112 1435 3457 DCA I TSTAD /RE-WRITE B/D LOCATION
113 1436 5626 JWP I ERRA1
114 1437 3000 ERRA1, 0
115 1440 3056 DCA TEMP
116 1441 1067 TAD B
117 1442 7040 CMA
118 1443 36 TAD TEMP
119 1444 1073 DCA BDATA /DATA READ
120 1445 1067 TAD B
121 1446 7040 CMA
122 1447 4254 JWS GERRC /GO TO ERRC SETUP ROUTINE
123 1450 1067 TAD L
124 1451 7040 CMA
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925 1452 3457 DCA I TSTAD
926 1453 5637 JWP I ERRA1
927 1454 3000 GERRC, 0 /GO TO ERRC
928 1455 3072 DCA BDATA /DATA WRITTEN
929 1456 1051 TAD PROFLO
930 1457 1176 TAD 16201
931 1460 3261 DCA -1
932 1461 6201 CCF 0 /PROGRAM OF
933 1462 4777 JWS ERRC /DATA OR CHPBD ERROR
934 1463 1052 TAD TSTFLO
935 1464 1176 TAD 16201
936 1465 3268 DCA -1
937 1466 6201 CCF 0 /TEST OF
938 1467 5654 JWP I GERRC
939
940
941 /CHECK FOR LEGAL FIELD SELECTION
942 /
943 1470 3000 LEGAL, 0
944 1471 7330 CMA CLL
945 1472 3064 DCA INSAFE /SAVE FIELD CONTROL
946 1473 7344 CLL CMA RAL /AC--2
947 1474 3065 DCA LEGALO /LEGAL SELECTION CONTROL
948 1475 3052 DCA TSTFLO
949 1476 1025 TAD 000FLG /GET ATBA INDICATOR
950 1477 7650 SNA CLA /SKIP IF ACTIVE
951 1500 5123 JWP NOEMB
952 1501 3776 DCA FLOCONT /START WITH FIELD 0
953 1502 1051 JWS 000FLD
954 1503 1340 JWS LEGALO /VALID FIELD SELECTION
955 1504 4777 JWS 152 FLOCONT /UPDATE PATTERN
956 1505 1778 TAD FLOCONT
957 1506 4611 TAD 0003 /REARRANGE INTO PROPER CONFIGURATION
958 1507 3052 DCA TSTFLO /AND THE NEW VALUE
959 1510 1776 TAD FLOCONT
960 1511 0375 AND 137 /ISOLATE FIELD
961 1512 7440 SZA CLA /SKIP IF RETURNED TO ZERO
962 1513 5302 JWP -1 /GO BACK AND TRY AGAIN
963 1514 2085 ILEGAL, 152 LEGALO /NO FIELD SELECTION
964 1515 5774 JWP NOFLD
965 1516 1064 TAD INSAFE
966 1517 7040 SZA CLA
967 1520 5773 JWP BINT /PRG IN SELECTED FIELD
968 1521 3056 DCA 000FLD /ONLY 1 FIELD SELECTED
969 1522 5370 JWP I LEGAL
970
971 /AND FUNCTIONS
972 /
973 1522 3776 NOEMB, DCA FLOCONT /START WITH FIELD ZERO
974 1524 4351 JWS 000FLD /GO TEST FOR VALID SELECTION
975 1525 4342 JWS LEGALO
976 1526 2776 JWS FLOCONT /UPDATE FIELD NUMBER
977 1527 1778 TAD FLOCONT /GET BACK FIELD SELECTION
978 1530 0035 AND 5901 /ISOLATE FIELD
979 1531 7106 CLL RTL
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980 1532 7004      PAL      /MOVE INTO POSITION
981 1532 5052      DCA      /SAVE NEW VALUE
982 1534 1052      TAD      /TSTFLD
983 1535 7640      SZA CLA  /COME BACK TO ZERO?
984 1536 5324      JMP      /GO BACK AND TRY AGAIN
985 1537 5314      JMP      /INVALID SELECTIONS OR NOT ENOUGH
986
987
988
989
990
991 1540 0000      LEGALA, 0
992 1541 2065      ISZ      LEGALO /FIELD SELECTED
993 1542 7410      SKP      /
994 1543 5670      JMP I    LEGAL  /AT LEAST 2 FIELDS SELECTED
995 1544 6224      RIF
996 1545 3251      DCA      PROFLD
997 1546 4772      JMS      /PROGRAM IN SELECTED FIELD?
998 1547 2064      ISZ      /YES
999 1550 5140      JMP I    LEGALA
1000
1001
1002 1551 0000      /SET UP FOR TESTING FIELD STATUS FOR LEGAL SELECTION
1003 1552 7200      LG.FLD, C
1004 1553 1776      CLA      /MAKE SURE /C IS CLEAR
1005 1554 0035      TAD      /GET FIELD TO TEST
1006 1555 1371      AND      /ISOLATE FIELDS
1007 1556 3050      TAD      /FSTAB
1008 1557 1456      DCA      /SAVE ROUTINE POINTER
1009 1558 3056      TAD I    /GET POINTER
1010 1559 4456      DCA      /
1011 1562 2351      JMS I    /EXECUTE FS ROUTINE
1012 1563 5751      ISZ      /INVALID FIELD
1013
1014
1015 1571 0740      /
1016 1572 1600
1017 1573 5044
1018 1574 4600
1019 1575 0037
1020 1576 2546
1021 1577 2600      PAGE
1022
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1028 1600 0000      /
1029 1601 1051      SAME, 0
1030 1602 7041      TAD      PROFLD
1031 1603 1052      CIA      /
1032 1604 7640      TAD      /TSTFLD
1033 1605 2200      SZA CLA  /
1034
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1038 1607 0000      /
1039 1610 4504      CSRO3, 0
1040 1611 0031      GETSR  /
1041 1612 7640      AND      /SR03
1042 1613 2207      SZA CLA  /
1043 1614 5607      ISZ      /CSRO3
1044
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1048
1049 1615 0000      /SETUP FIELD STATUS (FS)
1050 1616 7200      /INC FIELDS NOT PRESENT OR NOT SELECTED
1051 1617 2260      /STORE NUMBER OF FIELDS PRESENT IN FCNT
1052 1620 3077      FSSET, 0
1053 1621 1076      CLA      /
1054 1622 7700      CCA      /FCNT
1055 1623 5233      DCA      /NUMFLD
1056 1624 1021      TAD      /NOTTY
1057 1625 7710      SZA CLA  /SKIP IF NO TTY AVAILABLE
1058 1626 5233      JMP      /
1059 1627 1020      SPA CLA  /SKIP IF NO HARDWARE SWITCHES
1060 1630 0377      JMS      /
1061 1631 3063      TAD      /PSR
1062 1632 5247      AND      /137
1063 1633 4505      DCA      /ENDF
1064 1634 7402      JMP      /SET UP LAST FIELD TO 00
1065 1635 4504      CBCAL  /BYPASS SETUP
1066 1636 0377      HLT
1067 1637 3063      GETSR  /GET NEW SWITCH VALUE
1068 1640 4504      AND      /137
1069 1641 7002      DCA      /ISOLATE FIELD WIN
1070 1642 0377      GETSR  /SAVE THE STARTING FIELD
1071 1643 3062      AND      /GET NEW SWITCH VALUE
1072 1644 4776      JMS      /
1073 1645 4505      STARTF /SAVE END FIELD LIMIT
1074 1646 7402      SETPAR /PRINT "SELECT TEST PARAMETER"
1075
1076
1077
1078 1647 3040      /NOW DETERMINE WHICH FIELDS TO TEST
1079 1650 3041      DCA      /FS
1080 1651 3042      DCA      /FS1
1081 1652 3043      DCA      /FS2
1082 1653 25      DCA      /FS3
1083 1654 5500      TAD      /EXMPFLG
1084
1085 1655 5274      SZA CLA  /TEST TO SEE IF EXPANDED MODE FEATURES
1086 1656 3775      JMP      /ARE AVAILABLE TO BE USED.
1087 1657 1775      JMS      /NOT ENABLED. TEST AS THOUGH A KMBE/A
1088 1660 4511      DCA      /
1089
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1089 1681 1176 TAD [6201 /ESTABLISH DATA FIELD
1090 1682 3263 DCA [-1
1091 1683 7402 MVT
1092 1684 4774 JVS CFP /CHANGED TO DATA FIELD TO TEST
1093 1685 4314 JVS SPFLD /GO TEST FIELD FOR PRESENCE
1094 1686 2775 ISZ FLOCNT /SET APPROPRIATE BIT
1095 1687 1775 TAD FLOCNT /UPDATE PATTERN
1096 1670 0377 AND 127 /GO BACK TO ZERO YET
1097 1671 7640 SZA CLA /SKIP IF BACK TO FIELD ZERO
1098 1672 5257 JVP [-13
1099 1673 5307 JVP FSEND
1100 1674 1373 BANKO, TAD (7760
1101 1675 3041 DCA FS1
1102 1676 1373 TAD (7760
1103 1677 3042 DCA FS2
1104 1700 1373 TAD (7760
1105 1701 3043 DCA FS3
1106 1702 7200 BANKO, CLA
1107 1703 6240 LRR /CLEAR RELOCATION
1108 1704 6250 RRR
1109 1705 2775 DCA FLOCNT
1110 1706 4772 JVS TESBVK /TEST BANK ZERO
1111
1112 1707 1076 FSEND, TAD NOTTY /GET TTY FLAG
1113 1710 7710 SPA CLA /IS TTY AVAILABLE
1114 1711 5615 JMP I FSSET /NO, ABORT TTY MESSAGE AND RETURN
1115 1712 4771 JVS PNTFLD
1116 1713 5715 JMP I FSSET
1117
1118 /SET UP ROUTINE TO SET STATUS BIT
1119 /
1120 1714 0000 SPFLD, 0
1121 1715 1775 TAD FLOCNT
1122 1716 0035 AND SAG11 /ISOLATE FIELD
1123 1717 1370 TAD (SFSTAB
1124 1720 3056 DCA TEMP /SAVE POINTER
1125 1721 1458 TAD I TEMP
1126 1722 3056 DCA TEMP /SAVE THE POINTER
1127 1723 4458 JVS I TEMP /GO PERFORM SET FUNCTION
1128 1724 7030 NOP
1129 1725 5714 JMP I SPFLD
1130
1131 /
1132 /
1133 /PRINT SELECTED OPTION FOR TESTING WHETHER KMB OR KTB
1134 /
1135 1726 0000 PNTOPT, 0
1136 1727 1076 TAD NOTTY /TEST FOR NOTTY TO USE
1137 1730 7640 SZA CLA
1138 1731 5726 JMP I PNTOPT /NO TTY AVAILABLE
1139 1732 1025 TAD EMFLG
1140 1733 7640 SZA CLA /SKIP IF KT NOT ENABLED
1141 1734 1367 TAD (7
1142 1735 1366 TAD (1315
1143 1736 3341 DCA OPT /SAVE OPTION VALUE
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1144 1737 4755 JVS MES
1145 1740 4543 4543 /CRLF
1146 1741 0000 OPT, 0000
1147 1742 7040 TEXT *0 SELECTED FOR TESTING *
1743 2305
1744 1405
1745 0324
1746 0174
1747 1008
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1157 1774 2500
1158 1775 2348
1159 1776 5000
1160 1777 3037
1161 2000
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1163
1164 /RETURN+1 IF FIELD PRESENT IN SYSTEM & IS SELECTED
1165 /
1166 2000 0000 CFP, 0
1167 2001 7300 CLA CLL
1168 2002 6224 RIF
1169 2003 1176 TAD [6201
1170 2004 3212 DCA CFPD
1171 2005 1165 TAD [-1
1172 2006 3677 DCA I CHECK
1173 2007 1677 TAD I CHECK
1174 2010 7640 SZA CLA /SKIP IF NO" PRESENT
1175 2011 5214 JMP [-3
1176 2012 8201 CFPD, 0 /PROGRAM DP
1177 2013 5600 JMP I CFP
1178 2014 2060 ISZ PCNT /FIELD IS PRESENT
1179 2015 6214 RIF /START = ENDP
1180 2016 3301 DCA CFTMP /SAVE TEST FIELD
1181 2017 1025 TAD EMFLG
1182 2020 7650 SNA CLA /SKIP IF EXPANDED MODE ACTIVE
1183 2021 5228 JMP [-5
1184 2022 1301 TAD CFTMP /GET BACK SELECTED FIELD
1185 2023 4510 RACA /CHANGE INT. BINARY COUNT
1186 2024 3301 DCA CFTMP /RESTORE IT FOR TESTING
```

```
1107 2025 5000 JWP L+5
1108 2025 1201 TAD CFP1MP /GET BACK FIELD VALUE
1109 2027 7112 CLL RTR
1150 2030 7210 JWP /CHANGE TO BINARY COUNT NUMBER
1191 2031 5001 DCA CFP1MP /SAVE NUMBER FOR TESTING
1192 2032 1003 TAD ENDF
1193 2033 7041 CIA
1194 2034 1002 TAD STARTF
1195 2035 7000 SNA
1196 2036 5046 JWP CFP2
1197 2037 1301 TAD CFP1MP /GET BACK NUMBER
1198 2040 7041 CIA
1199 2041 1002 TAD STARTF
1200 2042 7000 SNA CLA
1201 2043 7000 CFP1, ISZ CFP /FIELD IS PRESENT & SELECTED
1202 2044 2077 ISZ CFP /FIELD PRESENT AND SELECTED
1203 2045 5212 JWP CFPD
1204 2046 7110 CFP2, SPA CLA /STARTF < ENDF
1205 2047 5207 JWP CFP4
1206 2050 1301 TAD CFP1MP
1207 2051 7041 CIA
1208 2052 1002 TAD STARTF
1209 2053 7000 SNA
1210 2054 5243 JWP CFP1 /DF = STARTF (SELECTED)
1211 2055 7710 SPA CLA
1212 2056 5243 JWP CFP1 /DF > STARTF (SELECTED)
1213 2057 1301 CFP3, TAD CFP1MP /DF < STARTF ***
1214 2058 7041 CIA ENDF
1215 2061 1003 TAD
1216 2062 7000 SNA
1217 2063 5243 JWP CFP1 /DF = ENDF (SELECTED)
1218 2064 7110 SPA CLA
1219 2065 9312 JWP CFPD /DF > ENDF (NOT SELECTED)
1220 2066 5243 JWP CFP1 /DF < ENDF (SELECTED)
1221 2067 1301 CFP4, TAD CFP1MP /STARTF < ENDF
1222 2070 7041 CIA
1223 2071 1002 TAD STARTF
1224 2072 7000 SNA
1225 2073 5243 JWP CFP1 /DF = STARTF (SELECTED)
1226 2074 7710 SPA CLA
1227 2075 5207 JWP CFP3 /DF > STARTF THIS TIME ***
1228 2076 5212 JWP CFPD /DF < STARTF (NOT SELECTED)
1229 2077 2100 CHECK, CHECKD
1230 2100 0000 CHECKD, 0
1231 2101 0000 CFP1MP, 0
1232
1233
1234
1235
1236
1237
1238 2102 0000 /
1239 2103 4777' /FIND OUT WHICH FIELDS HAVE BEEN SELECTED FOR TESTING
1240 2104 5307 FLDSEL, 0
1241 2105 1164 JWS TFS7
JWP . 3
TAD [267
```

```
1242 2106 4776' JWS TYPE /FIELD 7
1243 2107 4775' JWS TFS6
1244 2110 5313 JWP .-3
1245 2111 1160 TAD [266
1246 2112 4776' JWS TYPE /FIELD 6
1247 2113 4774' JWS TFS5
1248 2114 5317 JWP .-3
1249 2115 1162 TAD [265
1250 2116 4776' JWS TYPE /FIELD 5
1251 2117 4773' JWS TFS4
1252 2120 5323 JWP .-3
1253 2121 1161 TAD [264
1254 2122 4776' JWS TYPE /FIELD 4
1255 2123 4772' JWS TFS3
1256 2124 5327 JWP .-3
1257 2125 1160 TAD [263
1258 2126 4776' JWS TYPE /FIELD 3
1259 2127 4771' JWS TFS2
1260 2130 5333 JWP .-3
1261 2131 1157 TAD [262
1262 2132 4776' JWS TYPE /FIELD 2
1263 2133 4770' JWS TFS1
1264 2134 5337 JWP .-3
1265 2135 1156 TAD [261
1266 2136 4776' JWS TYPE /FIELD 1
1267 2137 4767' JWS TFS0
1268 2140 5343 JWP .-3
1269 2141 1157 TAD [260
1270 2142 1164' JWS TYPE /FIELD 0
1271 2143 5702 JWP I FLDSEL
1272
1273
1274 2167 3000 /
1275 2170 3007
1276 2171 3011
1277 2172 3027
1278 2173 3040
1279 2174 3051
1280 2175 3062
1281 2176 3075
1282 2177 3072
2200 PAGE
1283
1284
1285
1286
1287
1288 2200 0000 /
1289 2201 7000 SIXTY, 0
1290 2202 1000 CLA CLL
1291 2203 3000 TAD : SIXTY /ADDRESS OF OPERAND
1292 2204 1000 DCA :
1293 2205 1000 ISZ SIXTY
1294 2206 1000 TAD I SIXTY /STORE ADDRESS
1295 2207 2200 DCA 51
ISZ SIXTY
```

```
1296 2210 1155 TAD [77
1297 2211 7140 CMA
1298 2212 0635 AND I 50 /AC-7700
1299 2213 7002 BSW /FIRST 2 DIGITS OF OPERAND
1300 2214 4222 JMS CNV /CONVERT DIGITS FOR TYPEOUT
1301 2215 2236 ISZ 51 /INC STORAGE ADDRESS
1302 2216 1155 TAD [77
1303 2217 0735 AND I 52 /SECOND 2 DIGITS OF OPERAND
1304 2220 1122 JMS CNV
1305 2221 1600 JMP I SIXTY /CODE
1306 2222 0000 0
1307 2223 3237 DCA 52
1308 2224 1237 TAD 52
1309 2225 7106 CLL RTL
1310 2226 7004 RAL
1311 2227 0154 AND [707 /LEFT DIGIT
1312 2230 1237 TAD 52
1313 2231 0154 AND [707 /RIGHT DIGIT
1314 2232 1153 TAD [0000
1315 2233 3036 DCA I 51 /STORE CONVERTED DIGITS
1316 2234 5622 JMP I CNV
1317 2235 0000 SO, 0
1318 2236 0000 SI, 0
1319 2237 0000 S2, 0
1320 /
1321 /TELETYPE OUTPUT WITH BELL
1322 /
1323 2240 0000 MES, 0
1324 2241 7240 STA
1325 2242 3023 DCA INCODE /SET PRINTER ACTIVE INDICATOR /-CB-/
1326 2243 7240 STA
1327 2244 1240 TAD MES /FIRST WORD -1
1328 2245 3010 DCA 10
1329 /
1330 2246 1410 TAD I 10
1331 2247 3113 DCA M0
1332 2250 1313 TAD M0
1333 2251 7002 BSW
1334 2252 4256 JMS TYPCH /TYPEOUT FIRST CHARACTER
1335 2253 1313 TAD M0
1336 2254 4256 JMS TYPCH /TYPEOUT SECOND CHARACTER
1337 2255 5248 JMP MES+6 /CONTINUE
1338 /
1339 2256 0000 TYPCH, 0
1340 2257 0155 AND [77
1341 2260 7440 SZA
1342 2261 5264 JMP -+3
1343 2262 3023 DCA INCODE /CLEAR MESSAGE ACTIVE INDICATOR /-CB-/
1344 2263 5410 JMP I 10 /END OF MESSAGE RETURN
1345 2264 1152 TAD -134
1346 2265 7440 SZA
1347 2266 5271 JMP -+3
1348 2267 1151 TAD [207 /CODE IS BELL
1349 2270 5311 JMP MTP
1350 2271 1172 TAD [-4
```

```
1351 2272 7500 SMA
1352 2273 5276 JMP -+3 /CODE LESS THAN 40?
1353 2274 1150 TAD [340 /NO
1354 2275 5311 JMP MTP /YES, ADD 300, CODE IS ALPHA
1355 2276 1147 TAD [-3
1356 2277 7440 SZA
1357 2300 5303 JMP -+3
1358 2301 1146 TAD [212 /CODE IS LINE FEED
1359 2302 5311 JMP MTP
1360 2303 1145 TAD [-2
1361 2304 7440 SZA
1362 2305 5310 JMP -+3
1363 2306 1144 TAD [215 /CODE IS CR
1364 2307 7410 SKP
1365 2310 1143 TAD [245 /ADD 200 TO OTHERS > 40
1366 2311 4777 JMS TYPE
1367 2312 5656 JMP I TYPCH
1368 2313 0000 NO, 0
1369 /
1370 /
1371 /TEST THE SELECTED BANK FOR FIELDS AVAILABLE
1372 /
1373 2314 0000 TESTBANK, 0
1374 2315 6271 CDF 70
1375 2316 4776 JMS CFP /CHECK FIELD PRESENT
1376 2317 4775 JMS SFS7 /SET FIELD STATUS BIT 7
1377 2320 6261 CDF 60
1378 2321 4776 JMS CFP
1379 2322 4774 JMS SFS6
1380 2323 6251 CDF 50
1381 2324 4776 JMS CFP
1382 2325 4773 JMS SFS5
1383 2326 6241 CDF 40
1384 2327 4776 JMS CFP
1385 2330 4772 JMS SFS4
1386 2331 6231 CDF 30
1387 2332 4776 JMS CFP
1388 2333 4771 JMS SFS3
1389 2334 6221 CDF 20
1390 2335 4776 JMS CFP
1391 2336 4770 JMS SFS2
1392 2337 6211 CDF 10
1393 2340 4776 JMS CFP
1394 2341 4767 JMS SFS1
1395 2342 6201 CDF 00
1396 2343 4776 JMS CFP
1397 2344 4766 JMS SFS0
1398 2345 5714 JMP I TESTBANK
1399 2346 1140 FIDONT, 0
1400 /
1401 /MARCH TEST IN ERROR
1402 /
1403 2347 0000 TVAR, 0
1404 2350 4240 JMS 115
1405 2351 1501 TEXT MARCH -
```

```
2352 2203
2353 1042
2354 5548
2355 0000
2356 5747
1406 JWP I TWAR
1407
1408 2366 3277
1409 2367 3317
1410 2370 3340
1411 2371 3412
1412 2372 3432
1413 2373 3452
1414 2374 3472
1415 2375 3512
1416 2376 2000
1417 2377 5025
2400
1418 PAGE
1419 /TYPEOUT CHARACTER IN AC AND A SPACE
1420
1421 2400 0000 TYPSP, 0
1422 2401 4777 JMS TYPE
1423 2402 1142 TAD 1240
1424 2403 4777 JMS TYPE
1425 2404 5600 JWP I TYPSP
1426
1427
1428
1429 /ERROR ROUTINE (BELL ON ERROR HAS PRIORITY)
1430
1431 2405 0000 RETURN, 0 /PROGRAM RETURN ADDRESS
1432 2406 4405 CODERR, JMS I 1AFTER
1433 2407 4504 GETSR
1434 2410 0030 AND SMO2 /BELL ON ERROR?
1435 2411 7650 SNA CLA
1436 2412 5221 JWP 1-7
1437 2413 1076 TAD NOTTY /GET TTY FLAG
1438 2414 7710 SPA CLA
1439 2415 5605 JWP I RETURN /NO TELETYPE AVAILABLE DO NOT RING BELL
1440 2416 1151 TAD 1207
1441 2417 4777 JMS TYPE /RING BELL
1442 2420 5605 JWP I RETURN
1443 2421 4504 GETSR
1444 2422 0027 AND SMO1
1445 2423 7640 SZA CLA
1446 2424 5267 JWP STOP /INHIBIT TYPEOUT
1447 2425 1025 TAD ENWFLG /TEST FOR W'BA
1448 2426 7650 SNA CLA
1449 2427 5234 JWP 1-5
1450 2430 6224 RIF /GET PROGRAM FIELD
1451 2431 4510 RACA /BINARY COUNT NUMBER
1452 2432 3056 DCA TEMP
1453 2433 5240 JWP 1-5
1454 2434 6224 RIF
1455 2435 7112 CLL RTR
```

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1456 2436 7010 RAR
1457 2437 3356 DCA TEMP /MAKE A BINARY NUMBER
1458 2440 4776 JMS SIXTY /SAVE CHARACTER
1459 2441 0056 TEMP
1460 2442 2457 ERROR0-1
1461 2443 1141 TAD 14543
1462 2444 3257 DCA ERROR0-1
1463 2445 1175 TAD RETURN
1464 2446 1165 TAD [-1]
1465 2447 3056 DCA TEMP
1466 2450 4776 JMS SIXTY
1467 2451 0056 TEMP
1468 2452 2461 ERROR1
1469 2453 1076 TAD NOTTY /GET TTY FLAG
1470 2454 7710 SPA CLA /IS THERE A TTY ON SYSTEM
1471 2455 5656 JWP I ADDER /NO, SO HALT ON ERRORS INSTEAD-INFO IN AC
1472 2456 4775 JMS YES
1473 2457 4543 4543
1474 2460 0000 ERROR0, 0 /FIELD
1475 2461 0000 ERROR1, 0
1476 2462 0000
1477 2463 4040 4040 /PROGRAM LOCATION OF ERROR JMS
1478 2464 0000 0
1479 2465 5666 JWP I 1-1 /TYPEOUT ERROR
1480 2466 0000 ADDER, 0 /ADDRESS OF ERROR TYPEOUT
1481 2467 4504 STOP, GETSR
1482 2470 0026 AND SMO0 /HALT AFTER ERROR?
1483 2471 7650 SNA CLA
1484 2472 5276 JWP LIMIT /INHIBIT ERROR HALT
1485 2473 1205 TAD RETURN
1486 2474 1165 TAD [-1]
1487
1488 2475 7402 HLT /HALT WITH AC = ERROR JMS
1489 2476 4504 LIMIT, GETSR
1490 2477 0032 AND SMO4 /CHANGE FIELD LIMITS?
1491 2500 7640 SZA CLA
1492 2501 5774 JWP DATA /YES
1493 2502 5605 JWP I RETURN /NO
1494
1495
1496 /RELOCATION MOVE ERROR
1497
1498
1499 2503 0000 ERRN, 0
1500
1501 2504 2053 ISZ COUNT /MOVE ERROR OCCURRED
1502 2505 7410 SKP
1503 2506 5304 JWP 1-2
1504 2507 7200 CLA
1505 2510 1303 TAD EPRN
1506 2511 3205 DCA RETURN /RETURN ADDRESS
1507 2512 1373 TAD DEPRN
1508 2513 3266 CCA ADDER /ERROR TYPEOUT ADDRESS
1509 2514 5206 JWP CODERR
1510 2515 4772 PERM, JMS FLDAT /SET UP FIELD VALUE TO PRINT
```

```
1511 2516 2536 Z10 /WHERE TO PUT IT
1512 2517 4776 JMS SIXTY
1513 2520 0054 MOVE
1514 2521 2537 Z11
1515 2522 1076 TAD NOTTY /GET TTY FLAG
1516 2523 7710 SPA /IS THERE A TELETYPE AVAILABLE?
1517 2524 5345 JMP ERRNO /NO-MALT CN ERRORS INSTED-INFO IN AC
1518 2525 4775 JMS MES
1519 2526 2205 TEXT *RELO ERR AT *
      2527 1417
      2530 4005
      2531 2222
      2532 4001
      2533 2440
      2534 0000
1520 2535 4775 JMS MES
1521 2536 0000 Z10. 0
1522 2537 0000 Z11. 0
1523 2540 0000 0
1524 2541 0000 0
1525 2542 7240 STA
1526 2543 3055 DCA HEAD1
1527 2544 5267 JMP STOP
1528
1529
1530 //RELOCATION ERROR ROUTINE ON A SYSTEM WITHOUT A TELETYPE
1531
1532
1533 2545 7240 ERRNO, CLA CMA
1534 2546 1205 TAD RETURN /
1535 2547 7402 MLT /AC=PROGRAM LOCATION OF ERROR JMS
1536 2550 7200 CLA
1537 2551 1771 TAD FLD0CNT /GET TEST FIELD
1538 2552 0370 AND 137 /MASK TO FIELD BITS
1539 2553 7402 MLT /AC=FIELD BEING TESTED IN BITS 7-11
1540 2554 7200 CLA
1541 2555 1054 TAD MOVE
1542 2556 7402 MLT /AC=ADDRESS OF LOCATION IN ERROR
1543 2557 7240 CLA CMA
1544 2560 3055 DCA HEAD1
1545 2561 5267 JMP STOP /GO CHECK FOR MALT AFTER ERROR SWITCH
1546
1547 2570 0037
1548 2571 2346
1549 2572 5067
1550 2573 2515
1551 2574 0205
1552 2575 2710
1553 2576 1000
1554 2577 5025
      2600
      PAGE
1555
1556
1557 /
1558 /DATA OR CHECKERBOARD ERROR OCCURRED
```

```
1559 /
1560 2600 0000 ERRNO, 0
1561
1562 2601 2053 ISZ COUNT /ERROR OCCURRED
1563 2602 7410 SKP
1564 2603 5001 JMP -2
1565 2604 7200 CLA
1566 2605 1200 TAD ERRNO
1567 2606 3777 DCA RETURN /RETURN ADDRESS
1568 2607 1376 TAD IPERRNO
1569 2610 3775 DCA ADDR /ERROR TYPEOUT ADDRESS
1570 2611 4504 GETSR
1571 2612 0000 AND SR02 /BELL CN ERROR
1572 2613 7640 SZA CLA
1573 2614 5774 JMP RBELL /RING BELL
1574 2615 4504 GETSR
1575 2616 0027 AND SR01
1576 2617 7640 SZA CLA
1577 2620 5773 JMP STOP /INHIBIT TYPEOUT
1578 2621 205E ISZ HEAD1
1579 2622 7410 SKP
1580 2623 4772 JMS ERRNO /TYPEOUT ERROR HEADING
1581 2624 5771 JMP CODERR
1582
1583 2625 4770 PERRC, JMS FLD0DAT /SET UP FIELD INFORMATION
1584 2626 2644 Z1 /LOCATION TO PUT IT IN
1585 2627 4767 JMS SIXTY
1586 2630 0047 TSTAD
1587 2631 1045 Z2
1588 2632 4767 JMS SIXTY
1589 2633 0072 GOATA
1590 2634 2651 Z3
1591 2635 4767 JMS SIXTY
1592 2636 0073 BOATA
1593 2637 2654 Z4
1594 2640 1076 TAD NOTTY /GET TTY FLAG
1595 2641 7710 SPA /IS THERE A TELETYPE AVAILABLE
1596 2642 5307 JMP ERRNO /NO GO MALT CN ERRORS INSTEAD
1597 2643 4766 JMS MES
1598 2644 0000 Z1. 0
1599 2645 0000 Z2. 0
1600 2646 0000 0 /FAIL ADR
1601 2647 4040 4040
1602 2650 4040 4040
1603 2651 0000 Z3. 0
1604 2652 0000 0 /GOOD
1605
1606
1607 2653 4040 4040
1608 2654 0000 Z4. 0
1609 2655 0000 0 /BAD
1610 2656 4000 4000
1611 2657 4765 PARORC, JMS TTS
1612 2660 4135 JMS TN /NONE
1613 2661 5773 JMP STOP
```



```
1710 3222 6110 TEXT *0000-7777 WCP - *
      3223 2210
      3224 5257
      3225 6787
      3226 6740
      3227 7703
      3230 2640
      3231 5540
      3232 6600
1711 3233 5620 JKP I TC7
1712 3234 0000 T70, 0
1713 3235 4777 JMS MES
1714 3236 6767 TEXT *7777-0000 WCP - *
      3237 6767
      3240 5660
      3241 6060
      3242 6340
      3243 2703
      3244 2040
      3245 5540
      3246 0000
1715 3247 5634 JKP I T70
1716
1717 3250 0000 T25, 0
1718 3251 4777 JMS MES
1719 3252 0205 TEXT *2525-5252 WCP - *
      3253 6105
      3254 5145
      3255 1 45
      3256 1340
      3257 2703
      3260 2040
      3261 5540
      3262 0000
1720 3263 5650 JKP I T25
1721
1722 3264 0000 T52, 0
1723 3265 4777 JMS MES
1724 3266 6562 TEXT *5252-2525 WCP - *
      3267 6182
      3270 5562
      3271 6562
      3272 6540
      3273 2703
      3274 2040
      3275 5540
      3276 0000
1725 3277 5664 JKP I T52
1726
1727
1728 /
1729 /PARITY ERROR
1730 /
1731 3100 7200 PARINT. CLA
1732 3101 1376 TAD (INTR
```

```
1733 3102 3775 JCA RETURN
1734 3103 4774 JMS SIXTY
1735 3104 0000 0
1736 3105 3131 Z20
1737 3106 4774 JMS SIXTY
1738 3107 0057 TSTAD
1739 3110 3145 Z21
1740 3111 1 76 TAD NOTTY
1741 3112 1 10 SPA CLA
1742 3113 5354 JUP PARERR
1743 3114 4777 JMS MES
1744 3115 4543 TEXT *%PARITY ERR, LOC 0*
      3116 2001
      3117 2211
      3120 2431
      3121 4005
      3122 2122
      3123 5440
      3124 1 17
      3125 6340
      3126 6075
      3127 0000
1745 3130 4777 JMS YES
1746 3131 0000 Z20, 0
1747 3132 0000 0
1748 3133 4040 4040
1749 3134 2423 2423
1750 3135 2401 2401
1751 3136 0475 0475
1752 3137 0000 0000
1753 3140 6004 GTF
1754 3141 2136 AND 17
1755 3142 1167 TAD 1260
1756 3143 4773 JMS TYPE
1757 3144 4777 JMS MES
1758 3145 0000 Z21, 0
1759 3146 0000 0
1760 3147 4000 4000
1761 3150 0104 JUP
1762 3151 7540 STA
1763 3152 3555 JCA
1764 3153 5772 JUP
1765
1766 /PARITY ERROR ON A NON TTY SYSTEM- ERROR INFO IN THE AC FOR EACH HALT
1767
1768
1769 3154 1000 PARERR. TAD 0
1770 3155 7402 RLT
1771 3156 7402 CLA
1772 3157 6004 GTF
1773 3160 3371 AND 17
1774 3161 7402 RLT
1775 3162 7200 CLA
1776 3163 1657 TAD TSTAD
1777 3164 7402 RLT
```

/AC+ADDRESS IN FIELD BEING TESTED

```
1778 3165 7200      CLA
1779 3166 6104      CMP
1780 3167 5772'     JMP      PARCRC      /CLEAR MEMORY PARITY ERROR BIT
1781                                     /GO GET PATTERN BEING TESTED
1782 3171 0007
1783 3172 2657
1784 3173 5025
1785 3174 2200
1786 3175 2405
1787 3176 4256
1788 3177 2240
1789                                     PAGE
1790
1791 /UNWANTED INTERRUPT OCCURRED
1792 /
1793 3200 1377      BADINT, TAD      (BADINT
1794 3201 4405      JMS I      IAPTR      /GO TO APT IF NEED BE
1795 3202 1076      TAD      NOTTY      /GET THE TELETYPE FLAG
1796 3203 7700      SHA      CLA      /IS THERE A TELETYPE ON THE SYSTEM
1797 3204 5207      JMP      -43      /YES GO PRINT THE MESSAGE
1798 3205 7402      HLT
1799 3206 5227      JMP      BINTC      /UNWANTED INTERRUPT OCCURRED
1800 3207 4776'     JMS      MES      /GO CLEAR THE WORLD AND CONTINUE
1801 3210 4543      TEXT      *XUNWANTED INTERRUPT OCCURRED*
1802
1803
1804
1805
1806
1807
1808
1809 3233 0000      STS0, 0      /SET TS0 (ALL 0 TEST)
1810 3234 7~10      CLA STL RAR
1811 3235 1137      DCA      TS
1812 3236 5633      JMP I      STS0
1813 3237 0000      STS1, 0      /SET TS1 (ALL 1 TEST)
1814 3240 7332      CLA STL RTR
1815 3241 3037      DCA      TS
1816 3242 5637      JMP I      STS1
1817 3243 0000      STS2, 0      /SET TS2 (0000 - 7777 MCP TEST)
1818
1819
1820
1821
1822
1823
1824 3250 0000      STS3, 0      /SET TS3 (7777 - 0000 MCP TEST)
1825 3251 7332      CLA STL RTR
1826 3252 7012      RTR
1827 3253 2037      DCA      TS
1828 3254 5650      JMP I      STS3
1829 3255 0000      STS4, 0      /SET TS4 (2525 - 5252 MCP TEST)
1830 3256 7203      CLA IAC BSW
1831 3257 7104      CLL RAL
1832 3258 3037      DCA      TS
1833 3259 5655      JMP I      STS4
1834 3260 0000      STS5, 0      /SET TS5 (5252 - 2525 MCP TEST)
1835 3263 7203      CLA IAC BSW
1836 3264 3037      DCA      TS
1837 3265 5662      JMP I      STS5
1838
1839 3266 0000      SCS1, 0      /SET CS1 (1 COMPLEMENT)
1840 3267 7332      CLA STL RTR
1841 3270 3036      DCA      CS
1842 3271 5666      JMP I      SCS1
1843 3272 0000      SCS2, 0      /SET CS2 (2 COMPLEMENTS)
1844 3273 7332      CLA STL RTR
1845 3274 7010      RAR
1846 3275 136      DCA      CS
1847 3276 1672      JMP I      SCS2
1848
1849 /SET ALSO STATUS BIT SPECIFIED
1850 /
1851 3277 0000      SF50, 0      /SET FS0 (DON'T TEST FIELD 0)
1852 3300 7200      CLA
1853 3301 4506      SETFS
1854 3302 1456      TAD I      T_LVP      /SETUP BANK POINTER
1855 3303 7004      RAL
1856 3304 7130      STL RAR
1857 3305 3436      DCA I      TEMP      /SAVE STATUS WORD
1858 3306 5677      JMP I      SF50
1859 3307 0000      SRS0, 0      /SET RS0 (DON'T RELO TO FIELD 0)
1860 3310 7200      CLA
1861 3311 4507      SETRS
1862 3312 1456      TAD I      TEMP      /SETUP BANK POINTER
1863 3313 7004      RAL
1864 3314 7130      STL RAR
1865 3315 3456      DCA I      TEMP      /SAVE NEW WORD
1866 3316 5707      JMP I      SRS0
1867 3317 0000      SFS1, 0      /SET FS1 (DON'T TEST FIELD 1)
1868 3320 7200      CLA
1869 3321 4506      SETFS
1870 3322 1456      TAD I      TEMP      /SETUP BANK POINTER
1871 3323 7006      RTL
1872 3324 7132      STL RTR
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1818 3244 7332      CLA STL RTR
1819 3245 7010      RAR
1820 3246 3037      DCA      TS
1821 3247 5643      JMP I      STS2
1822
1823
1824 3250 0000      STS3, 0      /SET TS3 (7777 - 0000 MCP TEST)
1825 3251 7332      CLA STL RTR
1826 3252 7012      RTR
1827 3253 2037      DCA      TS
1828 3254 5650      JMP I      STS3
1829 3255 0000      STS4, 0      /SET TS4 (2525 - 5252 MCP TEST)
1830 3256 7203      CLA IAC BSW
1831 3257 7104      CLL RAL
1832 3258 3037      DCA      TS
1833 3259 5655      JMP I      STS4
1834 3260 0000      STS5, 0      /SET TS5 (5252 - 2525 MCP TEST)
1835 3263 7203      CLA IAC BSW
1836 3264 3037      DCA      TS
1837 3265 5662      JMP I      STS5
1838
1839 3266 0000      SCS1, 0      /SET CS1 (1 COMPLEMENT)
1840 3267 7332      CLA STL RTR
1841 3270 3036      DCA      CS
1842 3271 5666      JMP I      SCS1
1843 3272 0000      SCS2, 0      /SET CS2 (2 COMPLEMENTS)
1844 3273 7332      CLA STL RTR
1845 3274 7010      RAR
1846 3275 136      DCA      CS
1847 3276 1672      JMP I      SCS2
1848
1849 /SET ALSO STATUS BIT SPECIFIED
1850 /
1851 3277 0000      SF50, 0      /SET FS0 (DON'T TEST FIELD 0)
1852 3300 7200      CLA
1853 3301 4506      SETFS
1854 3302 1456      TAD I      T_LVP      /SETUP BANK POINTER
1855 3303 7004      RAL
1856 3304 7130      STL RAR
1857 3305 3436      DCA I      TEMP      /SAVE STATUS WORD
1858 3306 5677      JMP I      SF50
1859 3307 0000      SRS0, 0      /SET RS0 (DON'T RELO TO FIELD 0)
1860 3310 7200      CLA
1861 3311 4507      SETRS
1862 3312 1456      TAD I      TEMP      /SETUP BANK POINTER
1863 3313 7004      RAL
1864 3314 7130      STL RAR
1865 3315 3456      DCA I      TEMP      /SAVE NEW WORD
1866 3316 5707      JMP I      SRS0
1867 3317 0000      SFS1, 0      /SET FS1 (DON'T TEST FIELD 1)
1868 3320 7200      CLA
1869 3321 4506      SETFS
1870 3322 1456      TAD I      TEMP      /SETUP BANK POINTER
1871 3323 7006      RTL
1872 3324 7132      STL RTR
```

1873	3325	3456	DCA I	TEMP	
1874	3326	5717	JMP I	SFS1	
1875	3327	0000	SRS1,	0	
1876	3330	7200	CLA		
1877	3331	4507	SETRS		
1878	3332	1456	TAD I	TEMP	
1879	3333	1044	TAD	RS	
1880	3334	7116	RTL		
1881	3335	132	STL RTR		
1882	3336	1456	DCA I	TEMP	
1883	3337	5727	JMP I	SRS1	/SAVE NEW WORD
1884	3340	0000	SFS2,	0	
1885	3341	7200	CLA		/SET FS2 (DON'T TEST FIELD 2)
1886	3342	4506	SETFS		
1887	3343	1456	TAD I	TEMP	/SETUP BANK POINTER
1888	3344	7006	RTL		
1889	3345	7500	SMA		
1890	3346	1135	TAD	[4000]	
1891	3347	7112	RTR		
1892	3350	3456	DCA I	TEMP	/SAVE NEW WORD
1893	3351	5740	JMP I	SFS2	
1894			/		
1895	3352	3277	SF: TAB,	SFS3	
1896	3353	3317		SFS1	
1897	3354	3340		SFS2	
1898	3355	2412		SFS3	
1899	3356	3432		SFS4	
1900	3357	3452		SFS5	
1901	3360	3472		SFS6	
1902	3361	3512		SFS7	
1903	3375	4256			
1904	3376	2240			
1905	3377	3200	PAGE		
		3400			
1906	3400	0000	SRS2,	0	/SET RS2 (DON'T RELO TO FIELD 2)
1907	3401	7200	CLA		
1908	3402	4507	SETRS		/SETUP BANK POINTER
1909	3403	1456	TAD I	TEMP	
1910	3404	7006	RTL		
1911	3405	7500	SMA		
1912	3406	1135	TAD	[4000]	
1913	3407	7012	RTR		
1914	3410	3456	DCA I	TEMP	/SAVE NEW WORD
1915	3411	5800	JMP I	SRS2	
1916			/		
1917			/		
1918	3412	0000	SFS3,	0	/SET FS3 (DON'T TEST FIELD 3)
1919	3413	7200	CLA		
1920	3414	4506	SETFS		/SETUP BANK POINTER
1921	3415	1456	TAD I	TEMP	
1922	3416	0134	AND	[7360]	
1923	3417	1133	TAD	[400]	
1924	3420	3456	DCA I	TEMP	/SAVE NEW WORD
1925	3421	5612	JMP I	SFS3	
1926	3422	0000	SRS3,	0	/SET RS3 (DON'T RELO TO FIELD 3)

1927	3423	7200	CLA		
1928	3424	4507	SETRS		/SETUP BANK POINTER
1929	3425	1456	TAD I	TEMP	
1930	3426	0134	AND	[7360]	
1931	3427	1133	TAD	[400]	
1932	3430	3456	DCA I	TEMP	/SAVE NEW WORD
1933	3431	5622	JMP I	SRS3	
1934	3432	0000	SFS4,	0	/SET FS4 (DON'T TEST FIELD 4)
1935	3433	7200	CLA		
1936	3434	4506	SETFS		/SETUP BANK POINTER
1937	3435	1456	TAD I	TEMP	
1938	3436	0132	AND	[7560]	
1939	3437	1131	TAD	[200]	
1940	3440	3456	DCA I	TEMP	/SAVE NEW WORD
1941	3441	5632	JMP I	SFS4	
1942					
1943					
1944					
1945	3442	0000	SRS4,	0	/SET RS4 (DON'T RELO TO FIELD 4)
1946	3443	7200	CLA		
1947	3444	4507	SETRS		/SETUP BANK POINTER
1948	3445	1456	TAD I	TEMP	
1949	3446	0132	AND	[7560]	
1950	3447	1131	TAD	[200]	
1951	3450	3456	DCA I	TEMP	/SAVE NEW WORD
1952	3451	5642	JMP I	SRS4	
1953	3452	0000	SFS5,	0	/SET FS5 (DON'T TEST FIELD 5)
1954	3453	7200	CLA		
1955	3454	4506	SETFS		/SETUP BANK POINTER
1956	3455	1456	TAD I	TEMP	
1957	3456	0130	AND	[7660]	
1958	3457	1127	TAD	[100]	
1959	3460	3456	DCA I	TEMP	/SAVE NEW WORD
1960	3461	5652	JMP I	SFS5	
1961	3462	0000	SRS5,	0	/SET RS5 (DON'T RELO TO FIELD 5)
1962	3463	7200	CLA		
1963	3464	4507	SETRS		/SETUP BANK POINTER
1964	3465	1456	TAD I	TEMP	
1965	3466	0130	AND	[7660]	
1966	3467	1127	TAD	[100]	
1967	3470	3456	DCA I	TEMP	/SAVE NEW WORD
1968	3471	5662	JMP I	SRS5	
1969	3472	0000	SFS6,	0	/SET FS6 (DON'T TEST FIELD 6)
1970	3473	7200	CLA		
1971	3474	4506	SETFS		/SETUP BANK POINTER
1972	3475	1456	TAD I	TEMP	
1973	3476	0126	AND	[7720]	
1974	3477	1171	TAD	[30]	
1975	3500	136	DCA I	TEMP	/SAVE NEW WORD
1976	3501	5672	JMP I	SFS6	
1977	3502	0000	SRS6,	0	/SET RS6 (DON'T RELO TO FIELD 6)
1978	3503	7200	CLA		
1979	3504	4507	SETRS		/SETUP BANK POINTER
1980	3505	1456	TAD I	TEMP	
1981	3506	0126	AND	[7720]	

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1982 3507 1171 TAD [40
1983 3510 3456 DCA I TEMP /SAVE NEW WORD
1984 3511 5732 JMP I SR56 /SET FS7 (DON'T TEST FIELD 7)
1985 3512 0000 SFS7, 0
1986 3513 7200 CLA /SETUP BANK POINTER
1987 3514 4508 SETFS
1988 3515 1456 TAD I TEMP
1989 3516 0173 AND [7740
1990 3517 1170 TAD [20
1991 3520 3456 DCA I TEMP /SAVE NEW WORD
1992 3521 5712 JMP I SFS7 /SET RS7 (DON'T RELO TO FIELD 7)
1993 3522 0000 SRS7, 0
1994 3523 7200 CLA /SETUP BANK POINTER
1995 3524 4507 SETRS
1996 3525 1456 TAD I TEMP
1997 3526 0173 AND [7740
1998 3527 1170 TAD [20
1999 3530 3456 DCA I TEMP /SAVE NEW WORD
2000 3531 5722 JMP I SRS7
2001
2002 /TEST TEST STATUS
2003 /RETURN IF NO TEST
2004 /RETURN +2 IF ALL 0 TEST
2005 /RETURN +4 IF ALL 1 TEST
2006 /RETURN +6 IF 0000 - 7777 WCP
2007 /RETURN +8 IF 7777 - 0000 WCP
2008 /RETURN +10 IF 2525 - 5252 WCP
2009 /RETURN +12 IF 5252 - 2525 WCP
2010 /RETURN +14 IF MARCH PATTERN
2011
2012 3532 0000 TTS, 0
2013 3533 7200 CLA /CLEAR PATTERN NUM FOR NON TTY SYSTEMS
2014 3534 3355 DCA TSNUM
2015 3535 1037 TAD [0
2016 3536 0125 AND [7770
2017 3537 7450 SNA
2018 3540 5356 JMP TTYCHK /NO TEST
2019 3541 2355 ISZ TSNUM
2020 3542 2332 ISZ TTS
2021 3543 2332 ISZ TTS
2022 3544 7104 TTSO, CLL RAL
2023 3545 7421 ROL
2024 3546 7430 SLL /CHECK THIS TEST BIT
2025 3547 5356 JMP TTYCHK
2026 3550 2355 ISZ TSNUM
2027 3551 2332 ISZ TTS
2028 3552 2332 ISZ TTS
2029 3553 7521 SWP
2030 3554 5344 JMP TTSO /CHECK NEXT TEST BIT
2031
2032
2033 3555 0000 TSNUM, 0
2034
2035 3556 7200 TTYCHK, CLA
2036 3557 1076 TAD NOTTY /GET PROGRAM FLAG
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2037 3560 7710 SPA CLA /WAS THERE A TELETYPE AVAILABLE
2038 3561 5777 JMP PATERR /NO, GO HALT ON ERROR
2039 3562 5732 JMP I TTS /RETURN TO ERROR PRINTOUT
2040
2041 3577 2731 PAGE
2042 3600
2043
2044 /TEST FIELD STATUS
2045 /RETURN IF FIELD STATUS BIT SET (DON'T TEST FIELD)
2046 /RETURN +1 IF FIELD STATUS BIT RESET (TEST THIS FIELD)
2047
2048
2049 3600 0000 TFSO, 0
2050 3601 7200 CLA /SETUP BANK POINTER
2051 3602 4506 SETFS
2052 3603 1456 TAD I TEMP /FIELD 0
2053 3604 7200 SVA CLA
2054 3605 2700 ISZ TFSO
2055 3606 5600 JMP I TFSO
2056
2057 3607 0000 TFS1, 0
2058 3610 7200 CLA /SETUP BANK POINTER
2059 3611 4506 SETFS
2060 3612 1456 TAD I TEMP
2061 3613 7200 RAL /FIELD 1
2062 3614 7700 SVA CLA
2063 3615 2207 ISZ TFS1
2064 3616 5607 JMP I TFS1
2065
2066 3617 0000 TFS2, 0
2067 3620 7200 CLA /SETUP BANK POINTER
2068 3621 4506 SETFS
2069 3622 1456 TAD I TEMP
2070 3623 7200 RAL /FIELD 2
2071 3624 7700 SVA CLA
2072 3625 2217 ISZ TFS2
2073 3626 5617 JMP I TFS2
2074
2075 3627 0000 TFS3, 0
2076 3630 7200 CLA /SETUP BANK POINTER
2077 3631 4506 SETFS
2078 3632 1456 TAD I TEMP
2079 3633 7200 RAL /FIELD 3
2080 3634 7700 SVA CLA
2081 3635 2227 ISZ TFS3
2082 3636 5627 JMP I TFS3
2083
2084
2085 3640 0000 TFS4, 0
2086 3641 7200 CLA /SETUP BANK POINTER
2087 3642 4506 SETFS
2088 3643 1456 TAD I TEMP
2089 3644 7200 RAL
2090 3645 7000 RAL
```

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2291 3656 7700 SWA CLA /FIELD 4
2092 3647 2243 ISZ TFS4
2093 3650 5640 JMP I TFS4
2094 3651 0000 TFS5, 0
2095 3652 7200 CLA
2096 3653 4506 SETFS /SETUP BANK POINTER
2097 3654 1456 TAD I TEMP
2098 3655 7002 BSW
2099 3656 7010 RAL
2100 3657 7620 SWA CLA /FIELD 5
2101 3660 2251 ISZ TFS5
2102 3661 5651 JMP I TFS5
2103
2104 3662 0000 TFS6, 0
2105 3663 7200 CLA
2106 3664 4506 SETFS /SETUP BANK POINTER
2107 3665 1456 TAD I TEMP
2108 3666 7002 BSW
2109 3667 7700 SWA CLA /FIELD 6
2110 3670 2262 ISZ TFS6
2111 3671 5662 JMP I TFS6
2112
2113
2114 3672 0000 TFS7, 0
2115 3673 7200 CLA
2116 3674 4506 SETFS /SETUP BANK POINTER
2117 3675 1456 TAD I TEMP
2118 3676 7002 BSW
2119 3677 7004 RAL
2120 3700 7700 SWA CLA /FIELD 7
2121 3701 2272 ISZ TFS7
2122 3702 5672 JMP I TFS7
2123
2124 /
2125 /TEST RELOCATION STATUS
2126 /RETURN IF RELO STATUS BIT SET (DON'T RELO TO FIELD)
2127 /RETURN+1 IF RELO STATUS BIT RESET (RELO TO THIS FIELD)
2128 /
2129 3703 0000 TRS0, 0
2130 3704 7200 CLA
2131 3705 4507 SETRS /SETUP BANK POINTER
2132 3706 1456 TAD I TEMP
2133 3707 7700 SWA CLA /FIELD 0
2134 3710 2303 ISZ TRS0
2135 3711 5703 JMP I TRS0
2136
2137 3712 0000 TRS1, 0
2138 3713 7200 CLA
2139 3714 4507 SETRS /SETUP BANK POINTER
2140 3715 1456 TAD I TEMP
2141 3716 7004 RAL
2142 3717 7700 SWA CLA /FIELD 1
2143 3720 2312 ISZ TRS1
2144 3721 5712 JMP I TRS1
2145
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2146 3722 0000 TRS2, 0
2147 3723 7200 CLA
2148 3724 4507 SETRS /SETUP BANK POINTER
2149 3725 1456 TAD I TEMP
2150 3726 7006 RTL
2151 3727 7700 SWA CLA /FIELD 2
2152 3730 2322 ISZ TRS2
2153 3731 5722 JMP I TRS2
2154
2155 3732 0000 TRS3, 0
2156 3733 7200 CLA
2157 3734 4507 SETRS /SETUP BANK POINTER
2158 3735 1456 TAD I TEMP
2159 3736 7004 RAL
2160 3737 7006 RTL
2161 3740 7700 SWA CLA /FIELD 3
2162 3741 2332 ISZ TRS3
2163 3742 5732 JMP I TRS3
2164
2165 3743 0000 TRS4, 0
2166 3744 7200 CLA
2167 3745 4507 SETRS /SETUP BANK POINTER
2168 3746 1456 TAD I TEMP
2169 3747 7006 RTL
2170 3750 7006 RTL
2171 3751 7700 SWA CLA /FIELD 4
2172 3752 2343 ISZ TRS4
2173 3753 5743 JMP I TRS4
2174
2175 /*CB*/ PAGE
2176
2177
2178 3754 0000 TRS5, 0
2179 3755 7200 CLA
2180 3756 4507 SETRS /SETUP BANK POINTER
2181 3757 1456 TAD I TEMP
2182 3760 7202 BSW
2183 3761 7010 RAL
2184 3762 7620 SWA CLA /FIELD 5
2185 3763 2251 ISZ TRS5
2186 3764 5754 JMP I TRS5
2187
2188 3765 0000 TRS6, 0
2189 3766 7200 CLA
2190 3767 4507 SETRS /SETUP BANK POINTER
2191 3770 1456 TAD I TEMP
2192 3771 7202 BSW
2193 3772 7700 SWA CLA /FIELD 6
2194 3773 2365 ISZ TRS6
2195 3774 5765 JMP I TRS6
2196
2197 /PAGE
2198 /
2199
2200 4000 0000 TRS7, 0
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2201 4001 7200 CLA
2202 4002 4507 SETRS /SETUP BANK POINTER
2203 4003 1456 TAD I TEMP
2204 4004 7002 BSW
2205 4005 7004 RAL
2206 4006 7700 SMA CLA /FIELD 7
2207 4007 2200 ISZ TRS7
2208 4010 5700 JMP I TRS7
2209
2210
2211
2212
2213 /TYPEOUT ERROR HEADING
2214
2215 4011 0000 ERRND, 0
2216 4012 1076 TAD NOTTY /GET TTY FLAG
2217 4013 7710 SPA CLA
2218 4014 5611 JMP I ERRND /NO TELETYPE AVAILABLE DON'T PRINT
2219 4015 4777 JMS MES
2220 4016 4543 TEXT '*PR LOC FAIL ADR GOOD BAD PATTERN*'
2221 4017 2022
2222 4020 4014
2223 4021 1703
2224 4022 4040
2225 4023 0601
2226 4024 1114
2227 4025 4001
2228 4026 0422
2229 4027 4040
2230 4030 0717
2231 4031 1704
2232 4032 4040
2233 4033 0201
2234 4034 0440
2235 4035 4020
2236 4036 0124
2237 4037 2405
2238 4040 2216
2239 4041 0000
2240 4042 5611 JMP I ERRND
2241
2242
2243 /TYPEOUT PROGRAM TITLE
2244
2245 4043 0000 TITLE, 0
2246 4044 1076 TAD NOTTY /GET TTY FLAG
2247 4045 7710 SPA CLA /TTY AVAILABLE ?
2248 4046 5643 JMP I TITLE /NO. ABORT MESSAGE
2249 4047 4777 JMS MES
2250 4050 4543 TEXT '*PDP-0E EXT MEM DATA & CHKBD*'
2251 4051 4320
2252 4052 0420
2253 4053 5570
2254 4054 0540
2255 4055 0530
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4056 2440
4057 1505
4060 1540
4061 0401
4062 2401
4063 4046
4064 4003
4065 1013
4068 0204
4069 4300
2232 4070 5643 JMP I TITLE
2233
2234
2235 /TYPEOUT TO SET SWITCHES
2236
2237 4071 0000 SETSW, 0
2238 4072 1076 TAD NOTTY /GET TTY FLAG
2239 4073 7710 SPA CLA /IS THERE A TTY AVAILABLE
2240 4074 5671 JMP I SETSW
2241 4075 4777 JMS MES
2242 4076 4543 TEXT '*SELECT FIELD PARAMETERS*'
2243 4077 2309
2244 4100 1405
2245 4101 0324
2246 4102 4006
2247 4103 1105
2248 4104 1404
2249 4105 4020
2250 4106 0122
2251 4107 0115
2252 4110 0524
2253 4111 0522
2254 4112 2345
2255 4113 4300
2243 4114 5671 JMP I SETSW
2244
2245
2246
2247 /TYPEOUT 'NO RELOCATION'
2248
2249 4115 0000 PNDREL, 0
2250 4116 1076 TAD NOTTY /GET TTY FLAG
2251 4117 7710 SPA CLA /IS THERE A TTY ON SYSTEM
2252 4120 5715 JMP I PNDREL /NO. GO RUN TEST
2253 4121 4777 JMS MES
2254 4122 4543 TEXT '*NO RELOCATION, PROG IN FIELD*'
2255 4123 1217
2256 4124 22
2257 4125 0514
2258 4126 1703
2259 4127 0124
2260 4130 1117
2261 4131 1654
2262 4132 4020
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4133 2217
4134 0740
4135 1118
4136 4008
4137 1105
4140 1404
4141 4000
2256 4142 6224 RIF
2257 4143 7106 CLL RTL
2258 4144 7004 RAL
2259 4145 1124 TAD [6000
2260 4146 3350 DCA ZB
2261 4147 4777 JMS MES
2262 4150 0000 ZB. 0
2263 4151 7240 STA
2264 4152 3055 DCA HEAD1
2265 4153 5715 JMP I PNOREL
2266
2267 /
2268 /TYPEOUT 'RELOCATION'
2269 /
2270 4154 0000 PREL. 0
2271 4155 1076 TAD NOTTY /GET TELETYPE FLAG
2272 4156 7710 SPA CLA /PRINT MESSAGE 7
2273 4157 5754 JMP I PREL /NO TTY - DO NOT PRINT
2274 4160 4777 JMS MES
2275 4161 4543 TEXT *$PRCG WILL RELOCATE*
4162 3772
4163 07
4164 4027
4165 1114
4166 1440
4167 2205
4170 1417
4171 0301
4172 2405
4173 0000
2276 4174 7240 STA
2277 4175 3055 DCA HEAD1
2278 4176 5.54 JMP I PREL
2279
2280
2281
2282 /
2283 4177 2240 PAGE
4200
2284
2285 /
2286 /
2287 /
2288
2289
2290 /
2291 /RELOCATE THE PROGRAM
2292 /
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2293 4200 0000 RELO. 0
2294 4201 7200 CLA
2295 4202 3053 DCA COUNT /CLEAR ERROR COUNTER
2296 4203 3054 DCA MOVE /CLEAR MOVE COUNTER
2297 4204 2061 ISZ RELCNT /SEE IF ALL FIELDS DONE
2298 4205 5212 JMP .+5
2299 4206 1077 TAD NUMFLO
2300 4207 7711 CIA
2301 4210 0061 DCA RELCNT
2302 4211 4777 JMS ENDPAS
2303 4212 1176 TAD [6201
2304 4213 1051 TAD PROFLO
2305 4214 3225 DCA RELO2
2306 4215 1176 TAD [6201
2307 4216 1052 TAD TSTFLO
2308 4217 3227 DCA RELO3
2309 4220 1225 TAD RELO2
2310 4221 3232 DCA RELO4
2311 4222 7705 CLL CLA IAC RAL /AC+2
2312 4223 1227 TAD RELO3
2313 4224 3544 DCA RELO5
2314 4225 6201 RELO2, CDF 0 /MOVE FROM CF
2315 4226 1454 TAD 1 MOVE
2316 4227 6201 RELO3, CDF 0 /MOVE TO DF
2317 4230 3454 DCA 1 MOVE
2318 4231 1454 TAD 1 MOVE
2319 4232 6201 RELO4, CDF 0 /MOVE FROM DF
2320 4233 7041 CIA
2321 4234 1454 TAD 1 MOVE
2322 4235 7640 SZA CLA
2323 4236 4776 JMS ERRN /MOVE ERROR
2324 4237 2054 ISZ MOVE
2325 4240 5225 JMP RELC2
2326 4241 1053 TAD COUNT
2327 4242 7640 SZA CLA /SKIP IF MOVE ERROR
2328 4243 5600 JMP I RELO
2329 4244 6203 RELO5, CDF 0 /NEW PROGRAM FIELD
2330 4245 5600 JMP I RELO
2331
2332 /
2333 /INTERRUPT ROUTINE
2334 /
2335 4246 4300 INTROU, JMS SAVINT
2336 4247 6107 SPO /SKIP IF PARITY OPTION
2337 4250 5253 JMP .+3
2338 4251 6101 SWP
2339 4252 5775 JMP PARINT /PARITY ERROR
2340 4253 6031 KGF
2341 4254 5774 JMP EADINT /UNWANTED INTERRUPT
2342 4255 4773 JMS KBINT /KEYBOARD INTERRUPT
2343 4256 4772 INTR. JMS RESINT
2344 4257 7200 CLA
2345 4260 1075 TAD SMO
2346 4261 7421 MQL
2347 4262 6004 GTF /RESTORE MQ
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2348 4263 8005 RTF
2349 4264 7200 CLA
2350 4265 1974 TAD SAC /RESTORE AC
2351 4266 5400 JMP I C
2352
2353
2354
2355 /TURN INTERRUPT ON IF FIELD 0 AND PARITY OPTION INSTALLED
2356
2357 4267 0000 PAR, 0
2358 4270 7300 CLA CLL
2359 4271 6007 CAF
2360 4272 6107 SPD /SKIP ON PARITY OPTION
2361 4273 5657 JMP I PAR
2362 4274 6224 RIF
2363 4275 7850 SNA CLA /SKIP IF NOT FIELD 0
2364 4276 6001 ICN
2365 4277 5687 JMP I PAR
2366 4300 0000 SAVINT, 0
2367 4301 7200 CLA
2368 4302 1771 TAD SIXTY
2369 4303 3335 DCA A1
2370 4304 1770 TAD CNV
2371 4305 3336 DCA A2
2372 4306 1787 TAD S0
2373 4307 3337 DCA A3
2374 4310 1766 TAD S1
2375 4311 3340 DCA A4
2376 4312 1765 TAD S2
2377 4313 3341 DCA A5
2378 4314 1764 TAD YES
2379 4315 3342 DCA A6
2380 4316 1783 TAD TYPCH
2381 4317 3343 DCA A7
2382 4320 1762 TAD NO
2383 4321 3344 DCA A8
2384 4322 1761 TAD TYPE
2385 4323 3345 DCA A9
2386 4324 1760 TAD TYPSP
2387 4325 3346 DCA A10
2388 4326 1757 TAD RETURN
2389 4327 3347 DCA A11
2390 4330 1756 TAD ERRD90
2391 4331 3350 DCA A12
2392 4332 1755 TAD ERROR1
2393 4333 3351 DCA A13
2394
2395 4334 5754 JMP C80000 /CB/
2396
2397 4335 0000 A1, 0 /CB/
2398 4336 0000 A2, 0 /CB/
2399 4337 0000 A3, 0 /CB/
2400 4340 0000 A4, 0 /CB/
2401 4341 0000 A5, 0 /CB/
2402 4342 0000 A6, 0 /CB/
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```
2403 4343 0000 A7, 0 /CB/
2404 4344 0000 A8, 0 /CB/
2405 4345 0000 A9, 0 /CB/
2406 4346 0000 A10, 0 /CB/
2407 4347 0000 A11, 0 /CB/
2408 4350 0000 A12, 0 /CB/
2409 4351 0000 A13, 0 /CB/
2410
2411 4354 2400
2412 4355 2461
2413 4356 2460
2414 4357 2405
2415 4360 2400
2416 4361 5025
2417 4362 2313
2418 4363 2256
2419 4364 2240
2420 4365 2237
2421 4366 2236
2422 4367 2235
2423 4370 2222
2424 4371 2200
2425 4372 2192
2426 4373 1546
2427 4374 3200
2428 4375 3100
2429 4376 2503
2430 4377 5726
2431 PAGE /CB/
2432 C80000, /CB/
2433
2434
2435 4400 1777 TAD ERROR1+1
2436 4401 3315 DCA A14
2437 4402 1778 TAD A14
2438 4403 3316 DCA A15
2439 4404 1775 TAD TA
2440 4405 3317 DCA A16
2441 4406 1774 TAD TC
2442 4407 3320 DCA A17
2443 4410 1773 TAD T1
2444 4411 3321 DCA A18
2445 4412 1772 TAD T07
2446 4413 3322 DCA A19
2447 4414 1771 TAD T70
2448 4415 3323 DCA A20
2449 4416 1770 TAD T25
2450 4417 3324 DCA A21
2451 4420 1767 TAD T52
2452 4421 3325 DCA A22
2453 4422 1766 TAD T05
2454 4423 3326 DCA A23
2455 4424 1765 TAD T76
2456 4425 3327 DCA A24
```

```
2457
2458 4428 1754' TAD SAVINT /CB/
2459 4427 3231 DCA CBCCD1 /CB/
2460 4430 5831 JMP : CBCCD1 /CB/
2461
2462 4431 0000 CB00D1, 0 /CB/
2463
2464
2465
2466
2467 4432 0000 RESINT, 0
2468 4433 1200 CLA
2469 4434 1763' TAD A1
2470 4435 3782' DCA SIXTY
2471 4436 1761' TAD A2
2472 4437 3750' DCA C5V
2473 4440 1757' TAD A3
2474 4441 3755' DCA SC
2475 4442 1755' TAD A4
2476 4443 3754' DCA S1
2477 4444 1753' TAD A5
2478 4445 3752' DCA S2
2479 4446 1751' TAD A6
2480 4447 3750' DCA SES
2481 4450 1747' TAD A7
2482 4451 3746' DCA T/PCN
2483 4452 1745' TAD A8
2484 4453 3744' DCA MC
2485 4454 1743' TAD A9
2486 4455 3742' DCA TYPE
2487 4456 1741' TAD A10
2488 4457 3740' DCA TYPSP
2489 4460 1737' TAD A11
2490 4461 3736' DCA RETURN
2491 4462 1735' TAD A12
2492 4463 3734' DCA ENFCRD
2493 4464 1733' TAD A13
2494 4465 3732' DCA ERROR1
2495 4466 1315 T/D A14
2496 4467 3777' DCA ERROR1+1
2497 4470 1316 TAD A15
2498 4471 3776' DCA ADDER
2499 4472 1317 TAD A16
2500 4473 3775' DCA TN
2501 4474 1323 TAD A17
2502 4475 3774' DCA T0
2503 4476 1321 TAD A18
2504 4477 3773' DCA T1
2505 4500 1322 TAD A19
2506 4501 3772' DCA T07
2507 4502 1323 TAD A20
2508 4503 3771' DCA T70
2509 4504 1324 TAD A21
2510 4505 3770' DCA T25
2511 4506 1325 TAD A22
```

```
2512 4507 3767' DCA T52
2513 4510 1326 TAD A23
2514 4511 3766' DCA TCS
2515 4512 1327 TAD A24
2516 4513 3765' DCA T13
2517 4514 5832 JMP : RESINT
2518
2519 4515 0000 A14, 0 /CB/
2520 4516 0000 A15, 0 /CB/
2521 4517 0000 A16, 0 /CB/
2522 4520 0000 A17, 0 /CB/
2523 4521 0000 A18, 0 /CB/
2524 4522 0000 A19, 0 /CB/
2525 4523 0000 A20, 0 /CB/
2526 4524 0000 A21, 0 /CB/
2527 4525 0000 A22, 0 /CB/
2528 4526 0000 A23, 0 /CB/
2529 4527 0000 A24, 0 /CB/
2530
2531 /
2532 4532 2461
2533 4533 4351
2534 4534 2460
2535 4535 4350
2536 4536 2465
2537 4537 4347
2538 4540 2463
2539 4541 4348
2540 4542 2465
2541 4543 4345
2542 4544 2313
2543 4545 4344
2544 4546 2256
2545 4547 4343
2546 4550 2240
2547 4551 4342
2548 4552 2237
2549 4553 4341
2550 4554 2226
2551 4555 4340
2552 4556 2235
2553 4557 4337
2554 4558 1102
2555 4561 4336
2556 4562 2200
2557 4563 4335
2558 4564 4300
2559 4565 1332
2560 4566 13
2561 4567 1064
2562 4570 3050
2563 4571 3034
2564 4572 3020
2565 4573 3010
2566 4574 3000
```

```
2507 4575 2735
2508 4576 2466
2509 4577 2462
      4600
2570
2571 /
2572 /TYPEOUT 'NONE' FOR NO LEGAL FIELD SELECTION.
2573 /
2573 4600 1076 NOFLD, TAD NOTTY /GET THE TTY FLAG
2574 4601 7710 SPA CLA /WAS IT SET
2575 4602 5777 JMP PATA /YES NO TELETYPE DO NOT PRINT
2576 4603 4776 JMS MES
2577 4604 1617 TEXT "NONE"
      4605 1605
      4606 0000
2578 4607 5777 JMP PATA /SETUP SWITCHES AGAIN
2579
2580 /THIS ROUTINE ESTABLISHES THE PROPER ERROR ROUTINE TO GO TO
2581 /
2582
2583 4610 0000 SETERR, 0
2584 4611 7200 CLA
2585 4612 1775 TAD FLD CNT
2586 4613 0035 AND SR911
2587 4614 1374 TAD (ERRTAB /GET TO ERROR ROUTINE TO EXECUTE
2588 4615 3253 DCA SRS
2589 4616 1653 TAD I SRS /GET ROUTINE TO EXECUTE
2590 4617 3253 DCA SRS /SAVE IT
2591 4620 4F53 JMS I SRS /GO EXECUTE ROUTINE
2592 4621 1100 STA /ACA-1
2593 4622 1060 TAD FCNT /-1 TO NUMBER OF FIELDS TO DO
2594 4623 3060 DCA FCNT /SAVE NEW VALUE
2595 4624 1060 TAD FCNT
2596 4625 7640 SZA CLA /ANY FIELDS LEFT TO DO
2597 4626 8610 JMP I SETERR /YES CONTINUE TESTING
2598 4627 4776 JMS MES
2599 4630 4543 TEXT "X#DISCONNECTED"
      4631 0411
      4632 2303
      4633 1716
      4634 1005
      4635 0324
      4636 0504
      4637 0000
2600 4640 7402 HLT
2601 4641 5240 JMP -1 /DON'T CONTINUE
2602 4642 5610 JMP I SETERR
2603
2604 4643 3307 /
2605 4644 3327 ERRTAB, SR50
2606 4645 3400 SR51
2607 4646 3422 SR52
2608 4647 3442 SR53
2609 4650 3462 SR54
2610 4651 3502 SR55
2611 4652 3522 SR56
      SR57
```

```
2612
2613 4653 0000 SRS, 0
2614
2615 /
2616 4654 0000 PNTFLD, 0
2617 4655 4776 JMS MES
2618 4656 4543
2619 4657 1100 0
2620 4660 1060 TAD FCNT
2621 4661 0034 AND SR68 /ISOLATE BANK INFORMATION
2622 4662 7112 CLL RTR
2623 4663 7010 RAR /MOVE INTO POSITION
2624 4664 1167 TAD [760
2625 4665 4773 JMS TYPE /TYPE BANK SELECTION
2626 4666 1060 TAD FCNT /GET FLD CNT
2627 4667 0035 AND SP911
2628 4670 1167 TAD [280 /SET UP ASCII FOR FIELDS
2629 4671 4772 JMS TYPSP /TYPEOUT # OF FIELDS IN THIS SYSTEM
2630 4672 4776 JMS MES
2631 4673 0611 TEXT "FIELDS IN THIS SYSTEM"
      4674 0514
      4675 0423
      4676 4011
      4677 1640
      4700 2410
      4701 1123
      4702 4023
      4703 3123
      4704 2405
      4705 1500
2632 4706 4776 JMS MES
2633 4707 4543 TEXT "X#FIELDS SEL'D ARE "
      4710 0611
      4711 0514
      4712 0423
      4713 4023
      4714 0514
      4715 4703
      4716 4001
      4717 2205
      4720 4000
2634 4721 4771 JMS TQSEL
2635 4722 5654 JMP I PNTFLD
2636
2637 /
2638 /ROUTINE TO CHECK FOR CONSOLE PACKAGE ACTIVE
2639 /
2640 /IF CONSOLE PACKAGE ACTIVE, GO TO CONSOLE PACKAGE
2641 /RETURN CALL + 2
2642 /
2643 /IF CONSOLE PACKAGE NOT ACTIVE, RETURN CALL + 1
2644 /
2645 4723 0000 XCBAL, 0
2646 4724 3346 DCA CHRTMP /SAVE AC
2647 4725 1022 TAD 22 /GET MCW2
```



```
2731 5067 0000 FLD0AT, 0
2732 5070 1667 TAD I FLD0AT /GET LOCATION TO STORE IT IN
2733 5071 3016 DCA DATTMP /SAVE IT
2734 5072 2267 ISZ FLD0AT /UPDATE RETURN
2735 5073 1275 TAD EMMFLG
2736 5074 7650 SNA CLA /SKIP IF KTB ACTIVE
2737 5075 5002 JAP -5
2738 5076 112 TAD TSTFLD /GET FIELD BEING DONE
2739 5077 1110 RACA /BINARY NUMBER
2740 5100 3056 DCA TEMP /SAVE IT
2741 5101 5056 JAP -5 /PROCESS IT
2742 5102 1052 TAD TSTFLD
2743 5103 7112 CLL RTR
2744 5104 7010 RAR
2745 5105 3056 DCA TEMP /SAVE IT
2746 5106 4775 JMS SIXTY
2747 5107 0656 TEMP /LOCATION TO 00
2748 5110 5114 QQ /WHERE TO PUT IT
2749 5111 1115 TAD YY /GET DECODED VALUE
2750 5112 3716 DCA I DATTMP
2751 5113 5067 JMP I FLD0AT /AND EXIT
2752
2753 /
2754 5114 0000 CC, 0
2755 5115 0000 YY, 0
2756 5116 0000 DATTMP, 0
2757 /
2758
2759 /
2760 /MAKE A BINARY NUMBER OUT OF A FIELD CHANGE
2761 /
2762 5117 0000 XRACA, 0
2763 5120 3360 DCA RTEMP
2764 5121 1060 TAD RTEMP
2765 5122 0365 AND K104
2766 5123 3361 DCA RTEMP1 /SAVE BANK VALUE
2767 5124 1361 TAD RTEMP1
2768 5125 7002 SSW /5 INTO 11
2769 5126 7106 CLL RTL /MOVE INTO 9
2770 5127 7004 RAL /NOW 8
2771 5130 1261 TAD RTEMP1
2772 5131 7004 RAL
2773 5132 0374 AND 130 /ISOLATE BANK
2774 5133 3361 DCA RTEMP1
2775 5134 1360 TAD RTEMP
2776 5135 0373 AND 170 /ISOLATE FIELD
2777 5136 7112 CLL RTR
2778 5137 7010 RAR /INTO BIT 9-11
2779 5140 1361 TAD RTEMP1
2780 5141 5717 JMP I XRACA /EXIT WITH BINARY NUMBER IN THE AC
2781 /
2782 /MAKE A FIELD CHANGE OUT OF A BINARY NUMBER
2783 /
2784 5142 0000 XRACS, 0
2785 5143 3360 DCA RTEMP
```

```
2786 5144 1360 TAD RTEMP
2787 5145 7112 CLL RTR
2788 5146 7010 RAR /BANK IN 10-11
2789 5147 0372 AND 13
2790 5150 1371 TAD (BANKR /MAKE A POINTER
2791 5151 3217 DCA XRACA /SAVE THE POINTER
2792 5152 1360 TAD RTEMP
2793 5153 0370 AND 17 /ISOLATE FIELD
2794 5154 7106 CLL RTL
2795 5155 7004 RAL /NOW IN POSITION
2796 5156 1717 TAD I XRACA
2797 5157 5742 JMP I XRACS
2798 /
2799 5160 0000 RTEMP, 0
2800 5161 0000 RTEMP1, 0
2801 5162 0000 BANKR, 0
2802 5163 0004 4
2803 5164 0100 100
2804 5165 0104 K104, 104
2805 /
2806 5170 0007
2807 5171 5132
2808 5172 0003
2809 5173 0070
2810 5174 0030
2811 5175 0000
2812 5176 0005
2813 5177 2240
2814
2815 /
2816 /
2817 /
2818 /THE FOLLOWING TEST IS A MARCH PATTERN DEVELOPED FOR TESTING
2819 /THE MSB-C MOS MEMORY.
2820 /
2821 /
2822 /THE TEST SELECTED FOR THE MOS MEMORY TESTING IS A TYPICAL MARCH
2823 /PATTERN. THE TEST BEGINS BY LOADING THE ENTIRE MEMORY WITH
2824 /A 2525 PATTERN. THEN STARTING AT ADDRESS ZERO OF LOWEST POSSIBLE
2825 /FIELD THE TEST READS THE CONTENTS, COMPARES IT, AND THEN WRITES BACK
2826 /THE COMPLEMENT VALUE. THE PROCESS IS REPEATED THROUGHOUT THE ENTIRE
2827 /MEMORY.
2828 /
2829 /NEXT THE PROCESS REPEATS FROM MAXIMUM TO MINIMUM, COMPLEMENTING
2830 /AS IT IS BEING DONE.
2831 /
2832 /THE ENTIRE SEQUENCE IS THEN REPEATED USING A BACKGROUND OF
2833 /0122. THIS INSURES THAT A ONE AND A ZERO CAN BE WRITTEN INTO
2834 /EACH MEMORY CELL.
2835 /
2836 /
2837 /
2838 5200 0000 POSTST, 0
2839 5201 7244 CLL CLA CVA RAL /AC=2
```

```
2840 5202 3364 DCA PATCNT
2841 5203 7344 CLL CLA CVA RAL /AC=-2
2842 5204 3362 DCA TSTCNT
2843 5205 7336 CLL CLA
2844 5206 3777 DCA FLOCNT
2845 5207 3624 DCA BANK /CLEAR INDICATORS
2846 5210 7301 CLL CLA IAC
2847 5211 3360 DCA ADDINC
2848 5212 7301 CLL CLA IAC
2849 5213 3361 DCA FLDINC
2850 5214 4778 MOSLOD, JMS MUFLD /TEST FOR VALID FIELD SELECTION
2851 5215 5232 JVP MUUPD /GO UPDATE FIELD VALUE
2852 5216 1051 TAD MUFLD /GET CURRENT FIELD
2853 5217 1176 TAD [6201 /MAKE IT A CDF
2854 5220 3227 DCA MUFLD /SAVE FOR RETURN
2855 5221 1052 TAD TSTFLD
2856 5222 1176 TAD [6201 /MAKE TEST FIELD A CDF
2857 5223 3225 DCA -2
2858 5224 1385 VENLOD, TAD PAT1 /FILL MEMORY WITH BACKGROUND
2859 5225 6201 CDF /CHANGED TO LOAD FIELD
2860 5226 3457 DCA I TSTAD
2861 5227 6201 MOSFLD, CDF /MAKE DF=PROFLD
2862 5230 3057 ISZ TSTAD /SEE IF ALL DONE
2863 5231 5224 JMP MUFLD /GO BACK AND TRY IT AGAIN
2864 /UPDATE TEST FIELD VALUE AND TEST AGAIN
2865 /
2866 5232 7200 MUUPD, CLA
2867 5233 1177 TAD FLOCNT
2868 5234 1141 CIA
2869 5235 1060 TAD FCNT /TEST FOR MAX VALUE
2870 5236 7850 SNA CLA /SKIP IF NOT AT MAX
2871 5237 5242 JMP -3 /AT MAX START READING
2872 5240 2777 ISZ FLOCNT /UPDATE FIELD TO DO
2873 5241 5214 JMP MUFLD /GO BACK AND TEST THIS FIELD VALUE
2874 /
2875 /AT THIS POINT ALL MEMORY IS FILLED WITH BACKGROUND 2525
2876 /
2877 5242 4406 JMS I IAPTOX /NOTIFY APT IF REQUIRED.
2878 5243 3777 DCA FLOCNT
2879 5244 3624 DCA BANK /CLEAR INDICATORS AGAIN FOR READ CYCLE
2880 5245 4778 MOSRED, JMS MUFLD
2881 5246 5300 JMP MUUPD /NOT A VALID SELECTION.
2882 5247 1365 TAD PAT1 /SET UP COMPARISON
2883 5250 3672 DCA GOATA
2884 5251 1051 TAD PROFLD /GET CURRENT FIELD LOCATION
2885 5252 1176 TAD [6201 /MAKE IT A CDF
2886 5253 3264 DCA REDFLD /SET UP RETURN
2887 5254 1052 TAD TSTFLD /GET FIELD TO READ
2888 5255 1176 TAD [6201 /MAKE IT A CDF
2889 5256 3257 DCA -1 /AND SAVE IT FOR USE
2890 5257 6201 REDLUR, CDF /CHANGE TO TEST FIELD CDF
2891 5260 1457 TAD I TSTAD /GET VALUE IN SELECTED FIELD
2892 5261 3073 DCA BOATA /SAVE IT FOR COMPARISON
2893 5262 1365 TAD PAT2 /NOW WRITE BACK COMPLIMENT VALUE
2894 5263 3457 DCA I TSTAD /BACK INTO SELECTED FIELD
```

```
2895 5264 6201 REDFLD, CDF /CHANGED TO CURRENT CDF
2896 5265 1072 TAD GOATA
2897 5266 7041 CIA
2898 5267 1073 TAD BOATA /SET UP COMPARISON
2899 5270 7640 SZA CLA /SKIP IF EQUAL
2900 5271 4775 JMS MOSERR /GO REPORT ERROR
2901 5272 1057 TAD TSTAD
2902 5273 1176 TAD ADDINC /ADD IN ADDRESS OFFSET.
2903 5274 1176 DCA TSTAD /AND RESTORE NEW VALUE
2904 5275 1057 TAD TSTAD
2905 5276 7640 SZA CLA
2906 5277 5257 JMP REDLUR /GO BACK AND DO THE NEXT
2907 5300 1361 MUUPD, TAD FLDINC
2908 5301 7710 SPA CLA /SKIP IF READING LOW TO HIGH
2909 5302 5305 JMP -3 /BY PASS COMPARISON
2910 5303 1060 TAD FCNT
2911 5304 7641 CIA
2912 5305 1777 TAD FLOCNT
2913 5306 7750 SNA CLA /SKIP IF NOT AT MAX
2914 5307 5314 JMP -5
2915 5310 1361 TAD FLDINC /ADD IN FIELD OFF SET VALUE
2916 5311 1777 TAD FLOCNT /TO THE CURRENT FIELD POSITION
2917 5312 3777 DCA FLOCNT
2918 5313 5245 JMP MOSRED /GO BACK AND READ NEXT FIELD
2919 /
2920 /NOW UPDATE PATTERN TO LOAD AND READ BACK VALUE
2921 /
2922 5314 1361 TAD FLDINC
2923 5315 7041 CIA /NEGATE CURRENT FIELD INCREMENT VALUE
2924 5316 3261 DCA FLDINC /AND RESTORE IT
2925 5317 1360 TAD ADDINC /NOW DO SAME FOR THE ADDRESS OFFSET
2926 5320 7041 CIA
2927 5321 3360 DCA ADDINC
2928 5322 1360 TAD ADDINC
2929 5323 7700 SNA CLA /SKIP IF READING HIGH TO LOW
2930 5324 3231 JMP -5
2931 5325 7240 STA /AC TO -1
2932 5326 3257 DCA TSTAD /START AT ADDRESS 7777 OF HIGH FIELD
2933 5327 1583 TAD FCNT /START AT HIGHEST FIELD
2934 5330 3777 DCA FLOCNT
2935 5331 1265 TAD PAT1 /COMPLIMENT PATTERN
2936 5332 7640 DCA
2937 5333 3233 DCA PAT1
2938 5334 1366 TAD PAT2
2939 5335 7640 DCA
2940 5336 3266 DCA PAT2
2941 5337 4406 JMS I IAPTOX
2942 5340 2562 ISZ TSTCNT /ALL DONE ?
2943 5341 7610 SNA
2944 5342 5246 JMP -4
2945 5343 7336 CLL CLA CVA RTR /SET BIT ONE OF CS WORD
2946 5344 3236 DCA CS /ONES COMPLIMENT
2947 5345 5243 JMP MOSRED
2948 5346 3236 DCA CS
2949 5347 1365 TAD PAT1
```

```
2950 5350 7040 CMA
2951 5351 3385 DCA PAT1
2952 5352 1386 TAD PAT2
2953 5353 7040 CMA
2954 5354 3386 DCA PAT2
2955 5355 2384 ISZ PATCNT /SEE IF ALL PATTERNS DONE
2956 5356 5203 JMP MOSTST+3
2957 5357 5600 JMP I MOSTST
2958
2959 5360 0000 /
2960 5361 0000 ADDINC, 0
2961 5362 0000 FLDINC, 0
2962 5363 0000 TSTCNT, 0
2963 5364 0000 ADDCNT, 0
2964 5365 0000 PATCNT, 0
2965 5365 2525 PAT1, 2525
2966 5366 5252 PAT2, 5252
2967
2967 5375 5415 /
2968 5376 0707
2969 5377 2348
2970 5400 PAGE
2971 5400 0000 /
2972 5401 6040 XPRINT, 0
2973 5402 6041 TLS
2974 5403 5202 JMP -1
2975 5404 6042 TCF
2976 5405 7200 CLA
2977 5406 6031 NSF /IS KEY BOARD VIATING
2978 5407 5600 JMP I XPRINT /GET CHARACTER
2979 5410 6036 MRB /TEST FOR ACTIVE CONSOLE
2980 5411 4505 CBCL /NOT ACTIVE JUST IGNORE CHARACTER
2981 5412 7200 CLA /CLEAR FLAG
2982 5413 6032 NCC
2983 5414 5600 JMP I XPRINT
2984
2985
2986 /
2987 /
2988
2989 5415 0000 MOSERR, 0
2990 5416 2053 ISZ COUNT /UPDATE ERROR COUNT
2991 5417 1377 TAD (40
2992 5420 3037 DCA TS /SAVE TEST STATUS FOR PRINTOUT
2993 5421 3036 DCA CS
2994 5422 1072 TAD QDATA /DATA WRITTEN
2995 5423 4776 JMS GERRC
2996 5424 3053 DCA COUNT
2997 5425 1115 JMP I MOSERR
2998
2999
3000 /
3001 /
3002 /PRINT END OF PASS MESSAGE
3003
```

```
3004
3005 5426 0000 /
3006 5427 7200 CLAS
3007 5430 1076 TAD NOTTY
3008 5431 7710 SPA CLAS /SKIP IF TTY AVAILABLE
3009 5432 5626 JMP I ENDPAS
3010 5433 2255 ISZ PASSES
3011 5434 4775 JMS SIXTY
3012 5435 5455 PASSES
3013 5436 5451 ENDMES
3014 5437 4774 JMS MES
3015 5440 4345 TEXT *#END OF PASS *
3016 5441 0516
3017 5442 0440
3018 5443 1706
3019 5444 4020
3020 5445 0123
3021 5446 2340
3022 5447 0000
3023 5450 4774 JMS MES
3024 5451 0000 ENDMES, 0
3025 5452 0000 0
3026 5453 0000 0
3027 5454 5626 JMP I ENDPAS
3028
3029 /
3030 PASSES, 0
3031 0
3032
3033 /
3034 /APT/ ROUTINE TO HANDLE ERRORS UNDER CONTROL OF APT
3035 /
3036 APTER, 0
3037 IDP
3038 DCA APTIZ /APT/
3039 TAD 22 /SAVE ANYTHING IN THE AC
3040 SMA CLAS /GET MCW2
3041 JMP I APTER /SKIP IF APT ALIVE
3042 RIF
3043 TAD [G203 /APT/
3044 DCA I APTER1 /APTS CREATES A CDF INST.
3045 TAD I APTER1 /APT CDF IN PROM CODE
3046 DCA .+4
3047 TAD APTIZ /APT/MODIFY NEXT CDF INST.
3048 SNA 7450 /SEE IF ANYTHING WAS IN AC
3049 TAD ADDER /SKIP IF THERE WAS
3050 CDF /APT/AC=ERROR PC.
3051 NOP /APT/(MODIFIED CDF) DP=IF.
3052 JMP GS20 /APT/CALL APT - 'ERROR'.
3053
3054 /
3055 APTER1, GS23
```

```
3051 /
3052 /APT/ THIS ROUTINE INITIALIZES PROGRAM FOR APT
3053 /
3054 APT12, 0
3055 5622 0000 /MAKE SURE INTERRUPT IS OFF
3056 5623 0002 /GET APT CONTROL WORD
3057 5624 1022 /SKIP IF APT ENABLED.
3058 5625 7700 /
3059 5626 5774 /AC=1
3060 5627 140 /NOP CONSOLE TERMINAL
3061 5630 5076 /SET UP FOR AUTO SIZE
3062 5631 1375 /
3063 5632 3020 /APT ENABLED
3064 5633 5251 /
3065 5634 1076 /
3066 5635 7700 /SKIP IF NO TTY ON SYSTEM
3067 5636 5250 /
3068 5637 1021 /GET CONFIGURATION WORD 1
3069 5640 7710 /SKIP IF SOFTWARE SWITCHES TO BE USED
3070 5641 5747 /
3071 5642 1020 /GET PSEUDO SWITCH REGISTER
3072 5643 7840 /SKIP IF NO VALUE IN PSEUDO SWITCHES
3073 5644 5247 /
3074 5645 1375 /SETUP DEFAULT FOR AUTO SIZING
3075 5646 3020 /
3076 5647 7610 /BYPASS SAVING OS/B MONITOR
3077 5650 4774 /SAVE OSB MONITOR IN FIELD 1
3078 5651 5622 /AND EXIT
3079 /
3080 /
3081 /
3082 /
3083 /APT/ ROUTINE TO 'NOTIFY' APT THAT THE PROGRAM IS RUNNING OK.
3084 /
3085 5652 0000 APTOK, 0 /APT/
3086 5653 7200 CLA /APT/
3087 5654 1022 TAD /APT/UNDER APT CONTROL?
3088 5655 7700 SNA CLA /APT/SKP IF YES.
3089 5656 5274 JMP APTOK0 /
3090 5657 0002 IOF /APT/
3091 5660 6224 RIF /APT/AC=IF.
3092 5661 1123 TAD [6203 /APT/CREATE A CDF INST.
3093 5662 3873 DCA APTOK1 /SET UP APT CODE CDF
3094 5663 1673 TAD APTOK1
3095 5664 3265 DCA /
3096 5665 6201 CDF /APT/MODIFY NEXT LOC.
3097 5666 7000 NOP /APT/INCDIF ED CDF) OF=CURRENT IF.
3098 5667 4773 JMS 6500 /APT/CALL APT - 'PRG CK'.
3099 5670 5652 JMP APTOK /APT/RTN FROM APT - RTN TO CALL=1.
3100 5671 0000 APTCT, 0
3101 5672 0000 APTCTY, 0
3102 5673 6505 APTOK1, 6505 /LOCATION TO OVERLAY FOR PROPER FIELD
3103 /
3104 /SEE IF KEY BOARD WAITING
3105 /
```

```
3106 5674 0000 APTOK0, 0
3107 5675 6031 KSF
3108 5676 5852 JMP I APTOK
3109 5677 6036 KRB
3110 5700 4505 CBCAL /TEST FOR CONSOLE
3111 5701 7200 CLA /IGNORE CHARACTER
3112 5702 6832 KCC
3113 5703 5652 JMP I APTOK /EXIT
3114 /
3115 /
3116 /THIS ROUTINE DETERMINES IF MEMORY IS CONTIGUOUS IS LOWER 32K
3117 /OF MEMORY. IF NOT FIELD 7 IS NOT TESTED.
3118 /
3119 5704 0000 APTFL, 0 /APT/
3120 5705 1022 TAD /GET APT CONTROL WORD
3121 5706 7700 SNA CLA /SKIP IF APT ENABLED
3122 5707 5704 JMP I APTFL /EXIT IF NOT
3123 5710 7333 CLL CLA CML IAD RTR /AC=6000
3124 5711 3337 DCA APTMOV /SET UP STARTING ADDRESS
3125 5712 6271 APTLUP, CDF 70 /POINTER TO PROM CODE
3126 5713 1737 TAD I APTMOV /GET AN ADDRESS
3127 5714 6201 CDF /FIELD ZERO
3128 5715 3737 DCA I APTMOV /SAVE THE VALUE
3129 5716 1737 TAD I APTMOV /GET BACK VALUE JUST MOVED
3130 5717 7041 CIA
3131 5720 6271 CDF 70 /BACK TO FIELD 7 FOR COMPARE
3132 5721 1737 TAD I APTMOV /GET BACK ORIGINAL VALUE
3133 5722 6201 CDF /BACK TO FIELD ZERO
3134 5723 7640 SZA CLA /SKIP IF EQUAL
3135 5724 4340 JMS MOVFAL /MOVE FAILURE. SOME BAD STUFF
3136 5725 2337 ISZ APTMOV /UP DATE ADDRESS POINTER
3137 5726 5312 JMP APTLUP /GO BACK AND TRY AGAIN
3138 /
3139 /AT THIS POINT THE APT PROM CODE IS SITTING IN THE PROGRAM
3140 /FIELD. FROM THIS POINT OUT ALL APT PROCESSING WILL BE DONE
3141 /IN THE FIELD BEING EXERCISED.
3142 /
3143 5727 1040 TAD FS /TEST TO SEE IF LOWER 32K CONTIGUOUS
3144 5730 7650 SNA CLA
3145 5731 5704 JMP I APTFL /ALL MEMORY CONTIGUOUS
3146 5732 1040 TAD FS /GET BACK FIELD STATUS
3147 5733 0072 AND 17740 /MASK OUT 7
3148 5734 1371 TAD 120 /NOP FIELD 7 TESTING
3149 5735 3040 DCA FS /AND RESTORE FIELD STATUS WORD
3150 5736 5704 JMP I APTFL /AND EXIT.
3151 /
3152 5737 0000 APTMOV, 0
3153 /
3154 5740 30 MOVFAL, 0
3155 5741 7240 STA
3156 5742 1340 TAD MOVFAL /ERROR PC
3157 5743 6201 CDF /ERROR FIELD
3158 5744 6272 CIF 70 /TO PROM CODE
3159 5745 5776 JMP 6220 /REPORT THE ERROR
3160
```

```
3161 /
3162 /
3163 /THE FOLLOWING LOCATIONS FROM 6000 TO 7777 ARE USED AS THE COMMUNICATIONS
3164 /INTERFACES FOR APT SHOULD APT BE AVAILABLE.
3165 /
3166 5771 5C23
3167 5772 7740
3168 5773 65CC
3169 5774 6036
3170 5775 6037
3171 5776 6620
3172 5777 2468
3173 5C00
3174 6000 4777 LOOP1, JMS SAVCF
3175 6001 1076 TAD NGTTY /GET THE TELETYPE FLAG
3176 6002 7710 SPA CLA /IS THERE CN ON THE SYSTEM
3177 6003 5223 JMP LOOP1A-1 /NO ABORT MESSAGE AND GALT
3178 6004 4776 JMS WES
3179 6005 4542 TEXT *%LOOP ON ADDRESS SET IN SR*
6006 1417
6007 1720
6010 4017
6011 1640
6012 6164
6013 0422
6014 0523
6015 2740
6016 05
6017 1440
6020 1116
6021 4023
6022 3200
3180 6023 4775 LOOP1A, JMS KLSDF
3181 6024 4504 GETSR
3182 6025 3235 DCA SR
3183 6026 1635 TAD I SR
3184 6027 7040 CMA
3185 6030 3635 DCA I SR
3186 6031 1135 TAD I SR
3187 6032 7040 CMA
3188 6033 3635 DCA I SR
3189 6034 5224 JMP LOOP1A
3190 6035 6000 SR, 0
3191 /
3192 /C0/ ROUTINE TO SAVE PAGE 37 OF FIELD 1
3193
3194 6036 0000 COSM, 0
3195 6037 7200 CLA
3196 6040 6224 RIF /READ THE INSTRUCTION FIELD
3197 6041 1374 TAD (6201 /ADD CDF 0 TO IT
3198 6042 3251 DCA COSM0 /MODIFY THE CDF INSTR AT LOC COSM0
3199 6043 1373 TAD (7577 /SET UP PAGE 37 POINTER -1
3200 6044 3010 DCA 10 /SAVE IN AUTO INDEX 10
3201 6045 1372 TAD (COSM-1 /GET ADDRESS -1 OF STORAGE AREA
```

```
3202 6046 3011 DCA 11 /SAVE IN AUTO INDEX 11
3203 6047 6211 COSM1, CDF 10 /CHANGE DATA FIELD TO 1
3204 6050 1410 TAD I 10 /GET THE WORD
3205 6051 6201 COSM0, CDF /CHANGE DATA FIELD TO PRG FIELD
3206 6052 3411 DCA I 11 /SAVE IN STORE AREA
3207 6053 1010 TAD 10 /CHECK TO SEE IF PAGE DONE
3208 6054 7040 CMA
3209 6055 7 0 SZA CLA /DONE SAVING PAGE
3210 6056 1447 JMP COSM1 /NO-DO NEXT WORD
3211 6057 5636 JMP I COSM /YES-RETURN TO CALL+1
3212
3213
3214 /C0/ ROUTINE TO RESTORE PAGES 37 OF FIELD 0 AND 1
3215
3216 6060 7200 COSM, CLA
3217 6061 6224 RIF /GET THE PRESENT DATA FIELD
3218 6062 1374 TAD (6201 /GET THE CDF INSTRUCTION
3219 6063 1367 DCA COSM0 /SAVE THE NEW CDF INSTRUCTION
3220 6064 1367 TAD COSM0
3221 6065 3276 DCA COSM01
3222 6066 1373 TAD (7577 /SET UP AUTO INDEX FOR RESTORE OF 0
3223 6067 3010 DCA 10 /SAVE IN AUTO INDEX 10
3224 6070 1372 TAD (COSM-1 /SETUP STORAGE POINTER
3225 6071 3011 DCA 11 /SAVE IN AUTO INDEX 11
3226 6072 1373 TAD (7577 /SETUP AUTO INDEX OF RESTORE OF FIELD 1
3227 6073 3012 DCA 12 /SAVE IN AUTO INDEX 12
3228 6074 1373 TAD (7577 /SETUP NEXT POINTER
3229 6075 3013 DCA 13 /SAVE IN AUTO INDEX 13
3230 6076 6201 COSM01, CDF
3231 6077 1013 TAD 13
3232 6100 7040 CMA
3233 6101 7652 SNA CLA /ALL DONE
3234 6102 5307 JMP COSM0
3235 6103 3411 TAD I 11 /GET DATA TO RESTORE
3236 6104 6211 CDF 10 /CHANGE DATA FIELD TO 1
3237 6105 3413 DCA I 13 /PUT IT IN FIELD 1
3238 6106 5175 JMS COSM01 /GO DO NEXT WORD
3239 6107 6201 COSM0, CDF /MODIFIED CDF INSTRUCTION TO PRG FIELD
3240 6110 1010 TAD 10 /RESTORATION DONE
3241 6111 7040 CMA
3242 6112 7650 SNA CLA /SKIP IF NO
3243 6113 5307 JMP COSM1 /DONE-GO TO MONITOR AT 7600
3244 6114 1410 TAD I 10 /GET DATA FROM PROSPAN FIELD
3245 6115 6201 CDF 00
3246 6116 3412 DCA I 12 /RESTORE 0
3247 6117 5307 JMP COSM0
3248 6120 6223 COSM1, CDF /CHANGE DATA AND INSTR FIELD TO 0
3249 6121 5722 JMP I 1
3250 6122 7600
3251
3252 6172 6177
3253 6173 7577
3254 6174 6201
3255 6175 5670
```

```
3257 6176 2240
3258 6177 6664
3259 6200 *C100 /C8/
3253 6200 4777 LCOP2, JMS SAYDF
3254 6201 1C76 TAD NCTTY /GET TELETYPE STATUS
3255 6202 7710 SPA CLA /IS THERE ONE ON THE SYSTEM
3256 6203 5234 JMP LCOP2A-2 /NO-ABORT MESSAGE AND HALT FOR INFO
3257 6204 4776 JMS MES
3258 6205 4543 TEXT *%LCOP ONLY THE 2 ADDRESSES INPUT FROM THE SR*
3259 6206 1417
3260 6207 1720
3261 6210 4017
3262 6211 1614
3263 6212 3140
3264 6213 2410
3265 6214 0540
3266 6215 6240
3267 6216 01C4
3268 6217 0422
3269 6220 0523
3270 6221 2305
3271 6222 2340
3272 6223 1116
3273 6224 0025
3274 6225 4440
3275 6226 0622
3276 6227 1715
3277 6230 4024
3278 6231 1035
3279 6232 4023
3280 6233 22C0
3281 6234 4245 JMS IN12
3282 6235 4775 JMS RESDF
3283 6236 1731 LCOP2A, TAD I FIRST
3284 6237 7040 CMA
3285 6240 3731 DCA I FIRST
3286 6241 1732 TAD I SECOND
3287 6242 7040 CMA
3288 6243 3732 DCA I SECOND
3289 6244 5236 JMP LCOP2A
3290 6245 0C00 IN12, 0
3291 6246 1076 TAD NCTTY /GET TELETYPE FLAG
3292 6247 7710 SPA CLA /IS THERE ONE ON THE SYSTEM
3293 6248 8274 JMP IN12A /NO-ABORT MESSAGE AND HALT FOR INFO
3294 6251 4776 JMS MES
3295 6252 4543 TEXT *%SET SR TO FIRST ADDRESS & CONT*
3296 6253 2175
3297 6254 1440
3298 6255 1322
3299 6256 4024
3300 6257 1740
3301 6260 0611
3302 6261 2223
3303 6262 2440
```

```
6263 0104
6264 0422
6265 0523
6266 2340
6267 4640
6270 0317
6271 1824
6272 0000
3281 6273 4505 IN12A, C8CAL
3282 6274 74C2 HLT
3283 6275 4504 GETSR
3284 6276 3331 DCA FIRST
3285 6277 0276 TAD NCTTY /GET FLAG STATUS AGAIN
3286 6278 7710 SPA CLA /TELETYPE AVAILABLE?
3287 6279 9325 JMP FIRST-4 /NO-ABORT MESSAGE AND HALT FOR INFO
3288 6282 4776 JMS MES
3289 6283 4543 TEXT *%SET SR TO SECOND ADDRESS & CONT*
3290 6284 2329
3291 6285 2440
3292 6286 2322
3293 6287 4024
3294 6290 1740
3295 6291 2005
3296 6292 0317
3297 6293 16C4
3298 6294 4001
3299 6295 0174
3300 6296 14C5
3301 6297 1323
3302 6298 4046
3303 6299 4023
3304 6300 1716
3305 6303 24C0
3306 6304 4105 C8CAL
3307 6305 74C2 HLT
3308 6306 4504 GETSR
3309 6307 3732 DCA SECOND
3310 6308 3145 JMP I IN12
3311 6309 0000 FIRST, 0
3312 6310 0000 SECOND, 0
3313 6311 0676
3314 6312 2240
3315 6316 6664
3316 6317 6400 *6400 /C8/
3317 6400 4777 LCOP3, JMS SAYDF
3318 6401 1C76 TAD NCTTY /GET THE TELETYPE STATUS
3319 6402 7710 SPA CLA /IS THERE A TELETYPE AVAILABLE?
3320 6403 5234 JMP LCOP3A-6 /NO-ABORT MESSAGE AND HALT FOR INFO
3321 6404 4776 JMS MES
3322 6405 4543 TEXT *%LOOP FROM FIRST ADDRESS "HRL" SECOND ADDRESS*
3323 6406 1417
```

```
6407 1720
6410 4008
6411 2217
6412 1540
6413 6611
6414 2223
6415 2440
6416 0 14
6417 6422
6420 0523
6421 2340
6422 2410
6423 2225
6424 4023
6425 0503
6426 1716
6427 6440
6430 0104
6431 0122
6432 0523
6433 2300
3310 6434 4775' JMS IN12
3311 6435 1774' TAD FIRST
3312 6436 3263 DCA SRL1
3313 6437 1773' TAD SECOND
3314 6440 3264 DCA SRL2
3315 6441 4772' JMS RESDF
3316 6442 1263 LOOP3A, TAD SRL1
3317 6443 3262 LOOP3B, DCA SRL
3318 6444 1662 TAD I SRL
3319 6445 7040 CMA
3320 6446 3662 DCA I SRL
3321 6447 1662 TAD I SRL
3322 6450 7040 CMA
3323 6451 3662 DCA I SRL
3324 6452 1262 TAD SRL
3325 6453 7041 CIA
3326 6454 1264 TAD SRL2
3327 6455 7650 SNA CLA
3328 6456 5242 JMP LOOP3A
3329 6457 2262 ISZ SRL
3330 6460 5244 JMP LOOP3B
3331
3332
3333 6461 5200 JMP LOOP3
3334 6462 0000 SRL, 0
3335 6463 0000 SRL1, 0
3336 6464 0000 SRL2, 0
3337 6500 +6500 /CB/
3338
3339 8500 4777' LOOP5, JMS SAVDF
3340 8501 1076 TAD NOTTY
3341 8502 7710 SPA CLA
3342 8503 5336 JMP LOOP5A-6
3343 8504 4776' JMS MES
```

```
/GET TTY FLAG
/IS THERE A TELETYPE AVAILABLE
/NO-ABORT MESSAGE AND HALT FOR INFO
```

```
3344 6505 4543 TEXT *%LOOP DATA IN THE SR THRU THE ADDRESS SELECTION*
6506 1417
6507 1720
6510 4004
6511 0124
6512 0140
6513 1116
6514 4024
6515 1005
6516 4023
6517 2240
6520 2410
6521 2225
6522 4024
6523 1005
6524 4001
6525 0404
6526 2205
6527 2323
6530 4023
6531 0514
6532 0503
6533 2411
6534 1716
6535 0000
3345 6536 4775' JMS IN12
3346 6537 1774' TAD FIRST
3347 6540 3364 DCA SR5A
3348 6541 1773' TAD SECOND
3349 6542 3365 DCA SR5B
3350 6543 4772' JMS RESDF
3351 6544 1264 LOOP5A, TAD SR5A
3352 6545 3365 LOOP5B, DCA SR5C
3353 6546 4504 GETSR
3354 6547 3766 DCA I SR5C
3355 6550 1766 TAD I SR5C
3356 6551 3766 DCA I SR5C
3357 6552 1366 TAD SR5C
3358 6553 7041 CIA
3359 6554 1365 TAD SR5B
3360 6555 7650 SNA CLA
3361 6556 9344 JMP LOOP5A
3362 6557 2366 ISZ SR5C
3363 6560 9346 JMP LOOP5B
3364
3365 6561 4505 CBCL
3366 6562 7402 HLT
3367 6563 5700 JMP LOOP5
3368 6564 0 SR5A, 0
3369 6565 6000 SR5B, 0
3370 6566 0000 SR5C, 0
3371
3372 6572 6676
3373 6573 6332
3374 6574 6331
```

/START AGAIN WITH FIRST ADDRESS

/DO NEXT ADDRESS

/HALT RESULTED FROM ILLEGAL LIMITS

/FIRST ADDRESS OF GROUP

/LAST ADDRESS OF GROUP

/ADDRESS COUNTER

```
3375 6575 6245
3376 6576 2240
3377 6577 6664
3378 6600
3379 3800 4264 LQDP4, JMS SAVDF
3380 6601 1076 TAD NOTIFY /GET TTY STATUS
3381 6602 7710 SPA CLA /IS THERE CIE ON THE SYSTEM
3382 6603 5253 JMP LQDP4A-4 /NO-ABORT MESSAGE AND HALT FOR INFO
3383 6604 4777 JMS MES
3384 6605 4543 TEXT "X#LOOP DATA IN THE SR ON THE INPUT ADDRESS"
6606 1417
6607 1720
6608 4004
6609 0124
6610 0140
6611 1118
6612 4024
6613 1005
6614 4023
6615 2240
6616 1718
6617 4024
6618 1005
6619 4011
6620 1620
6621 2524
6622 4771
6623 034
6624 2205
6625 2223
6626 0000
3385 6627 4777 JMS MES
3386 6628 4543 TEXT "X#SET SR TO ADDRESS & CONT"
6629 2105
6630 2440
6631 2322
6632 4024
6633 1740
6634 0.04
6635 0422
6636 0523
6637 2340
6638 4640
6639 0317
6640 1624
6641 0000
3387 6652 4505 CBCAL
3388 6653 7402 HLT
3389 6654 4504 GETSR
3390 6655 3263 DCA SR4
3391 6656 4276 JMS RESDF /RESTORE DATA FIELD TO NEW
3392 6657 4504 LQDP4A, GETSR
3393 6660 3663 DCA I SR4
```

```
3395 6661 1663 TAD I SR4
3396 6662 5257 JMP LQDP4A
3397 6663 0000 SR4, 0
3398 6664 0000 SAVDF, 0
3399 6665 7200 CLA
3400 6666 6214 RDF
3401 6667 3275 DCA SAVE
3402 6670 6774 RIF
3403 6671 76 TAD [6201
3404 6672 3273 DCA .+1
3405 6673 6201 CDF 00 /PROGRAM DF
3406 6674 5664 JMP I SAVDF
3407 6675 0000 SAVE, 0
3408
3409 6676 0000 RESDF, 0
3410 6677 1275 TAD SAVE
3411 6700 76 TAD [6201
3412 6701 3302 DCA .+1
3413 6702 6701 CDF 00 /LOOP DF
3414 6703 5676 JMP I RESDF
3415
3416 6777 2240 /
3417 PAGE 7000
3418 COSA=.
3419
3420 /
3421 /
3422 /
3423 /IF ENTERED WITH AC=0000 THE SWITCH REGISTER
3424 /MODIFICATION ROUTINE IS ENTERED AUTOMATICALLY.
3425 /IF ENTERED WITH AC NOT EQUAL TO 0000, THE
3426 /KEYBOARD INPUT DECODER IS ENTERED AND IT IS ASSUMED
3427 /THAT THE AC CONTAINS THE ASCII CODE TO BE
3428 /CHECKED FOR A VALID CONTROL CHARACTER.
3429 /
3430 /
3431 /
3432 0205 RSTART-PATA
3433 0000 OFFSET=0
3434 /
3435 /
3436 7200 *7200
3437 /
3438 7200 0000 CBTEMP, 0 /TEMPORARY WORK AREA
3439 7201 6203 CBCDI, CIF CDF /USED TO CREATE CDI TO PROGRAM FIELD
3440 7202 6201 CBCDF, CDF /USED TO CREATE CDF TO CONSOLE FIELD
3441 7203 0000 CBSWR, 0 /SWITCH REGISTER SAVE AREA
3442 7204 0000 CBWDE, 0 /PRINT MODE SWITCH
3443 7205 0000 CBCNT, 0 /USED AS COUNTER
```

```
3444 7206 7775 CBW3. -3 /CONSTANT
3445 7207 7774 CBW4. -4 /CONSTANT
3446 7210 7773 CBW5. -5 /CONSTANT
3447 7211 7770 CS+10. -10 /CONSTANT
3448 7212 7520 CBW250. -250 /CONSTANT
3449 7213 0007 CBW7. 0007 /CONSTANT
3450 7214 0240 CBW240. 0240 /CONSTANT
3451 7215 0260 CBW260. 0260 /CONSTANT
3452 7216 0275 CBW275. 0275 /CONSTANT
3453 7217 0277 CBW277. 0277 /CONSTANT
3454 7220 0322 CBW322. 0322 /CONSTANT
3455 7221 0323 CBW323. 0323 /CONSTANT
3456 /
3457 /
3458 7222 0000 CBENR. 0
3459 7223 3200 DCA CBTEMP /SAVE AC
3460 7224 6214 RDF /READ PROGRAM FIELD
3461 7225 1201 TAD CBW3 /ADD CBI INSTRUCTION
3462 7226 3205 DCA CBENR /SAVE CBI TO PROGRAM FIELD TEMPORARILY
3463 7227 6224 RIF /READ CONSOLE FIELD
3464 7230 1202 TAD CBW3 /ADD CBI INSTRUCTION
```

```
3465 7231 3241 DCA CBFLD /SAVE CDF TO CONSOLE FIELD
3466 7232 1777 TAD 1 121 /GET MCM FROM PROGRAM FIELD
3467 7233 7710 SPA CLA /SKIP IF USING PSEUDO SWR
3468 7234 7614 LAS SWR /GET HARDWARE SWR AND SKIP
3469 7235 1776 TAD 1 120 /GET PSEUDO SWR
3470 7236 3203 DCA CBW3R /SAVE SWITCH REGISTER
3471 7237 1775 TAD 1 (INCODE) /GET MESSAGE ACTIVE FLAG
3472 7240 3204 DCA CBW3DE /SAVE MESSAGE ACTIVE FLAG
3473 7241 7602 CBFLD, MLT /MODIFIED CDF TO CONSOLE DATA FIELD
3474 7242 1222 TAD CBENR /GET RETURN ADDRESS
3475 7243 3774 DCA CBW3R /SAVE FOR EXIT
3476 7244 1205 TAD CBENR /GET CBI TO PROGRAM FIELD
3477 7245 3773 DCA CBWFLD /SAVE CBI TO PROGRAM FIELD FOR EXIT
3478 7246 1200 TAD CBTEMP /GET AC UPON ENTRY
3479 7247 7240 SZA /SKIP IF IT WAS ZERO
3480 7250 3772 JMP CBW3R /AC NOT ZERO, GO CHECK CTRL CHAR
3481 /
3482 /PRINT OUT SR*XXXX WHERE XXXX IS THE CURRENT CONTENTS
3483 /OF THE SWITCH REGISTER BEING USED (EITHER PSEUDO OR HARDWARE)
3484 /
3485 7251 4771 CBPSW, JMS CBW3R /DD A <CR> AND <LF>
3486 7252 1221 TAD CBW323 /GET ASCII CODE FOR "S"
3487 7253 4770 JMS CBW3R /PRINT "S"
3488 7254 1220 TAD CBW322 /GET ASCII CODE FOR "R"
3489 7255 4770 JMS CBW3R /PRINT "R"
3490 7256 1216 TAD CBW325 /GET ASCII CODE FOR "A"
3491 7257 4770 JMS CBW3R /PRINT "A"
3492 7258 1207 TAD CBW3R /AC--4
3493 7259 1206 DCA CBW3R /SET UP OCTAL DIGIT COUNTER
3494 7260 1203 TAD CBW3R /GET SWITCH REGISTER
3495 7261 7204 RAL /EXTRA ROTATE FOR LINK
3496 7262 7204 RAL /EXTRA ROTATE FOR LINK
3497 7263 7204 RAL /EXTRA ROTATE FOR LINK
3498 7264 7204 RAL /EXTRA ROTATE FOR LINK
3499 7265 7204 RAL /EXTRA ROTATE FOR LINK
3500 7266 1203 DCA CBW3R /SAVE ROTATED SWR
```

```
3499 7267 1203 TAD C85AR /GET ROTATED SWR
3500 7270 0213 AND C8K7 /MASK OFF DIGIT TO PRINT
3501 7271 1215 TAD C8A200 /ADD ASCII BASE CODE
3502 7272 4770 JMS C8TYP /PRINT AN OCTAL DIGIT
3503 7273 1203 TAD C85AR /GET SWR
3504 7274 2205 ISZ C8CNTR /INCREMENT LOOP COUNTER
3505 7275 5264 JMP C8LOPA /GO PRINT NEXT DIGIT
3506
3507 /ACCEPT KEYBOARD INPUT OF OCTAL DIGITS, <CR>, <LF>
3508 /CTRL/C OR CTRL/G. ALL OTHER CHARACTERS ARE INVALID
3509 /AND WILL BE ECHOED, FOLLOWED BY A "?".
3510 /A CARRIAGE RETURN, LINE FEED, AND A RESTART OF
3511 /THE SRXXXX ROUTINE
3512 /
3513
3514 7276 7300 CLA CLL
3515 7277 1210 TAD C875 /AC=5
3516 7278 3205 DCA C8CNTR /SET UP TO ACCEPT 5 CHARACTERS
3517 7279 3787 DCA C8BLD /CLEAR SWITCH REG. BUILD AREA
3518 7280 3786 DCA C8FLG /CLEAR SWR CHANGE SWITCH
3519 7281 1214 TAD C8K240 /GET ASCII CODE FOR SPACE
3520 7282 4770 JMS C8TYP /SPACE OVER ONE POSITION
3521 7283 4765 JMS C8TYP /GO WAIT FOR KEYBOARD INPUT
3522 7284 3200 DCA C8TEMP /SAVE INPUT CHARACTER
3523 7285 1200 TAD C8TEMP /GET CHARACTER
3524 7286 1264 TAD (-203
3525 7287 7490 SNA /SKIP IF NOT CTRL/C
3526 7288 5783 JMP C8CTLG /GO TO CTRL/C ROUTINE
3527 7289 1207 TAD C8M4 /AC=4
3528 7290 7450 SNA /SKIP IF NOT CTRL/G
3529 7291 5762 JMP C8CTLG /GO TO CTRL/G ROUTINE
3530 7292 1206 TAD C8M3 /SUBTRACT 3
3531 7293 7450 SNA /SKIP IF NOT LINE FEED
3532 7294 5761 JMP C8EXT1 /GO TO LINE FEED EXIT
3533 7295 1206 TAD C8M3 /SUBTRACT 3
3534 7296 7650 SNA CLA /SKIP IF NOT CARRIAGE RETURN
3535 7297 5760 JMP C8EXT2 /GO TO CARRIAGE RETURN EXIT
3536 7298 1200 TAD C8TEMP /GET CHARACTER
3537 7299 4770 JMS C8TYP /ECHO IT
3538 7300 1200 TAD C8TEMP /GET CHARACTER
3539 7301 1212 TAD C8M260
3540 7302 7510 SPA /SKIP IF >= TO ASCII CODE FOR ZERO
3541 7303 5351 JMP C8ERR /INVALID CHARACTER NOT OCTAL DIGIT
3542 7304 1211 TAD C8M10
3543 7305 7700 SNA CLA /SKIP IF <= ASCII CODE FOR SEVEN
3544 7306 5351 JMP C8ERR /INVALID CHARACTER NOT OCTAL DIGIT
3545 7307 7240 STA /AC=7777
```

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3546 7336 3766 DCA C8FLG /SET SWR CHANGE FLAG
3547 7337 1200 TAD C8TEMP /GET CHARACTER
3548 7338 0213 AND C8K7 /MASK TO 3 BITS
3549 7339 3200 DCA C8TEMP /SAVE OCTAL DIGIT
3550 7340 1787 TAD C8BLD /GET SWR BUILD AREA CONTENTS
3551 7341 7108 CLL RTL
3552 7342 7004 RAL /ROTATE TO BUILD SWR
3553 7343 1200 TAD C8TEMP /ADD NEXT OCTAL DIGIT
3554 7344 3787 DCA C8BLD /SAVE NEW SWR
3555 7345 2205 ISZ C8CNTR /INCREMENT OCTAL DIGIT COUNTER
3556 7346 5305 JMP C8SRLP /CONTINUE ACCEPTING OCTAL DIGITS
3557
3558 7351 7300 C8ERR, CLA CLL
3559 7352 1217 TAD C8K277 /GET ASCII CODE FOR "?"
3560 7353 4770 JMS C8TYP /PRINT "?"
3561 7354 4771 JMS C8CRLF /DO A <CR> AND <LF>
3562 7355 5261 JMP C8PSW /GO START OVER
3563 /
3564 /
3565 7360 7538
3566 7361 7525
3567 7362 7457
3568 7363 7485
3569 7364 7575
3570 7365 7510
3571 7366 7703
3572 7367 7400
3573 7368 7476
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3601 /
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3603 7420 1377 CBCTL, TAD (-203
3604 7421 7450 SNA /SKIP IF NOT CTRL/C
3605 7422 5265 JMP CBCTLC /CTRL/C TYPED EXIT TO MONITOR
3606 7423 1376 TAD (-4
3607 7424 7450 SNA /SKIP IF NOT CTRL/G
3608 7425 5257 JMP CBCTLG /CTRL/G TYPED GO PRINT "G"
3609 7426 1375 TAD (-12
3610 7427 7450 SNA /SKIP IF NOT CTRL/O
3611 7430 5255 JMP CBCTLO /CTRL/O TYPED
3612 7431 1374 TAD (-2
3613 7432 7450 SNA /SKIP IF NOT CTRL/S
3614 7433 5237 JMP CBCTLS /CTRL/S TYPED
3615 7434 3773 DCA /SET MESSAGE ACTIVE FLAG
3616 7435 2204 ISZ /TEST CTRL/S ACTIVE FLAG
3617 7436 5273 JMP CBECNO /GO ICHO CHARACTER AND RETURN TO PROGRAM
3618 /
3619 /CTRL/S HANDLER
3620 /
3621 7437 7240 CBCTLS, STA /AC=7777
3622 7440 3204 DCA /SET CTRL/S ACTIVE FLAG
3623 7441 1773 TAD /GET MESSAGE ACTIVE FLAG
3624 7442 7650 SNA CLA /SKIP IF CTRL/S TYPED WHILE MESSAGE ACTIVE
3625 7443 5337 JMP CBPFLD /RETURN TO PROGRAM
3626 /
3627 7444 7240 CBWAIT, STA /SET CONTROL S ACTIVE INDICATOR
3628 7445 74 J4 DCA /WAIT FOR KEYBOARD INPUT
3629 7446 3310 JMS CBTTY
3630 7447 1377 TAD (-203
3631 7450 7450 SNA /SKIP IF NOT CTRL/C
3632 7451 5265 JMP CBCTLC /CTRL/C TYPED EXIT TO MONITOR
3633 7452 1372 TAD (-16
3634 7453 7640 SNA CLA /SKIP IF CTRL/O
3635 7454 5244 JMP CBWAIT /NOT CTRL/C OR CTRL/O CONTINUE WAITING
3636 7455 3204 CBCTLO, DCA /CLEAR CTRL/S ACTIVE FLAG
3637 7456 5337 JMP CBPFLD /RETURN TO MAIN PROGRAM
3638 /
3639 /CONTROL G HANDLER
3640 /
3641 7457 4317 CBCTLG, JMS CBCTLF /DO A <CR> AND <LF>
3642 7460 1216 TAD /GET ASCII CODE FOR UP ARROW
3643 7461 4276 JMS CBTYP /PRINT UP ARROW
3644 7462 1215 TAD /GET ASCII CODE FOR "G"
3645 7463 4276 JMS CBTYP /PRINT "G"
3646 7464 5771 JMP CBPSW /GO TO "SR=XXXX" ROUTINE
3647 /
3648 /CONTROL C HANDLER
3649 /
3650 7465 3204 CBCTLC, DCA /SET CTRL/S ACTIVE FLAG
3651 7466 1216 TAD /GET ASCII CODE FOR UP ARROW
3652 7467 4276 JMS CBTYP /PRINT UP ARROW
3653 7470 1214 TAD /GET ASCII CODE FOR "C"
3654 7471 4276 JMS CBTYP /PRINT "C"
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3655 7472 5770 JMP CBPM /RESTORE MONITOR AND EXIT
3656 /
3657 7473 1767 CBECNO, TAD /GET CHARACTER
3658 7474 4276 JMS CBTYP /ECHO IT
3659 7475 5337 JMP CBPFLD /RETURN TO PROGRAM
3660 /
3661 /
3662 /
3663 /
3664 /PRINT ONE CHARACTER
3665 /
3666 7476 0000 CBTP, 0
3667 7477 2204 ISZ /TEST CTRL/S ACTIVE FLAG
3668 7500 7410 SKP /SKIP IF CTRL/S NOT ACTIVE
3669 7501 5244 JMP CBWAIT /GO WAIT FOR CTRL/O OR CTRL/C
3670 7502 6046 TLS /TRANSMIT CHARACTER
3671 7503 6041 TSF /TEST TTY FLAG
3672 7504 5303 JNP /WAIT FOR TTY FLAG
3673 7505 6042 TCF /CLEAR TTY FLAG
3674 7506 7200 CLA /CLEAR AC DO NOT CLEAR LINK
3675 /
3676 7507 5676 JMP I CBTYP /RETURN
3677 /
3678 /WAIT FOR KEYBOARD INPUT THEN EXIT WITH ASCII CODE IN AC
3679 /
3680 7510 0000 CBTTY, 0
3681 7511 6031 MSF /SKIP IF KEYBOARD FLAG SET
3682 7512 5311 JMP /WAIT FOR KEYBOARD INPUT
3683 7513 6036 KRB /READ KEYBOARD BUFFER CLEAR FLAG
3684 7514 0205 AND /MASK TO 7 BITS
3685 7515 1206 TAD /SET BIT 4
3686 7516 5710 JMP I CBTTY /RETURN
3687 /
3688 /EXECUTE A CARRIAGE RETURN AND LINE FEED
3689 /
3690 7517 0000 CBCTLF, 0
3691 7520 1212 TAD /GET ASCII CODE FOR CARRIAGE RETURN
3692 7521 4276 JMS CBTYP /GO EXECUTE THE CARRIAGE RETURN
3693 7522 1213 TAD /GET ASCII CODE FOR LINE FEED
3694 7523 4276 JMS CBTYP /GO EXECUTE THE LINE FEED
3695 7524 5717 JMP I CBCTLF /RETURN
3696 /
3697 /CONSOLE PACKAGE EXIT IF TERMINATED WITH LINE FEED
3698 /
3699 7525 4317 CBEXT1, JMS CBCTLF /DO A <CR> AND <LF>
3700 7526 1337 TAD /GET MODIFIED CDT TO PROGRAM FIELD
3701 7527 3230 DCA /SAVE FOR EXECUTION
3702 7528 7402 MLI /MODIFIED CDT TO PROGRAM FIELD
3703 7531 2203 ISZ /TEST SWR CHANGE FLAG
3704 7532 5601 JMP I CBSTR /RESTART PROGRAM WITHOUT CHANGE OF SWR
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3705	7533	1200	TAD	CSLD	/GET NEW SWITCH REGISTER
3706	7534	3768	CCA I	120	/SAVE IT IN PROGRAM FIELD
3707	7535	5601	JMP I	CBSTRY	/RESTART PROGRAM WITH NEW PSEUDO SWR
3708					
3709					/EXIT FROM CONSOLE PACKAGE IF TERMINATED WITH CARRIAGE RETURN
3710					/
3711	7536	4317	CBEXT2, JVS	C9CRLF	/ADD A KORN AND KLF
3712	7537	7402	CBPFLD, MLT		/MODIFIED C91 TO PROGRAM FIELD
3713	7540	7300	CLA CLL		/CLEAR AC AND LINK FOR RETURN
3714	7541	2203	ISZ	CBFLG	/TEST SWR CHANGE FLAG
3715	7542	5602	JMP I	CBRTN	/RETURN TO PROGRAM WITHOUT CHANGE OF SWR
3716	7543	1200	TAD	CSLD	/GET NEW SWITCH REGISTER
3717	7544	3768	CCA I	120	/SAVE IT IN PROGRAM FIELD
3718	7545	5602	JMP I	CBRTN	/RETURN TO PROGRAM
3719					
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A10	4348	C83LD	7400	C85A	8034	FIVE	9100
A11	4347	C8CAL	4835	C85M0	8051	FLOCNT	2348
A12	4350	C8C3F	7202	C85M1	8047	FLODAT	8067
A13	4351	C8C3J	7201	C85RLP	7301	FLOINC	5361
A14	4351	C8C4TL	7420	C85RTT	7401	FLOSEL	2102
A15	4351	C8C4T2	7425	C85WR	7200	FS	0040
A16	4357	C8C4LF	7417	C85TNP	7200	FS1	0041
A17	4350	C8C4LC	7465	C85TY	7510	FS2	0042
A18	4351	C8C4L3	7457	C85TP	7476	FS3	0043
A19	4352	C8C4LO	7455	C84AIT	7444	FSEND	1707
A2	4336	C8C4LS	7437	CAF	6007	FSSET	1615
A20	4323	C8ECM3	7473	CD:	6207	GOATA	0072
A21	4324	C8EXT1	7423	CFP	2020	GERRC	1454
A22	4325	C8EXT2	7331	CFP0	2012	GETSR	4504
A23	4326	C8EXT1	7525	CFP1	204J	GTF	6004
A24	4327	C8EXT2	7536	CFP2	2040	HCW1	0021
A3	4337	C8FLD	7241	CFP3	2057	HCW2	0022
A4	4340	C8FLG	7403	CFP4	2057	HEAD1	0055
A5	4341	C8K100	7411	CFPTNP	2107	IAPTR	0005
A6	4342	C8K177	7405	CHECK	2077	IAPTRM	0006
A7	4343	C84200	7406	CHECK0	2100	LEGAL1	1514
A8	4344	C8K212	7413	CMTAP	4740	IN12	8245
A9	4345	C8K215	7412	CMP	6104	IN12A	8274
ACL	7701	C8K240	7214	CMV	2222	INMODE	0023
ADDCNT	5363	C8K250	7215	COERR	2436	INNAME	0084
ADDER	2716	C8K275	7216	COUNT	0053	INTR	4256
ADDINC	1160	C8K277	7217	CRELQ	0030	INTROU	4246
APCTCX	1671	C8K303	7414	CS	0036	K104	5165
APCTCY	5672	C8K307	7415	CSME	0422	K400	4745
APTE08	0817	C8K322	7220	CSRO3	1607	KBINT	1346
APIER	5600	C8K323	7221	CUF	6254	KBINTC	1384
APIER1	5621	C8K336	7216	DATTNP	5116	KMBREL	0404
APIEX	5651	C8K7	7213	DFE1F	0057	KTB	0001
APIFL	5704	C8K77	7207	ENWFLG	0020	KTBREL	0334
APTIZ	5622	C8LCC	4744	ENDF	0063	KTEST	0552
APTLUP	5712	C8LOPA	7264	ENDHLT	4502	LEGAL	1470
APTRGV	5737	C8M10	7211	ENDNES	5457	LEGAL0	0085
APTOX	5652	C8M250	7212	ENDPAS	5420	LEGALA	1540
APTOX0	5674	C8M3	7208	ERRA	1400	LGLFLO	1551
APTOX1	5673	C8M4	7207	ERRA1	1417	LIMIT	2476
B	0067	C8M40	7410	ERRS	1420	LOOP1	6000
BADINT	3200	C8M5	7210	ERRB1	1437	LOOP1A	6024
BANK	0024	C8MCE	7204	ERRC	2600	LOOP2	6200
BANK0	1674	C8FFLC	7537	ERRCC	2707	LOOP2A	6238
BANK0A	1702	C8PSW	7291	ERRPD	4017	LOOP3	6400
BANKR	5162	C8PM	6060	ERRM	2503	LOOP3A	6442
BDATA	0073	C8RVO	6107	ERRM7	2645	LOOP3B	6444
BINTC	3227	C8RMO1	6076	ERRMO	2680	LOOP4	6600
B5W	7002	C8RM1	6120	ERRM1	2681	LOOP4A	6657
C80000	4400	C8RTN	7402	ERRTA	4843	LOOP5	6500

LOOP5A	6544	PNTOPT	1728	SETPAR	5000	ST55	3262
LOOP5B	6546	PREL	4194	SETREL	0400	SUF	8274
LRR	6240	PRINT	4503	SETAS	4537	SWP	7521
LUSR	6260	PROFLD	0351	SETSW	4077	TO	3000
LXW	6200	PSR	0020	SFSC	3277	T07	3020
MO	2313	QO	5114	SFS1	3317	TOUPD	0681
MOUPD	5232	RACA	4510	SFS2	3340	T1	3010
MIUPD	5710	RACE	4511	SFS3	3412	T1UPD	0700
MEMLOD	5124	RBELL	2413	SFS4	3432	T25	3050
MES	2240	RD42	1210	SFS5	3452	T52	3064
MINS	0101	RD4C	1215	SFS6	3472	T70	3034
MOSERR	5415	RD82	1283	SFS7	3512	TCS	0503
MOSFLD	5227	RD8C	1270	SFSTAB	3352	TEMP	0056
MOSLGD	5214	RDFLA	1200	SIATY	2230	TES9NK	2314
MOSRED	5245	RDFLCA	1213	SKCN	6000	TEST	0600
MOSTST	5200	RDFLCB	1258	SMP	6107	TEST0	0653
MOV	0054	RE4D	1327	SMQ	6076	TEST1	0670
NOVVAL	5740	REDFLD	5264	SPFLD	1714	TEST8	1000
NOL	7721	REOLUP	5257	SPO	6107	TFS	0737
NTP	2311	RELCNT	0061	SR	6035	TFS0	3600
NCEUB	1523	RELJ	4230	SRC0	0020	TFS1	3607
NCEW4	0732	RELQ2	4225	SRC1	0027	TFS2	3617
NDFLC	4600	RELQ3	4227	SRC2	0030	TFS3	3627
NOTAPT	5634	RELQ4	4232	SRC3	0037	TFS4	3640
NOTTY	0376	RELQ5	4244	SRC4	0030	TFS5	3651
MUFLD	0707	REM	6239	SRC5	0030	TFS6	3662
MUWFLD	0077	RESDF	6376	SRC6	6663	TFS7	3672
OFFSET	0000	RESINT	4222	SRC8	6564	TFSTAB	0740
OPT	1741	RETURN	2403	SRC9	6565	TITLE	4043
P2	0070	RFR	6250	SRC	6566	TWAR	2347
PAR	4267	RS	0044	SRC8	0034	TN	2735
PARERR	3154	RS1	0045	SRC11	0035	TSEL	1106
PARINT	3100	RS2	0046	SRL	6462	TSEL1	1136
PARORC	2687	RS3	0047	SRL1	6463	TR	0472
PASSES	9455	RSTANT	6120	SRL2	6464	TR0	3703
PAT1	5365	RTEVP	5163	SRS	4953	TR1	3712
PAT2	5366	RTEVP1	5161	SRS0	3077	TR2	3722
PAT3	5368	RTEVP2	5162	SRS1	3227	TR3	3732
PATCNT	5364	SC	2235	SRS2	3400	TR4	3743
PATERR	2731	S1	2235	SRS3	3422	TR5	3754
PATM	0081	S2	2237	SRS4	3442	TR6	3765
PATVO	0264	SAC	0074	SRS5	3462	TR7	4000
PATN	0270	SAVE	1803	SRS6	3500	TRSTAB	0473
PATNG	0073	SAVEF	6654	SRS7	3522	TS	0037
PATJ	0301	SAVE	6075	START	0324	TSNMJ	3155
PATCO	0314	SAVINT	4230	STARTF	0080	TSTAD	0057
PERIC	2625	SCS	3233	STCP	2487	TSTCNT	5362
PERCO	2677	SCS2	3272	STSC	3230	TSTFLD	0052
PERR	2616	SECCNT	6332	ST51	3237	T5	3532
PINF	5044	SETEPR	4210	ST52	3243	T50	3544
PNOREL	4115	SETFLD	0725	ST53	3250	TYCHK	3558
PNTFLD	4654	SETFS	4506	ST54	3256	TYPCN	2256

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TYPE 5025
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WRA1 1050
WRB 1063
WRB1 1072
WRFLD 1605
XCDCAL 4723
XENDML 0341
XGTSR 4747
XPRINT 5400
XRACA 5117
XRACB 5142
XSETFS 0522
XSETRS 0532
YY 5115
Z1 2644
Z10 2538
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ZB 4150

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ERRORS DETECTED: 0

LINKS GENERATED: 349

RUN-TIME: 8 SECONDS

3K CORE USED

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A13     2391  2408#  2491
A14     2393  2409#  2493
A15     2406  2495  2519#
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A26     2456  2515  2529#
A27     2373  2399#  2473
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A30     2379  2403#  2479
A31     2381  2404#  2481
A32     2383  2405#  2483
A33     2385  2406#  2485
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APTMGV   3124  3126  3128  3129  3132  3136  3152#
APTOK    314  3085#  3099  3108  3113
APTOKQ   3089  3106#
APTKK1   3093  3094  3102#
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BANK    78#  523  539
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C8CD1	3439#	3451							
C8CHTL	3400	3603#							
C8CHTR	3443#	3462	3476	3493	3504	3516	3555		
C8CRLF	3405	3561	3641	3670#	3695	3699	3711		
C8CTLG	3526	3509	3632	3650#					
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C8CTL5	3614	3621#							
C8ECMO	3617	3657#							
C8ENFR	3663	3458#	3474						
C8ERR	3541	3544	3558#						
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C8EXT2	3535	3711#							
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C8M4	3445#	3492	3527						
C8M40	3590#								
C8MS	3446#	3515							
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C8PFLD	3477	3625	3637	3659	3700	3712#			
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C8RTN	3475	3584#	3715	3718					

SEQ 0104

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C8SRLP	3521#	3556							
C8STRT	3583#	3704	3707						
C8SWR	3441#	3470	3494	3498	3499	3523	3538	3547	3549 3553 3657
C8TEMP	3438#	3459	3478	3522	3523	3536	3538	3547	3549 3553 3657
C8TTY	3521	3629	3680#	3686					
C8TYP	3467	3469	3431	3502	3520	3537	3560	3643	3645 3652 3654 3658 3666# 3676
	3692	3694							
C8WAIT	3627#	3635	3665						
CAF	52#	1802	2359						
CD1	41#	2329							
CFP	1092	1166#	1177	1201	1375	1378	1381	1384	1387 1390 1393 1396
CFP0	1170	1176#	1203	1219	1228				
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CFP3	1213#	1227							
CFP4	1205	1221#							
CFPTMP	1180	1184	1186	1188	1191	1197	1206	1213	1221 1231#
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CHECK0	1229	1230#							
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CHP	44#	1761	1779						
CHV	1300	1304	1306#	1316	2370	2472			
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CS	88#	185	424	758	805	1841	1846	2046	2948 2993
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CSRO3	298	254	279	1036#	1042	1043			
CUF	33#								
DATTMP	2733	2750	2756#						
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ENDMES	3013	3017#							
ENDPAS	2302	3005#	3009	3020					
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ERRA1	774	852#	904						
ERRB	811	831	845#	913					
ERRB1	821	914#	926						
ERRC	933	1560#	1586						
ERRCC	1586	1639#							
ERRHD	1580	2215#	2218	2221					
ERRM	1499#	1505	2223						
ERRM1	1517	1533#							
ERRORO	1460	1462	1474#	2390	2392	2496			
ERROR1	1468	1475#	2392	2435	2494				
ERRTAB	2587	2604#							

SEQ 0105

FCNT	125	199	203	329	548	1051	1178	2193	2594	2595	2620	2626	2669	2910	SEQ 0106
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	532	533	536	551	554	575	583	106	952	955	956	959	973	976	
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FLODNT	7872	2278	2912	2916	2917	2934									
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FLODNT	2849	2537	2915	2922	2924	2980									
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FS	121	187	254		457	709	1078	3143	3148	3149					
FS1	102	256	1101												
FS2	103	258	1103												
FS3	104	270	1105												
FSND	1099	1112													
FSSET	106	1049	1114	1116											
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GTF	45	177	1772	2347											
HCW1	71	3267													
HCW2	72	294	296	3055	3087	3120									
HEAD1	123	195	872	1526	1544	1578	1763	1804	2264	2277					
IAPTR	63	1452	1794												
IAPTR	64	745	2877	2941											
LEGAL	963	985													
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LEGAL	3278	3283													
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LEGAL	130	945	965	998											
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LEGAL	2765	2504													
LEGAL	2648	2504													
LEGAL	862	873	2342												
LEGAL	867	870													
LEGAL	346	354	371												
LEGAL	35														
LEGAL	338	376	393	398											
LEGAL	520	623													
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LEGAL	131	947	963	992											
LEGAL	954	975	991	999											
LEGAL	953	974	1002	1011	1012										
LEGAL	1484	1489													
LEGAL	3174														
LEGAL	3177	3181	3189												
LEGAL	3260														
LEGAL	3263	3268	3274												
LEGAL	3304	3333													
LEGAL	3307	3316	3328												

LOOP38	3318	3330														
LOOP4	3379															
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LOOP5	3399	3367														
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LOOP5B	3353	3363														
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LUSR	36															
LXM	36	177														
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MSERR	2500	2539	2927													
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MSLDD	2550	2573														
MSRED	2890	2878	2947													
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SEQ 0107

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QO	2748	2754																		
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RBELL	1437	1573																		
ROA2	750	751	753	797																
ROAC	758	788																		
ROB2	752	800	838																	
ROSC	805	835																		
ROFLD	542	743	798																	
ROFLDA	756	837																		
ROFLDB	790	803																		
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SEO 0117

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